Regulating Competing Payment Networks

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Abstract

Payment networks fund consumer rewards through merchant fees. Because merchants rarely surcharge, consumers fail to internalize costs and overuse credit cards relative to the social optimum. I develop a quantitative model of platform competition to compare policy solutions. Capping merchant fees reduces rewards and credit card use, increasing total welfare by $27 billion. Because consumers are sensitive to rewards but merchants are insensitive to fees, network competition inflates rewards and exacerbates the costs of over-adoption. Dual-routing mandates that increase consumer multi-homing redirect competition from rewards toward merchant fees, increasing welfare. The direction of competition matters, not just its intensity.

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I Introduction

In the U.S., Visa and Mastercard (MC) process 80% of card transactions and earn profit margins above 60% (**Visa2020**). At the same time, merchants pay around $120 billion in fees every year to accept cards (**Nilson2020a**). These facts have motivated two decades of litigation and legislation built on a view that high merchant fees reflect weak competition. Key policy interventions include the Durbin Amendment's cap on debit interchange fees, the proposed Credit Card Competition Act (**Andriotis2022**), and the Department of Justice (DOJ) monopolization lawsuit against Visa (**Au-Yeung2024**).

I find that merchant fees are indeed too high, but weak competition is the wrong diagnosis. Networks set rewards for consumers and fees for merchants. The division of costs matters because merchants typically charge the same price for all payment methods, a phenomenon known as "price coherence" (**Frankel1998**; **Stavins2018**).[1] Under price coherence, card users capture rewards while the cost of higher merchant fees is spread across all consumers through higher retail prices. I estimate that consumers are ten times more sensitive to rewards than merchants are to fees. This incentivizes networks to charge high merchant fees to fund generous consumer rewards. High-merchant-fee credit cards thus proliferate because of, not in spite of, intense network competition.

Three forces explain why networks tax merchants to subsidize consumers. First, consumer payment choices respond strongly to subsidies. Networks charge merchants interchange fees and pay the revenue to banks, giving them incentives to promote the networks' cards through consumer rewards and other forms of steering. A regulatory reduction in debit interchange fees — the Durbin Amendment — reduced debit card spending at regulated issuers by around 25% relative to exempt issuers. Second, accepting cards significantly increases merchant sales. Event-study evidence shows that, for merchants on the margin of acceptance, beginning to accept credit cards raises sales from credit users by around 12%. Third, while almost all merchants accept every network's cards, around 40% of consumers carry credit cards from only one network (i.e., they "single-home"). Together, these facts produce a partial "competitive bottleneck" (**Armstrong2006**): because enough consumers carry only one network's cards, merchants risk losing substantial sales by dropping a network, tilting competition toward rewards rather than fees.

Existing theory cannot resolve whether network competition raises or lowers fees and welfare. The U.S. market sits between two theoretical extremes. When all consumers single-home, each network is a monopoly gateway to its cardholders, so competi-

[1] This occurs even though cash discounts and card surcharges are largely legal. I explore surcharging both theoretically and empirically in Online Appendix **??**.

tion raises per-transaction merchant fees (**Armstrong2006**; **Edelman.Wright2015**). When consumers multi-home (i.e. carry cards from multiple networks), merchants can drop expensive networks without losing sales, so competition lowers fees (**Anderson.etal2018**; **Bakos.Halaburda2020**; **Teh.etal2022**).

I thus build a structural model in which payment networks compete as two-sided platforms. A mix of single- and multi-homing consumers choose payment methods based on rewards, merchant acceptance, and exogenous non-price characteristics. Merchants choose which cards to accept and set retail prices to maximize profits net of merchant fees. Multiproduct networks maximize profits by setting fees and rewards, balancing merchant acceptance against consumer adoption.

I estimate consumer and merchant preferences by matching the reduced-form facts, moments from payment surveys, and aggregate market-level data. The estimated parameters match several untargeted moments, including the acceptance response to American Express (AmEx) fee cuts, profit margins, and network costs from accounting data. A 1-bp increase in Visa credit rewards raises Visa's consumer market share by 3.7%, while a 1-bp increase in Visa credit merchant fees reduces merchant acceptance by only 0.4%. Networks thus face strong incentives to raise merchant fees to fund rewards.

Credit card adoption is socially excessive because of price coherence. Because merchants do not surcharge, each consumer who adopts a credit card raises retail prices for all other consumers. Consumers individually face incentives to distort their payment choices to capture cross-subsidies, but collectively prefer lower credit card use. A revealed preference argument illustrates the inefficiency. Since credit cards pay rewards whereas cash and debit do not, the marginal user of credit cards must prefer the non-price characteristics of cash and debit. High rewards reduce consumers' non-price utility from payment choices, but the rewards themselves are merely transfers funded by higher retail prices.

The structural model quantifies the magnitude of this coordination failure. I simulate a counterfactual in which credit card merchant fees are capped at 120 basis points (bp), roughly half their current level. Capping merchant fees reduces per-transaction margins, giving networks less reason to compete for consumers through rewards. Networks cut rewards, and some consumers switch away from credit cards. By revealed preference, these marginal switchers prefer cash or debit absent rewards, so the total welfare gain is the eliminated cross-subsidy. This reduction in credit card use ultimately raises total welfare by $27 billion annually. The gains from capping merchant fees exceed the $12 billion consumer welfare gain from the CARD Act (**Agarwal.etal2015**), suggesting that regulating networks is at least as important as regulating issuers.

The gains from credit fee caps are also progressive. Reduced merchant fees pass through to lower retail prices for all consumers, whereas the reduction in rewards falls mostly on high-income consumers who are more likely to use credit cards. Consumer surplus rises by 48 bp of consumption for low-income consumers, compared to only 15 bp for high-income consumers. My results show that there is no trade-off between equity and efficiency in regulating payment fees.

Two additional counterfactuals illustrate that the largest welfare losses in payment markets stem from the overuse of credit cards, not market power. First, repealing the Durbin Amendment's debit fee cap would raise total welfare by $7 billion per year. Under the usual market power story (**Cuesta.Sepulveda2021**), removing the debit fee cap should reduce welfare. But higher debit merchant fees fund higher debit rewards, drawing consumers away from high-reward credit cards onto a lower-fee network. This shrinks the cross-subsidy and increases total welfare. This substitution effect matches prior work on the Durbin Amendment (**Mukharlyamov.Sarin2025**) and illustrates that the current U.S. regime of capping debit but not credit fees is worse than laissez-faire.

Second, a merger to monopoly raises total welfare by $15 billion per year. Without competitive pressure to attract consumers, a monopoly network cuts rewards. Because consumers are highly sensitive to rewards, credit card use collapses. Per-transaction merchant fees rise, but credit volume falls so much that the total merchant fee burden shrinks. Consumers lose $11 billion as network markups increase sharply, but the reduction in credit card use raises total welfare. While a merger is not a realistic policy proposal, the counterfactual shows how network competition inflates the total merchant fee burden by funding rewards. These predictions align with rising U.S. interchange fees despite increasing network competition (**GAO2009**) and the global expansion of high-fee Buy Now, Pay Later products (**Berg.etal2024**). The problem is not market power stifling output but rewards competition generating excessive credit card adoption.

In the absence of credit fee caps, policy can also redirect competition by increasing consumer multi-homing. Dual-routing requirements, such as those proposed in the Credit Card Competition Act, mandate that issuers put multiple networks on each card. Every cardholder thus multi-homes by technological fiat, without having to sign up for a second card. I model these requirements as an increase in consumer multi-homing. When consumers multi-home, merchants can decline expensive networks without losing sales, shifting competition toward merchant fees rather than consumer rewards. Networks respond by cutting credit card fees by 13 bp and rewards by 15 bp. Total welfare rises. These routing requirements are effective even when the secondary network is a large player. Increasing multi-homing is also much more effective than introducing a public option, such as a Central Bank Digital Currency (CBDC), that competes as a debit card (**Berg.etal2024a**). High rates of consumer multi-homing, not low market shares, force networks to compete for merchants rather than consumers.

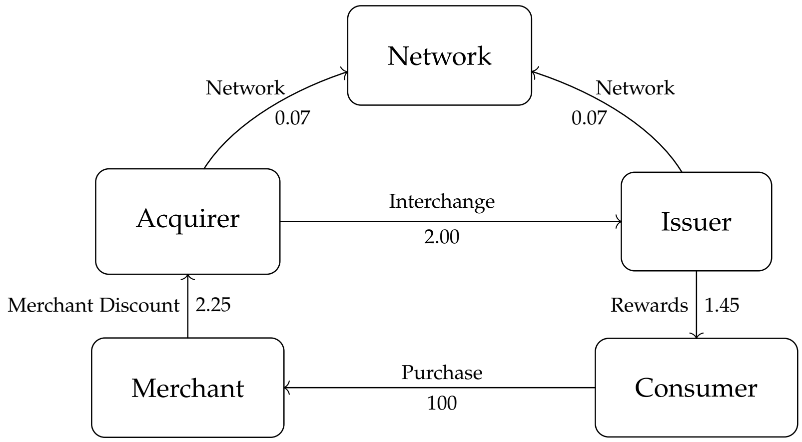
Related Literature The closest empirical work is **Huynh.etal2022<empty citation>**, which estimates a structural two-sided model of consumer and merchant card adoption. I build on their work by modeling merchant and network competition. Merchant competition explains how credit card rewards inflate retail prices and redistribute consumption. Network competition endogenizes merchant fees and consumer rewards, enabling me to assess how price controls and competition affect total welfare.

The effects of network competition on fees and welfare is a quantitative question, but common modeling assumptions predetermine the answer. Homogeneous merchants guarantee that competition reduces fees (**Guthrie.Wright2007**). Omitting merchant competition understates merchants' incentives to accept cards (**Rochet.Tirole2002**; **Wright2012**; **Li.etal2020**) and misses how fees pass through to retail prices. My model relaxes both assumptions and lets the data determine whether competition raises or lowers fees and welfare.

My paper contributes to the empirical IO literature on two-sided markets (**Rysman2004**; **Lee2013**; **Song2021**). My results on dual routing support the finding in **Rosaia2020<empty citation>** that consumer multi-homing intensifies platform competition. My paper also contributes to the literature on how platforms shape off-platform outcomes (**Bergemann.etal2024**; **Chen2024**; **Bursztyn.etal2025**). Under price coherence, merchant fees strongly influence retail prices, explaining the welfare gains from credit fee caps. **Sullivan2025<empty citation>** finds that commission caps reduce welfare in food delivery, where price coherence is absent.

My paper also relates to a literature on price regulation in credit card markets. Whereas **Nelson2020**; **Cuesta.Sepulveda2021<empty citation>** show how interest rate caps increase welfare by expanding quantities, my paper shows that credit card merchant fee caps can increase welfare by reducing the use of credit cards. **Knittel.Stango2003<empty citation>** document that non-binding state interest rate caps served as focal points for tacit coordination in credit card markets during the 1980s. The credit fee caps I study are binding constraints that push fees below their competitive levels, so they function as direct price regulation rather than as coordination devices.

Figure 1: Payment flows in a payment network.



Notes: Representative credit card fees in 2019. The merchant discount fee comes from **Nilson2020<empty citation>**. The average network fee comes from example rate sheets from acquirers and from dividing the non-foreign exchange fees from Visa's 10k by the total payment volumes (**Visa2020**; **Helcim2021**). I split the network fees evenly between the two sides as in (**FederalReserve2010**). The interchange is derived from Visa's interchange schedule as the average of the rates for Visa Signature and Visa Infinite cards at a large retailer (**Visa2019**). The rewards are from large banks' annual reports in 2019.

II Institutional Details and Data

Networks charge merchants fees and pay the revenue to issuers, who return most of it to consumers as rewards. The first subsection traces this fee flow, showing how merchant fees and rewards are linked. The second subsection describes the data sources used to identify how sensitive consumers and merchants are to rewards and fees.

II.A Network Pricing: Merchant Fees and Consumer Rewards

Payment networks are two-sided platforms that simultaneously set prices for merchants and consumers. While AmEx sets merchant fees and consumer rewards directly, "open-loop" networks like Visa and MC do so indirectly, through the *interchange fee* and *network fee*.

Visa and MC connect four parties: merchants, acquirers (merchants' banks), issuers (consumers' banks), and consumers (**Benson.etal2017**). Figure **??** shows the payment flows with representative prices. When a consumer uses her credit card to buy $100 of product at a large retailer, the merchant pays a $2.25 merchant discount fee to her acquiring bank to process the transaction. The acquirer can be a bank like Wells Fargo or a fintech firm like Square. Of the merchant discount, around $2 goes to the issuing bank (e.g., Chase) as interchange. The issuer and the acquirer collectively pay around $0.14 in network fees to Visa. Financial reports show that cardholder rewards average around $1.45 per $100 of transaction value. [2]

[2]These figures are similar to the regulatory numbers reported in **Drechsler.etal2025<empty citation>**,

Regulatory shocks confirm that interchange strongly affects merchant fees and rewards, but not borrowing costs. When the E.U. and Australia reduced interchange by regulation, merchant fees declined roughly one-for-one (**Gans2007**; **Valverde.etal2016**; **EuropeanCommission2020**). Appendix Figure **??** shows that after Australia capped credit card interchange, rewards fell and consumer annual fees on rewards cards rose, while annual fees on non-reward cards and interest rates were unchanged. Interchange has little effect on interest rates because balance-carriers pay 70% of interest but generate only 10% of purchase volume (**Adams.etal2022a**).

II.B Data

I combine five data sources to measure consumer reward sensitivity, how card acceptance affects sales, and multi-homing rates. Appendix **??** provides data construction details.

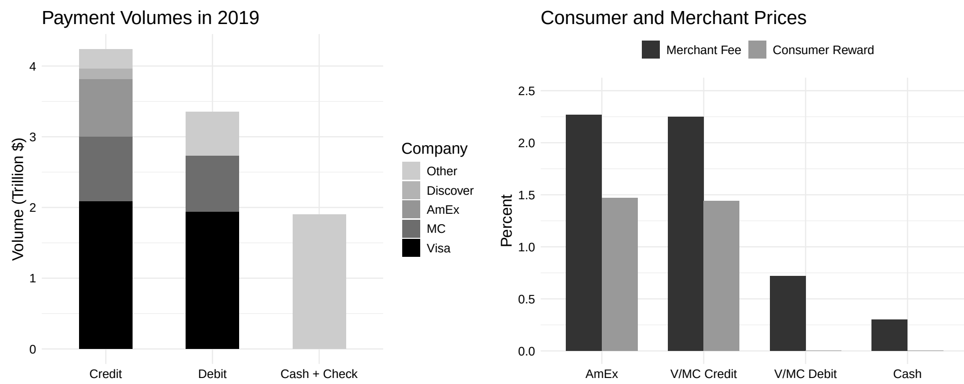
Aggregate Prices and Shares (2019): I use the Nilson Report's aggregate payment volumes and merchant fees in 2019. I collect average rewards data from the annual reports of the 6 largest banks, whose cards cover 80% of credit card purchase volumes. These data pin down baseline market shares and fee levels in the model. Figure **??** shows payment volumes, merchant fees, and rewards. Visa, MC, and AmEx process 90% of all card payments. All three major credit card networks charge merchant fees of around 2.25%, whereas debit networks charge around 0.7% due to the Durbin Amendment. Credit cards pay rewards around 1.4%, whereas aggregate debit rewards are effectively zero.

Issuer Payment Volumes (2007–2014): The Nilson Report publishes the dollar payment volumes of the top credit and debit card issuers annually. These issuers include both banks and large credit unions. I construct an annual panel of 39 issuers from the Nilson Report covering 2007–2014 to study how the Durbin Amendment affected payment volumes.

Homescan (2013–2023). The NielsenIQ Homescan panel tracks the payment decisions of around $100,000$ households at large consumer packaged goods stores from 2013–2023. I use this panel to measure how consumers split spending across the cards in their wallets and how card acceptance affects sales. Appendix Table **??** reports summary statistics. Appendix Table **??** compares Homescan payment shares to aggregate shares. The Homescan sample is broadly representative, with the exception that AmEx is underrepresented (about 4% versus 10% in aggregate Nilson data) because Homescan excludes sectors like travel where AmEx is more dominant. For this reason, AmEx-

which finds that banks' interchange income averages 1.82% of purchase volume while rewards costs average 1.57%.

Figure 2: Aggregate payment volumes, merchant fees, and rewards in 2019



Notes: The left chart shows payment volumes measured in trillions from **Nilson2020a**; **Nilson2020b<empty citation>**. Visa and MC operate credit and debit card networks, whereas AmEx primarily offers credit and charge cards. Discover is much smaller than the other three networks. The right chart shows merchant fees from **Nilson2020<empty citation>** and rewards from banks' annual financial reports. Aggregate debit rewards fell to effectively zero after the Durbin Amendment, though some small, exempt issuers continued to offer them (**Hayashi2012**). The cost of cash is from **Felt.etal2020<empty citation>**.

specific calibration moments (merchant fees, rewards) are sourced externally rather than from Homescan.

Consumer Payment Surveys. I combine the 2017–2019 waves of the Atlanta Federal Reserve's Diary of Consumer Payment Choice (DCPC) and its companion Survey (SCPC) to build a transaction-level dataset over three-day windows (**Greene.Stavins2021**). The rich demographic detail in these surveys identifies income gradients in payment preferences. The diary and survey share a common panel of respondents. Whereas Homescan oversamples large retailers, the DCPC captures transactions at small stores, allowing me to better estimate overall card acceptance rates. Table **??** shows summary statistics on consumers' payment preferences. Cards are accepted for around 95% of transactions.[3] Debit cards are the most popular payment method. Credit card users have higher incomes than cash or debit card users, and around one-quarter of transactions are online. I also conduct a second-choice survey in 2024 to estimate substitution patterns across payment methods (**Berry.etal2004**).

III Reduced-Form Facts

Three reduced-form facts characterize the two-sided structure of the payment market. Issuers' incentives drive consumer payment choices; card acceptance increases merchant sales; and a mix of single- and multi-homing consumers limits merchants' ability to

[3]Online Appendix Table **??** shows that diary-based acceptance rates closely match those from Yelp business listings.

Table 1: Summary statistics by consumer type in the Diary of Consumer Payment Choice (DCPC)

	Cash	Debit	Credit
Share	0.20	0.43	0.36
Owns credit card	0.70	0.79	1.00
Owns rewards credit card	0.47	0.53	0.90
Owns debit card	0.73	1.00	0.82
Owns bank account	0.86	1.00	0.99
Credit utilization	0.18	0.27	0.06
Household income (000's)	68.29	81.39	116.23
Card acceptance	0.93	0.96	0.96
Credit score above 650	0.65	0.70	0.96
Online transaction	0.19	0.27	0.28
N	1552	3265	2824

Notes: Consumers are split into three groups by preferred non-bill payment instrument. Share reports the fraction of the sample in each column. Card acceptance is the expenditure share at merchants that accept cards. All other variables report consumer-level averages. Online transaction reports the share of transactions done online.

steer between networks. These facts point toward a parital competitive bottleneck, but the degree of consumer multi-homing leaves the net effect of competition on fees and welfare an empirical question.

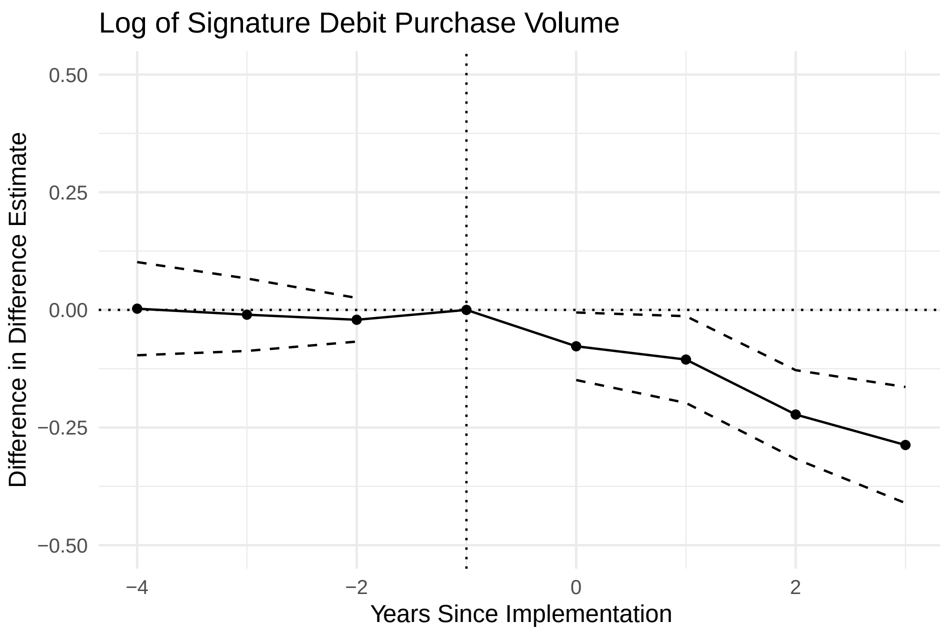
III.A Issuer Incentives Drive Consumer Payment Choices

The Durbin Amendment provides quasi-experimental evidence that issuers' incentives strongly affect consumer payment choices. Enacted as part of the 2010 Dodd-Frank Act, it capped debit interchange fees for banks and credit unions with over $10 billion in assets starting in October 2011, reducing large issuers' debit interchange revenue by roughly 75 bp, leaving credit interchange untouched. Large issuers raised checking account fees (**Kay.etal2018**; **Mukharlyamov.Sarin2025**), ended debit rewards programs (**Hayashi2012**; **Schneider.Borra2015**), cut debit card advertising (**Hayashi2012**), and scaled back bank teller sales incentives (**Johnson2010**). Many small issuers, by contrast, continued debit rewards programs and increased debit marketing (**CUOnline2012**; **Orem2016**).

I estimate the effect on payment volumes by comparing debit card volumes at large and small issuers.[4]

[4] Small issuers are an imperfect control group because they may have been indirectly affected by the Durbin Amendment through competitive spillovers. However, such spillovers are likely to attenuate, rather than amplify, the estimated treatment effect. If small issuers also reduced debit promotion in response to competitive pressure from large issuers shifting toward credit, the difference between treated and control groups would understate the true effect of losing debit interchange revenue.

Figure 3: The effect of the Durbin Amendment on debit card volumes



Notes: Data are from the Nilson Report. The points show the difference-in-differences estimates of the effects of the Durbin Amendment on debit card volumes. Dashed lines show 95% confidence intervals. The vertical line marks $t = -1$, which is 2010, the year before the policy took effect in Q3 2011. Standard errors are clustered at the issuer level.

$$y_{it} = \sum_{k \neq -1} \beta_k I\{t = k\} \times T_i + \delta_i + \delta_t + \epsilon_{it} \tag{1}$$

where y_{it} is the log of signature debit purchase volume at issuer i in year t, T_i indicates whether issuer i had more than \$10 billion in assets in 2010, and δ_i and δ_t are issuer and year fixed effects. I define $t = 0$ as 2011 and focus on institutions with between 2 and 200 billion in assets, excluding issuers that made large acquisitions during the sample period. This comparison of large and small issuers differences out the effects of the Durbin routing requirements, the CARD Act, and potential changes in merchant acceptance, all of which affect debit and credit card use symmetrically across issuer size.

Weakening large issuers' incentives to promote debit led to around a 25% relative decline in signature debit volumes. Figure **??** plots the estimated effects of being above the cutoff on signature debit card volumes. To evaluate how much of the decline could be explained by lost debit rewards, Online Appendix **??** restricts the sample to the 15 issuers that paid debit rewards pre-Durbin and compares issuers that cut rewards against those that kept them. Debit volumes at the banks that cut rewards grow about 25% more slowly than at those that kept rewards, a gap that matches the baseline estimate and informs the calibration of the rewards elasticity in Section **??**.

Online Appendix **??** presents robustness checks on the estimate and the mechanism. I focus on signature debit because it has the highest coverage in the Nilson Report. The relative decline drops to 17% for total debit volumes, although estimates are noisier because fewer banks report total debit early in the sample. Credit card volumes at large banks appear to rise relative to small banks after Durbin, though pre-trends are not cleanly zero. Treated and control banks had similar pre-Durbin payment mixes, the estimates are stable across asset cutoffs and merger exclusions, and the effect reflects within-bank payment switching rather than cross-bank switching or a post-Durbin increase in credit card rewards at large issuers.

III.B Card Acceptance Increases Merchant Sales

Merchants face strong incentives to accept cards because doing so increases sales. I identify the causal effect of credit card acceptance on sales using changes in merchant acceptance policies. Between 2015 and 2023, I identify multiple instances in the Homescan data where a grocery store begins or stops accepting credit cards. Homescan does not report acceptance directly. It records the payment method that consumers self-report for each trip. I infer acceptance changes from large shifts in credit card payment shares, validated with news sources, and classify a merchant as accepting credit cards only if more than 5% of its sales are paid by credit card.[5] I then compare the shopping behavior of consumers with high and low credit card usage at treated merchants relative to control grocers that did not change their acceptance policy.

The merchants in these event studies are highly selected. Around 98% of all Homescan trips occur at stores that already accept credit cards, so merchants that change acceptance are unusual. This pattern is expected if credit card acceptance increases sales for the bulk of merchants. When extrapolating these sales gains to the broader merchant population in the model, I account for this selection.

I compare spending by credit card users versus non-users, at merchants that changed acceptance versus those that did not, before versus after the change. I estimate

$$y_{hjte} \sim \text{Poisson}(\lambda_{hjte}) \tag{2}$$

$$\log \lambda_{hjte} = \delta_{ejtqz} + \phi_{ej} C_h + \sum_{k \neq -1} \beta_{ek} C_h \mathbf{1}\{t = k\} + \sum_{k \neq -1} \gamma_k C_h T_{je} \mathbf{1}\{t = k\} \tag{3}$$

where y_{hjte} is the dollars spent by household h at retailer j in period t relative to the

[5]Consumers occasionally misreport signature debit transactions as credit card payments (**Rysman.etal2025**), which can produce a small positive credit share even at non-accepting merchants.

acceptance change in event e, δ_{ejtqz} are event-retailer-period-income quintile-3-digit zip code fixed effects, $T_{je} = 1$ for treated grocers that begin to accept credit cards, $T_{je} = -1$ for those that stop, and C_h is household h's credit card share of payments measured in the year prior to the event. The coefficients of interest, γ_k, capture the dynamic effects of credit card acceptance on sales to credit card users.[6] I use a Poisson specification following **Cohn.etal2022<empty citation>**, who show that Poisson regression is preferred for difference-in-differences designs with non-negative outcomes that include many zeros.[7] Standard errors are clustered at the household level.

This strategy allows the focal merchants to experience other unobserved shocks contemporaneous with the acceptance change. The event-retailer fixed effects ϕ_{ej} absorb baseline spending differences between credit card users and non-users at each store, while the event-period interactions β_{ek} allow the credit-user spending gap to evolve over time in a way that is common across treated and control retailers. The key identifying assumption is that merchants do not differentially target credit card users relative to other consumers of similar income levels, so the strategy accommodates situations in which merchants target higher-income consumers at the same time that they start accepting credit cards. A violation would occur if the acceptance change were accompanied by a co-branded credit card campaign that simultaneously shifts both merchant acceptance and consumer card holdings. I verified that no such campaigns accompanied the events I study.[8]

Figure **??** shows that over the first two years of acceptance, sales to credit card users rise by approximately 12%. Online Appendix Table **??** reports the regression coefficients for both trips and spending. Even though most credit users own debit cards (Table **??**), accepting credit still increases sales. The Homescan data do not identify the mechanism directly, but merchant and network testimony from antitrust trials points to the credit line providing additional purchasing power (Online Appendix **??**). Credit is strongly preferred for larger purchases, and merchants report that debit does not substitute for it.

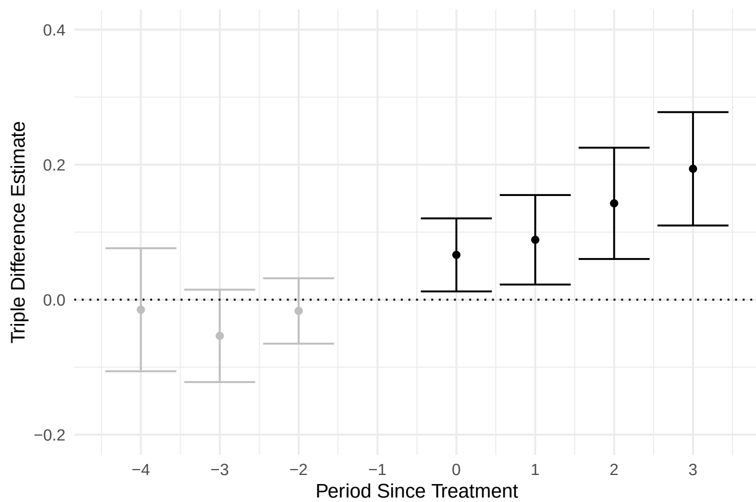
An alternative explanation is that credit card acceptance increases sales by low-

[6]Because the treatment groups are very small relative to the large control group, and treated stores in one event are excluded from the control group of another, the staggered-treatment complications addressed by recent difference-in-differences estimators do not arise here.

[7]A log transform of dollar spending is not feasible here because many household-retailer-period observations have zero spending, so taking logs would require either dropping zeros, which eliminates the extensive margin response, or adding an arbitrary constant, either of which biases the estimate.

[8]I excluded events involving warehouse stores for this reason. Changes in which credit cards a warehouse store accepts are often accompanied by co-branded card campaigns designed to shift consumers' card holdings along with the acceptance change, a coordinated shift rather than a change in merchant acceptance holding fixed consumer payment behavior.

Figure 4: Triple-difference estimates of the effect of credit card acceptance on sales to credit card consumers



Notes: Data are from Homescan. The graph presents estimated coefficients from a triple-difference model of the effect of changes in credit card acceptance on dollar sales to credit card users compared to other consumers, aggregated across retailers. The coefficients reflect percentage changes in sales, and the error bars show 95% confidence intervals. Periods are six months.

ering consumers' effective prices through rewards. Online Appendix **??** studies Discover's quarterly rewards programs and finds that rewards influence which credit card consumers use but not which merchants they visit.

III.C Merchants Multi-home More Than Consumers

Merchants overwhelmingly accept all cards, yet only around 60% of consumers use credit cards from two or more networks. Because declining a network's cards risks losing single-homers, merchants have little incentive to drop any network. This asymmetry limits merchants' bargaining power and reduces competitive pressure on merchant fees.

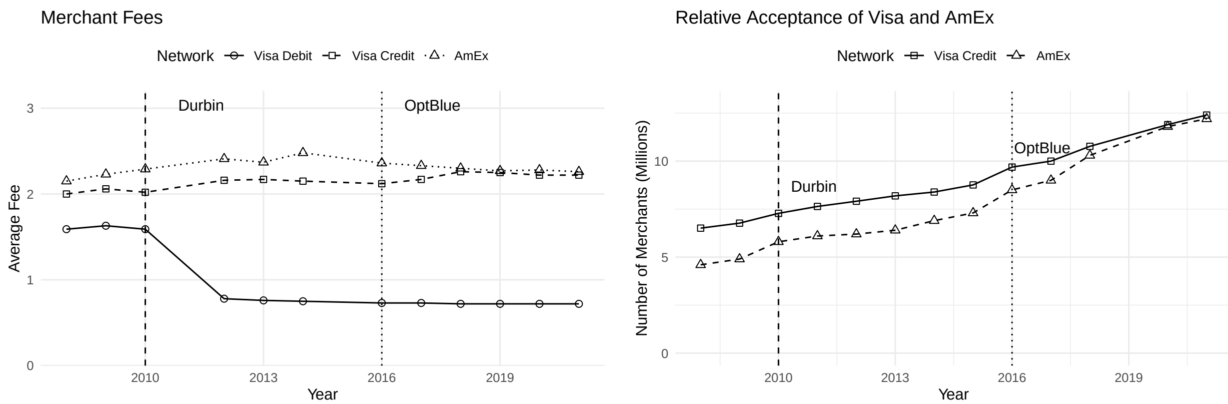
III.C.1 Almost All Merchants Multi-home

As of 2019, most merchants accept either all credit cards or none at all. The most natural margin for selective acceptance is AmEx, given its historically higher merchant fees. Figure **??** shows that the gap between AmEx's and Visa's credit card merchant fees fell by around 15 bps over the past decade, driven by AmEx's OptBlue program that cut fees for small businesses (**Glasheen2020**). Figure **??** shows that as AmEx's fees converged with Visa's, so did merchant acceptance. By 2019, the number of merchants accepting AmEx approximately equalled the number accepting Visa.[9]

[9]Similar aggregate acceptance counts could be consistent with merchants specializing in disjoint sets of networks. Historical Yelp reviews, primarily from 2010–2018 before the convergence in Visa and AmEx

Figure 5: Merchant fees and credit card acceptance

(a) Merchant fees **(b)** Credit card acceptance



Notes: Data are from the Nilson Report. The left panel plots average merchant fee levels for Visa Debit, Visa Credit, and AmEx. The right panel plots the number of merchants accepting Visa and AmEx. The dotted lines mark the imposition of the Durbin Amendment and the start of AmEx's OptBlue program.

III.C.2 Many Consumers Single-home

Whether merchants can steer consumers between networks depends on how many consumers carry cards from more than one. If most consumers single-home, a merchant that declines a network loses those consumers entirely. If most multi-home, the merchant can redirect them to a rival.

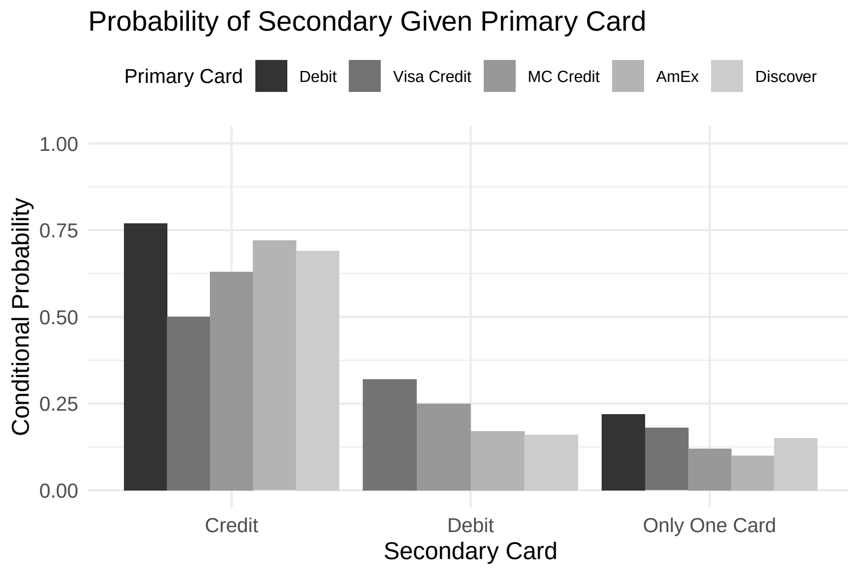
I study consumer multi-homing using the Homescan shopping data, defining a network as Visa credit, MC credit, AmEx, or any debit card. I define "carrying" a card based on observed usage, not self-reported ownership. A consumer is classified as carrying a card type if they use it during that year. I characterize household-years by their primary (most-used) and secondary (second-most-used) networks. Consumers put around 95% of their card spending on these top two networks (Online Appendix Table **??**). These usage shares reflect consumer preferences rather than merchant acceptance policies, since most Homescan merchants are large and accept all major payment methods.[10]

Around 40% of primary credit card consumers carry credit cards from only one network (Figure **??**). This usage-based measure aligns closely with ownership-based data. In the Diary of Consumer Payment Choice, 58% of credit card users report holding cards from two or more networks (Appendix Table **??**). Usage-based classification does

acceptance, suggest otherwise. Merchants progressively add payment methods rather than specialize in particular networks (Online Appendix **??**).

[10]Appendix **??** details the sample restrictions, including the exclusion of merchants with a low share of transactions from any network and the aggregation of Visa and MC debit into a single network for the consumer multi-homing moments.

Figure 6: Consumer multi-homing behavior



Notes: Data are from Homescan. The figure shows the probability that consumers use a secondary credit card, secondary debit card, or no secondary card, conditional on different primary cards.

not mechanically understate multi-homing relative to ownership surveys.

When a merchant declines a network's cards, it risks losing these single-homers, who cannot substitute to a rival. The remaining 60% multi-home. Visa users multi-home least at 50%, while AmEx users multi-home most at around 73%. This mix of single- and multi-homing consumers limits but does not eliminate merchants' ability to steer consumers between credit card networks.

III.D The Competitive Bottleneck

These patterns resemble a "competitive bottleneck" in which networks compete primarily for consumers, not merchants. Standard theoretical models assume either pure single-homing, under which competition unambiguously raises merchant fees, or pure multi-homing, which reverses the result (**Anderson.etal2018**; **Bakos.Halaburda2020**). Given that some consumers single-home and others multi-home, neither benchmark applies cleanly, and the net effect of competition on fees and welfare must be estimated.

IV Model

To quantify how network competition shapes fees, rewards, and welfare, I build a model with two-sided multi-homing, merchant competition, and price coherence.

Figure 7: Illustration of how multi-homing consumers choose payment methods at the point of sale.

	w.p. π	w.p. $1-\pi$
(a) Primary V / Secondary MC	V $\rightarrow$ MC $\rightarrow$ Cash	MC $\rightarrow$ V $\rightarrow$ Cash
(b) Primary V / Secondary D	V $\rightarrow$ Cash	D $\rightarrow$ Cash

Notes: With probability π, a consumer tries their primary card first. With probability $1-\pi$, they try their secondary card first. Consumers who multi-home on two credit cards substitute between them when one is declined. Consumers who multi-home on credit and debit do not substitute between them.

IV.A Structure of the Game

I model competition between card networks as a static game with three stages between networks, consumers, and merchants. In the first stage, profit-maximizing networks set per-transaction fees for merchants and promised adoption utility for consumers. In the second stage, consumers and merchants make adoption and pricing decisions. In the third stage, consumers make consumption decisions over merchants. The second and third stages micro-found demand for payments. Section **??** discusses key assumptions.

IV.B Stage 3: Consumer Shopping and Payments

Consumers' payment decisions depend on merchants' acceptance decisions, and their shopping decisions depend on their ability to use preferred payment methods. These choices give merchants incentives to accept cards in the second stage.

IV.B.1 Payment Behavior at the Point of Sale

At the point of sale, consumers' primary and secondary cards and merchant acceptance determine payment behavior. Consumers may use zero, one, or two cards. Those without cards pay cash. Those with one card pay with it if the merchant accepts it and otherwise pay cash.

Consumers with two cards can substitute between them (Figure **??**). One card is the primary. With probability π, the consumer tries the primary card first, then the secondary if it is the same card type as the first, and then cash. With probability $1-\pi$, the roles reverse.[11] Credit and debit cards are segmented. A consumer who wishes to use credit does not substitute to debit if her credit card is declined (and vice versa).

Formally, define the set of all inside payment methods (i.e., cards) as $\mathcal{J}_1 = \{1, \ldots, J\}$, and the set of all payment methods as $\mathcal{J} = \{0\} \cup \mathcal{J}_1$, where 0 refers to cash. A wallet

[11] Consumers may switch between the two cards due to transaction-level shocks that I do not model. For example, a credit card consumer may use the card with the best reward for the merchant.

$w = (w_1, w_2)$ has primary and secondary payment methods, w_1 and w_2. Let $\mathcal{W}$ denote the set of all possible wallets. It is

$$\mathcal{W} = \underbrace{(0,0)}_{\text{Cash}} \cup \underbrace{\{(w_1, 0) : w_1 \in \mathcal{J}_1\}}_{\text{One Card}} \cup \underbrace{\{(w_1, w_2) : w_1, w_2 \in \mathcal{J}_1, w_1 \neq w_2\}}_{\text{Two Cards}}$$

Let $\pi^w_{M,j}$ be the probability that a consumer with wallet w pays with card $j > 0$ when the merchant accepts cards $M \subset \mathcal{J}_1$. Define $\pi^w_{M,0} = 0$ for all w.[12]

IV.B.2 Consumption Decisions Over Merchants

Consumers have constant elasticity of substitution (CES) preferences over a continuum of merchants, each selling a differentiated variety. Consumers observe merchants' acceptance decisions and prices before choosing where to shop. Merchants are vertically differentiated by a type $\gamma \sim G$ that governs the importance of card acceptance.

A consumer with wallet w that consumes $q^w(\omega)$ of each variety ω has utility

$$\left(\int \left(1 + \gamma(\omega)\, v^w_{M^*(\omega)}\right)^{\frac{1}{\sigma}} q^w(\omega)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}\omega \right)^{\frac{\sigma}{\sigma-1}} \tag{4}$$

where σ is the elasticity of substitution between merchants, $v^w_M = \sum_{j=1}^{2} \pi^w_{M,w_j}$ is the payment surplus from wallet w at a merchant accepting cards M, and $\left(1 + \gamma v^w_M\right)$ is the quality of merchant ω for consumer w. Consumers maximize utility by spending their income $y(1 + f^w)$, where f^w is the percentage reward from adopting wallet w. Card acceptance raises quality, reallocating spending toward card-accepting merchants rather than increasing total spending. Rewards increase income but do not affect relative consumption across merchants (consistent with the Discover evidence in Online Appendix **??**).

Let all other merchants charge prices $p^*(\gamma)$ and accept payment methods $M^*(\gamma) \subset \mathcal{J}_1$. Standard CES results yield that a type γ merchant who sets price p and accepts $M \subset \mathcal{J}_1$ sells q^w to a consumer with wallet w:

$$q^w(\gamma, p, M, P^w, y, f^w) = (1 + \gamma v^w_M) \times \frac{p^{-\sigma}}{(P^w)^{1-\sigma}} \times y(1 + f^w) \tag{5}$$

$$(P^w)^{1-\sigma} = \int \left(1 + \gamma v^w_{M^*(\gamma)}\right) p^*(\gamma)^{1-\sigma} \, \mathrm{d}G(\gamma) \tag{6}$$

where P^w is a CES price index that depends on other merchants' actions and on the

[12] Online Appendix **??** explicitly works out these probabilities in a simplified model with two credit cards and one debit card.

consumer's wallet.

Consumers spend more at merchants that accept their most-used cards. This distinction between ownership and active use helps explain why credit card acceptance can increase sales even when all credit cardholders also own debit cards.

In equilibrium, a consumer with income y optimally consumes $y \times q^{w*}(\gamma)$ from each merchant type γ, given equilibrium pricing p^* and acceptance M^*:

$$q^{w}\left(\gamma, p^{*}\left(\gamma\right), M^{*}\left(\gamma\right), y\right) = y \times q^{w*}\left(\gamma\right) \tag{7}$$

IV.C Stage 2: Pricing, Acceptance, and Adoption

Merchants maximize profits by choosing prices and payment acceptance. Profits equal per-dollar quantities q^w times margins, weighted by spending power across consumer payment types. Appendix **??** shows that profits equal

$$\begin{aligned} \Pi\left(\gamma, p, M\right) &= \sum_{w \in \mathcal{W}} \tilde{\mu}^{w} \times q^{w}\left(\gamma, p, M, 1\right) \times \left[p\left(1 - \frac{\left(1+\gamma\right)\tau_{M}^{w}}{1+\gamma v_{M}^{w}}\right) - 1\right] \\ \tilde{\mu}^{w} &= \int \tilde{\mu}_{y}^{w} y \,\mathrm{d}F\left(y\right) \end{aligned} \tag{8}$$

where $\tilde{\mu}_y^w$ is the market share of wallet w among consumers with income y, F is the income distribution, τ_j is the per-dollar fee on card j, and $\tau_M^w = \sum_{j=1}^{J} \pi_{M,j}^w \tau_j$ is the average fee incurred by wallet w when the merchant accepts M. Because I normalize $\tau_0 = 0$ for cash, fees τ_j represent the cost of card j relative to cash. The income-weighted market share $\tilde{\mu}^w$ weights payment choices by income.

IV.C.1 Merchant Pricing

Conditional on acceptance M, merchants pass fees into prices. Under price coherence, merchants set uniform prices for all consumers and thus creating the externality that makes credit card use socially excessive. Appendix **??** shows the optimal uniform price is

$$\hat{p}\left(\gamma, M, \tau\right) = \frac{\sigma}{\sigma - 1} \times \frac{1}{1 - \hat{\tau}}, \hat{\tau} = \frac{\sum_{w} \mu^{w} \tau_{M}^{w}\left(1+\gamma\right)}{\sum_{w} \mu^{w}\left(1+\gamma v_{M}^{w}\right)} \tag{9}$$

where the average fee uses *demand shares* μ^w, normalized weighted sums of $\tilde{\mu}^w$:

$$\mu^{w} = \frac{1+f^{w}}{\left(P^{w}\right)^{1-\sigma}} \times \frac{\tilde{\mu}^{w}}{C}, C = \sum_{w} \frac{1+f^{w}}{\left(P^{w}\right)^{1-\sigma}} \times \tilde{\mu}^{w} \tag{10}$$

Demand shares reflect how other merchants' acceptance decisions affect consumer composition at each merchant through P^w.

In equilibrium, merchants set optimal prices $p^*(\gamma)$ given other merchants' pricing and adoption strategies:

$$\hat{p}(\gamma, M^*(\gamma), \tau) = p^*(\gamma) \tag{11}$$

IV.C.2 Merchant Acceptance

Acceptance decisions trade off higher sales against higher fees. Let $\widehat{\Pi}(\gamma, M)$ be profits from accepting $M \subset \mathcal{J}_1$ at the optimal price. Appendix **??** derives a fast approximate algorithm for optimizing $\widehat{\Pi}$ over M, yielding optimal acceptance $\widehat{M}$:

$$\widehat{M}(\gamma, \tau) = \underset{M \subset \mathcal{J}_1}{\operatorname{argmax}} -a_M + b_M \gamma \tag{12}$$

$$a_M = \sum_{w \in \mathcal{W}} \mu^w \tau_M^w,\ b_M = \frac{1}{\sigma} \sum_{w \in \mathcal{W}} \mu^w (v_M^w - \sigma \tau_M^w) \tag{13}$$

Adding a more expensive card incurs fees from all consumers who use it but raises sales only among single-homers. The incremental fee cost a_M covers all consumers who use the card, whereas the incremental sales benefit b_M is driven only by single-homers who cannot substitute. If all consumers who use credit cards multi-home across credit networks, this incremental sales gain vanishes and merchants accept only the lowest-fee network. Online Appendix **??** verifies these intuitions in a two-card economy. Merchants' fee sensitivity therefore depends on the type distribution G and consumer behavior.

In equilibrium, merchants adopt optimal bundles given other merchants' behavior:

$$\widehat{M}(\gamma, \tau) = M^*(\gamma) \tag{14}$$

IV.C.3 Consumer Adoption

Consumers choose up to two cards to maximize utility. The utility from a wallet w for a consumer i is

$$\log V_i^w = \log U^w + \frac{1}{\alpha_i} (\Gamma_i^w + \epsilon_i^w) \tag{15}$$

where U^w is pecuniary utility, α_i is rewards-sensitivity, Γ_i^w is mean non-pecuniary utility, and ϵ_i^w are wallet-level T1EV shocks.

Pecuniary Utility: Consumers prefer cards with high rewards and wide acceptance. The pecuniary utility for single-homing consumers is:

$$\log U^w = f^j - \log P^w, w = (j, 0), j \in \mathcal{J}_1 \tag{16}$$

where f^j represents percentage, lump-sum rewards and P^w is the CES price index from

Equation **??**. At adoption, consumers are fully informed about merchant acceptance, and greater acceptance lowers P^w. This expression can be derived from the log indirect utility of the CES problem (**??**) by dropping baseline income and using the approximation $\log(1+f) \approx f$.

Because pecuniary utility is derived from the CES problem, a 1 pp. increase in rewards exactly offsets a 1% increase in retail prices, so higher rewards and fees do not mechanically reduce welfare. By not normalizing the pecuniary utility of the outside option to zero, the model allows for externalities in which one consumer's payment choice affects the prices paid and thus welfare of other consumers.

For multi-homing agents, pecuniary utility is the weighted average of single-homing utilities:

$$\log U^w = \pi \log U^{(w_1,0)} + (1-\pi)\log U^{(w_2,0)}, w = (w_1, w_2) \in \mathcal{J}_1 \times \mathcal{J}_1, w_1 \neq w_2 \tag{17}$$

This specification implies consumers do not multi-home to increase acceptance coverage. I adopt this because even as AmEx acceptance has converged with Visa Credit, multi-homing rates between them have not changed (Online Appendix **??**). The expression closely approximates one that directly uses the CES price index of wallet w whenever merchant acceptance depends only on card type (credit or debit) rather than network identity (Online Appendix **??**).

Non-pecuniary Utility: Non-pecuniary utility captures within-wallet complementarities and adoption costs. Let card j have characteristics $X^j = \left(X_k^j\right)_{k=1}^K \in \mathbb{R}^K$. Mean non-pecuniary utility for consumer i is

$$\Gamma_i^w = \omega\left(\Xi^{w_1} + \beta_i X^{w_1}\right) + (1-\omega)\left(\Xi^{w_2} + \beta_i X^{w_2}\right) + \sum_{l=1}^K \sum_{m=1}^K \chi_i^{lm} X_l^{w_1} X_m^{w_2} \tag{18}$$

where Ξ^j is mean unobserved utility for card j, β_i is consumer i's value from characteristics, ω is the weight on primary-card characteristics, and χ_i^{lm} are consumer-specific interaction terms capturing within-wallet complementarity or substitution motives.[13]

Consumer Heterogeneity: Preferences vary with income and exhibit unobserved heterogeneity. Let $\log \tilde{y} \equiv \log y - \mathbb{E}[\log y]$ denote demeaned log income. The consumer-

[13] l and m index the two characteristics of the primary and secondary card: whether it is an inside good and whether it is credit. Thus χ^{12} measures the interaction between any primary card and a credit secondary. χ^{21} measures the interaction between a credit primary and any secondary. χ^{22} measures an additional interaction specific to credit in both positions. Online Appendix **??** microfounds these parameters in a two-stage adoption-and-usage model.

specific parameters are:

$$\log \alpha_i = \log \alpha_0 + \alpha_y \log \tilde{y} \tag{19}$$

$$\beta_i = \beta_y \log \tilde{y} + \tilde{\beta}_i, \quad \tilde{\beta}_i \sim N(0, \Sigma) \tag{20}$$

$$\chi_i^{lm} = \chi^{lm} + \chi_y^{lm} \log \tilde{y} \tag{21}$$

Demeaning ensures that α_0, Ξ, and χ^{lm} represent preferences at median income. Here α_y is the income elasticity of reward-sensitivity, β_y captures how card preferences vary with income, Σ is the covariance of unobserved preference heterogeneity, and χ_y^{lm} governs how within-wallet complementarities vary with income. Income dependence in χ_i^{lm} allows higher-income consumers to derive greater value from holding both a rewards credit card and a debit card for distinct transaction types.

Choice Probabilities The logit choice probability gives that

$$\tilde{\mu}_i^w = \frac{\exp\left(\alpha_i \log U^w + \Gamma_i^w\right)}{\sum_{m \in \mathcal{W}} \exp\left(\alpha_i \log U^m + \Gamma_i^m\right)} \tag{22}$$

$$\tilde{\mu}_y^w = \int \tilde{\mu}_i^w \, \mathrm{d}H(\beta_i), \quad \beta_i \sim N\left(\beta_y \log \tilde{y}, \Sigma\right) \tag{23}$$

IV.D Stage 1: Network Competition

Multiproduct payment networks maximize profits, anticipating consumer and merchant actions.

IV.D.1 Profits

Network profits equal transaction fees minus costs and rewards. Fee profits T_j equal the transaction margin times total dollar volume.

$$T_j = \left(\tau_j - c_j\right) d_j \tag{24}$$

$$d_j = \sum_{w \in \mathcal{W}} \tilde{\mu}^w \times \int \frac{(1+\gamma)\, \pi^w_{M^*(\gamma),j}}{1 + \gamma v^w_{M^*(\gamma)}} q^{w*}(\gamma)\, p^*(\gamma)\ \mathrm{d}G(\gamma) \tag{25}$$

where c_j is the cost of processing \$1 on method j.

Total reward costs equal

$$S_j = f^j \left(\tilde{\mu}^{(j,0)} + \sum_{k>0, k \neq j} \pi \tilde{\mu}^{(j,k)} + (1-\pi)\, \tilde{\mu}^{(k,j)} \right) \tag{26}$$

where f^j is the reward paid to a consumer single-homing on j. The three terms count single-homers on j, multi-homers who primarily use j, and multi-homers who use j as a secondary card.

For a network n that owns cards $\mathcal{O}_n \subset \mathcal{J}_1$, it earns profits:

$$\Psi_n = \sum_{j \in \mathcal{O}_n} \left(T_j - S_j\right) \tag{27}$$

IV.D.2 Conduct and Equilibrium Determinacy

Networks set transaction fees τ_j and pecuniary adoption benefits A_j, taking other networks' actions as given. Equivalently, networks set consumers' expectations of their own acceptance, fees, and rewards given expectations for rival networks. The adoption benefit A_j compares rewards and payment convenience relative to cash.

$$A_j = \log U^{(j,0)} - \log U^{(0,0)} = f^j - \left(\log P^{(j,0)} - \log P^0\right) \tag{28}$$

Networks set A_j^* and τ_j^* for their cards $\mathcal{O}_n$ to maximize expected profits under small trembles in all actions:

$$\left(A_j^*, \tau_j^*\right)_{j \in \mathcal{O}_n} = \underset{\left(\overline{A}_j, \overline{\tau}_j\right)_{j \in \mathcal{O}_n}}{\operatorname{argmax}} \ \mathbb{E}\left[\Psi_n\left(A_j, \tau_j, A_{-j}, \tau_{-j}\right)\right] \tag{29}$$

$$A_j \sim N\left(\overline{A}_j, \nu_x^2\right), \tau_j \sim N\left(\overline{\tau}_j, \nu_x^2\right), A_{-j} \sim N\left(A_{-j}^*, \nu_x^2\right), \tau_{-j} \sim N\left(\tau_{-j}^*, \nu_x^2\right)$$

Small trembles select a unique equilibrium when profits are not differentiable in fees (**Teh.etal2022**). I set $\nu_x = 10^{-4}$. The trembles can be interpreted as small fee heterogeneity across merchants. Online Appendix **??** discusses the conduct assumption.

IV.E Equilibrium

An equilibrium is a tuple $\left(\tau^*, A^*, \tilde{\mu}_y^w, p^*(\gamma), M^*(\gamma), q^{w*}(\gamma)\right)$ satisfying: consumption is optimal (**??**); merchant pricing and acceptance maximize profits (**??**, **??**); consumers choose optimal wallets (**??**); and networks maximize profits (**??**). Online Appendix **??** details the solution algorithm.

IV.F Discussion of Key Assumptions

The welfare results depend on several modeling assumptions, of which credit-debit segmentation at the point of sale is the most consequential. This section defends that assumption and addresses a set of smaller ones that matter for interpretation.

Segmentation Between Debit and Credit The baseline treats credit and debit as segmented at the point of sale. The assumption is imperfect: antitrust testimony and consumer surveys support it, but Discover's rewards experiment shows some substitution at the margin (Appendix **??**). Under substitution, credit acceptance would depend on credit-debit multi-homing rates, and debit fee caps would discipline credit fees. Antitrust testimony in *Ohio v. AmEx* contradicts the first, and Durbin halved debit interchange without moving credit interchange. Online Appendix **??** estimates two alternatives that incorporate credit-debit substitution. Welfare results are close to the baseline for every counterfactual other than uncapping debit fees.

Two-Card Restriction Consumers choose up to two cards and treat cards of the same type as interchangeable at the point of sale (e.g. both credit cards). In the data, consumers put around 95% of their card spending on just two networks (Online Appendix Table **??**). This setup understates consumer flexibility by excluding a potential third fallback option, but potentially overstates the willingness to substitute among the top-two cards. A consumer who puts 99% of their spending on one card and 1% on another may be less willing to substitute than the model implies. **Sullivan2025<empty citation>** takes an alternative approach and infers substitution patterns from within-consumer persistence in platform choice.

Fixed Costs I omit fixed costs of card acceptance because they cannot be separately identified from heterogeneity in sales benefits γ without exogenous shocks to consumer adoption as in **Higgins2022<empty citation>**. The absence of fixed costs rules out a feedback loop in which lower consumer adoption reduces merchant acceptance. I therefore focus on an intermediate credit fee cap resembling Canadian levels, well within the range that has not triggered acceptance cascades elsewhere.

Merchant Types Merchants differ only in the sales benefit γ. This one-dimensional type rationalizes hierarchical acceptance but predicts identical relative shares among accepted card networks at all card-accepting stores. If consumers with heterogeneous payment preferences sort across stores, the redistributive effects of fees would be dampened (**Gans2018**). Online Appendix **??** shows such sorting is quantitatively small in retail.

Non-Rewards Credit Card Characteristics The data lack non-rewards credit card characteristics such as interest rates, credit limits, or annual fees. I assume these do not change when rewards change. Counterfactuals are best interpreted as short-run predictions holding these characteristics fixed. Australia's interchange caps did not raise non-rewards annual fees, and the spread between credit card rates and term loans was unchanged (Online Appendix Figure **??**), consistent with high-volume transactors accounting for little credit card borrowing (**Adams.etal2022a**).

Price Coherence Merchants charge the same price regardless of payment method (price coherence). This friction causes merchant fees and rewards to distort consumer payment choices. Online Appendix **??** discusses the history and theory. Only about 5% of U.S. transactions feature payment-specific pricing despite legal permission (**Stavins2018**). Merchants with the strongest gains from card acceptance are those whose consumers value card payments highly and respond little to surcharges, so even small menu costs deter surcharging.

Issuers and Acquirers The model combines issuers, acquirers, and networks into a single entity that maximizes joint profits. This is a valid assumption so long as the contracting space between these parties is rich enough. The DOJ's 2024 antitrust suit against Visa alleges payments to issuers and potential competitors to maintain dominance, evidence consistent with this view of the contracting environment. Double marginalization and other vertical frictions within the payment chain are left to future work.

V Estimation

Estimation links the reduced-form facts to quantitative predictions about regulation and competition. The key primitives are consumers' preferences over payment options, merchants' gains from accepting cards, and networks' marginal costs.

V.A Estimation Procedure

I estimate all parameters jointly, but the identification argument has three components. Consumer demand is identified by quasi-experimental price variation, second-choice surveys (**Berry.etal2004**), and demographic moments (**Petrin2002**). Network costs are identified by inverting first-order conditions on rewards. Merchant types are identified from event-study evidence on card acceptance effects and acceptance rates. The model approximates the market structure as of 2019. I estimate standard errors by bootstrapping the joint distribution of data moments. Appendix **??** contains the estimation details.

V.A.1 Consumer Demand

The key consumer demand parameters are price sensitivity (α_0), substitution patterns (Σ), income gradients ($\alpha_y, \beta_y, \chi_y^{lm}$), and multi-homing complementarities (χ^{lm}).

The effect of the Durbin Amendment on debit card volumes identifies the price-sensitivity coefficient α_0. I simulate a 25 bps decline in debit rewards, holding fixed merchant acceptance. I match the resulting percentage change in debit volumes to the difference-in-difference estimate. **Hayashi2012<empty citation>** estimates that pre-Durbin debit rewards were around 25 bps, which I confirm with archived bank websites. The simulated moment holds merchant acceptance fixed because cardholders at regulated and exempt issuers face the same merchants. Any change in merchant behavior after Durbin is differenced out by the control group.

Durbin-exposed banks cut rewards along with non-price steering. The baseline calibration nonetheless targets the full 25% relative decline. Among the subsample of banks that paid debit rewards pre-Durbin, those that ended rewards saw debit volumes grow more slowly than those that kept rewards by a margin similar to the baseline specification. Online Appendix **??** argues that if banks differ in their productivities across steering levers, rewards-paying banks did less non-price steering, so the within-subsample comparison reflects mainly the rewards change. Section **??** discusses how the counterfactuals change under a more conservative target that attributes only half of the debit decline to the rewards channel.

My second-choice survey reveals the covariance matrix Σ of the random coefficients (**Berry.etal2004**). The characteristic vector X_j is an indicator for an inside good and an indicator for being a credit card. The inside-good indicator equals 1 for all cards, whether credit or debit. Online Appendix **??** shows that consumers view credit and debit cards as two separate categories and that cash substitutes more effectively for debit cards than credit cards. The volatility of the random coefficients is correspondingly large.

Wallet complementarity parameters are identified by multi-homing patterns in the Homescan data. Three parameters measure wallet interactions. χ^{12} is credit's value in the secondary position, χ^{21} is a secondary card's value for credit-primary consumers, and χ^{22} is the cost of managing two credit accounts.[14]

Demographic data identify how preferences vary with income. Higher-income respondents in the second-choice survey are more willing to switch cards for better rewards, identifying α_y. The empirical income distribution pins down ν_y, and the DCPC data on how preferred payment methods vary with income identify the conditional pref-

[14] The fourth interaction χ^{11} (any-primary, any-secondary) is normalized to zero: the only wallet where it would be separately identified is (debit,debit), which is negligible in the data.

erence gradient β_y. Homescan multi-homing patterns by income (Figure **??**) identify both χ^{lm} and its income gradient χ^{lm}_y. The multi-homing rate is 4.3 pp. higher above median income than below (Online Appendix Table **??**), and the model matches this gradient exactly.

V.A.2 Network Costs

I recover network costs by inverting first-order conditions with respect to rewards. High rewards are profitable only when networks earn positive margins from merchants, so marginal costs must be low relative to observed fees.

V.A.3 Merchant Types

I recover the distribution of merchant types G by combining three inputs: event-study evidence on grocery chains' credit card adoption from the Homescan panel, acceptance rates from the DCPC, and networks' optimal pricing conditions. U.S. payment markets are mature. In 2013, 98% of Homescan trips occurred at stores already accepting credit cards, so recovering the merchant-type distribution is closer to a calibration than a conventional estimation.

I model grocery chains that changed their acceptance policies during this period as the lowest-type merchant that accepts credit cards. This ultimately reveals σ. Under the baseline substitution assumption, such merchants trade-off γ^*% more sales to credit card consumers against the cost of paying additional merchant fees of τ instead of the cost of cash. As σ increases, merchant margins decrease, and thus a greater sales gain γ^* is needed to justify accepting cards. I choose a value of σ so that γ^* matches the estimated sales effects from credit card acceptance.[15]

I parameterize G as a two-parameter Gamma distribution. Two moments identify its shape: the expenditure-weighted acceptance rate from DCPC respondents, and networks' first-order conditions on merchant fees (which depend on the density near the marginal type).[16] Merchant price-sensitivity at the observed equilibrium follows from a standard insight in two-sided markets. Networks charge high fees to merchants and use the revenue to fund rewards for price-sensitive consumers, so merchants must be

[15]The standard errors in my estimate of σ account for uncertainty in the event-study estimates but not for model error from extrapolating beyond the grocery sector. Restaurants, e-commerce, and service providers face different customer mixes and transaction sizes, and the data contain no comparable natural experiments outside grocery.

[16]The acceptance rate is the share of merchants above the marginal type, weighted by transaction volume. Online Appendix Table **??** compares DCPC acceptance rates to Yelp business attributes by sector; the two sources agree closely in aggregate.

relatively insensitive to fees compared to consumers. Given estimates of consumer sensitivity, merchant sensitivity comes from networks' first-order conditions.

V.A.4 Calibrated Parameters

I calibrate two sets of parameters. Aggregate spending shares from the 2019 Nilson Report (Figure **??**) pin down the unobserved characteristics Ξ. I normalize $\tau_0 = 0$ for cash using the 30 bps cost-of-cash estimate from **Felt.etal2020<empty citation>** for the U.S., then bootstrap from a distribution centered at that value with a standard deviation of 10 bps to incorporate uncertainty in the cost of cash into other parameter estimates.

V.B Merchant and Consumer Sensitivities and Credit Aversion

The estimated parameters support the competitive bottleneck: consumers are far more sensitive to rewards than merchants are to fees, so networks compete for consumers rather than merchants. The estimates also reveal substantial credit aversion. Consumers who take credit cards for the rewards still incur a non-pecuniary cost, so cutting back on credit card use generates real welfare gains. Table **??** reports parameter estimates.

The consumer substitution matrix in Table **??** shows that consumers view credit cards, debit cards, and cash as distinct segments. A 1-bp increase in Visa credit rewards raises Visa's share by 3.7%, drawn mostly from MC credit (down 3.5%), while MC debit falls only 0.3% and cash 0.3%. Consumers freely substitute between networks' credit cards but not between credit and debit, or between cards and cash.

This asymmetry is large. Consumers are roughly ten times as sensitive to rewards as merchants are to fees. A 1-bp increase in Visa's merchant fees, holding fixed consumer adoption, reduces merchant acceptance by 0.4% (S.E. 0.2%). This asymmetry underpins the competitive bottleneck. Merchants cannot credibly threaten to drop cards that consumers expect to use, so networks compete for consumers rather than for merchant acceptance.

The parameters reveal substantial consumer "credit aversion." The median consumer would pay with debit if credit cards did not pay rewards. A consumer with median income is indifferent between a Visa debit card and a Visa credit card paying 1.2% in rewards.[17] The model estimates a large non-pecuniary cost of credit card use, consistent with survey evidence that consumers who avoid credit cards cite budget control, distaste for carrying debt, and the hassle of managing revolving accounts (Online Appendix **??**). This credit aversion is central to the welfare analysis. Rewards competition induces

[17] This is the difference in non-pecuniary utility between single-homing on Visa debit versus Visa credit, scaled by the primary-card weight ω.

Table 2: Estimated parameters

Panel A: Consumer Preference Parameters

Parameter	Est	SE
Log Price Sensitivity ($\log \alpha_0$)	6.74	0.34
Card-Credit Complement (χ^{12})	4.60	0.59
Credit-Card Complement (χ^{21})	4.00	0.48
Credit-Credit Complement (χ^{22})	-4.80	0.86
Visa Debit Utility (Ξ)	-4.00	0.59
Visa Credit Utility (Ξ)	-5.90	0.50
MC Debit Utility (Ξ)	-4.20	0.64
MC Credit Utility (Ξ)	-6.20	0.55
Amex Utility (Ξ)	-6.30	0.57
Primary Weight (ω)	0.62	0.02
P(Use Primary) (π)	0.83	0.00

Panel B: Consumer Heterogeneity Parameters

Parameter	Est	SE
Card Volatility	0.90	0.37
Credit Volatility	2.40	0.96
Correlation of β	-0.78	0.06
Income Volatility (ν_y)	0.75	0.01
Income Elasticity (α_y)	0.20	0.06
Card Income Effect (β_y^C)	-1.50	0.43
Credit Income Effect (β_y^L)	0.60	0.46
Card-Credit Income (χ_y^{12})	0.60	0.21
Credit-Card Income (χ_y^{21})	0.50	0.18
Credit-Credit Income (χ_y^{22})	-0.40	0.17

Panel C: Merchant Parameters

Parameter	Est	SE
Merchant CES	5.61	1.25
Average γ	0.26	0.07
S.D. of γ	0.11	0.03

Panel D: Network Cost Parameters (bps)

Parameter	Est	SE
Cash	30	10
Visa Debit	40	16
MC Debit	50	9
Visa Credit	40	16
MC Credit	50	9
Amex	50	9

Notes: S.D. refers to the standard deviation, and R.C. refers to the random coefficients for having a credit card and not being cash. The Ξ are the unobserved characteristics. The characteristic vector has two components: an inside-good indicator ($X^1 = 1$ for all inside goods, credit and debit alike) and a credit indicator ($X^2 = 1$ for credit cards only). The standard deviation of R.C. and T1EV shocks, χ, Ξ are all measured in pp. of pecuniary utility for the median consumer. In Panel B, volatility parameters are the standard deviations of the random coefficients, ρ is the correlation between the card and credit random coefficients, ν_y is the standard deviation of the log-income distribution, and α_y, β_y, χ_y are the income gradients of reward sensitivity, preferences, and complementarities, respectively. Merchant types γ are distributed according to a Gamma distribution.

Table 3: Estimated consumer own-reward and cross-reward semi-elasticities.

Instrument	Visa Debit	Visa Credit	MC Debit	MC Credit	AmEx
Cash	-0.9	-0.3	-0.4	-0.1	-0.1
Visa Debit	3.1	-0.3	-1.7	-0.1	-0.1
Visa Credit	-0.6	3.7	-0.2	-1.4	-1.3
MC Debit	-4.3	-0.3	5.7	-0.1	-0.1
MC Credit	-0.5	-3.5	-0.2	5.8	-1.5
AmEx	-0.5	-3.5	-0.2	-1.6	6.1

Notes: Each entry shows the effect of a 1-bp change in the rewards of the column payment method on the market share of the row payment method. The change is measured as a percentage of the row payment method's market share.

consumers to use credit cards despite these non-pecuniary costs. Reductions in credit card use raise welfare by eliminating this distortion.

Reward sensitivity is increasing in income ($\alpha_y > 0$). While higher-income consumers tend to be less price-sensitive in retail markets (**Sangani2024**), greater financial sophistication and attention to product terms make higher-income consumers more responsive to differences in fees and rewards (**Hastings.etal2017**). Because high-income consumers are both more likely to use credit cards and more reward-sensitive, rewards competition among networks is particularly intense.

V.C Goodness of Fit

I assess fit by examining three sets of untargeted moments: adoption shares, the joint distribution of primary and secondary cards, and merchant acceptance rates.

V.C.1 Consumer Demand

The baseline equilibrium matches untargeted adoption shares from the DCPC. Around 41% of consumers have a primary debit card, around 33% have a primary credit card, and the rest use cash for all transactions (Table **??**). This match is not automatic because I target spending shares, not adoption shares. Recovering correct adoption shares confirms the estimated correlation between payment preferences and income. Online Appendix Figure **??** also shows that the model matches the joint distribution of primary and secondary cards in consumers' wallets.

Table 4: Baseline Equilibrium

	Cash	Debit	Credit
Market Share (%)	25	41	33
Share of Spending (%)	20	36	44
Merchant Fees (bps)	30	72	225
Rewards (bps)	0	0	145

Notes: The market share of a card type is the share of consumers whose primary card is that type. The reward rate is shown as the lump sum reward on a card divided by the dollars spent on that card for a consumer who single-homes on that card.

V.C.2 Merchant Demand

I validate merchant parameter estimates against three types of evidence. AmEx's 2016–2019 OptBlue program cut merchant fees by 15 bps relative to Visa (Figure **??**), and the share of Visa merchants not accepting AmEx shrank from around 9–17 pp. to almost zero (Figure **??**). [18] Simulating this shock in the model, the gap shrinks by 14.1 pp. This out-of-sample prediction is the most direct test of the model's ability to predict merchant responses to fee changes, which is central to the counterfactual analysis.[19] The estimated average sales effect of 26% is consistent with experimental evidence: **Higgins2022<empty citation>** find that debit card adoption at corner stores increases sales by 20–60%, and **Berg.etal2024<empty citation>** find that accepting Buy Now, Pay Later raises sales by around 20%. The estimated retail margin of 17.8% is similar to the aggregate margins of 13–17% implied by macro studies of misallocation (**Edmond.etal2022**; **Sraer.Thesmar2023a**).

V.C.3 Network Parameters

Network cost parameters match accounting data. The estimated debit marginal cost for the combination of issuers, acquirers, and the network averages around 45 bps. Accounting estimates put issuer costs at 20–40 bps, acquirer costs at 5–10 bps, and network costs at around 5 bps (**Lowe2005**; **Mukharlyamov.Sarin2025**; **NACHA2017**; **Visa2020**).

VI Counterfactuals

I use the estimated model to show that equilibrium fees and rewards are too high, but that increasing network competition without breaking the competitive bottleneck makes the problem worse. I simulate five counterfactual policies: capping credit card fees, repealing the Durbin Amendment, merging all three networks, increasing consumer multi-homing through dual-routing mandates, and introducing a central bank digital currency. All counterfactuals fix consumers' preferences β_i, baseline income y, and merchants' sales benefits to card consumers γ. Consumer adoption, merchant acceptance, retail prices, and network prices adjust endogenously.[20] The baseline is the observed 2019 equilibrium, where networks maximize profits subject to the Durbin Amendment's debit fee cap while credit fees remain unregulated. Welfare comparisons are relative to this status quo.

[18] AmEx's 2019 annual report confirms its U.S. network went from covering 90% to 99% of card spending during this period. The count of merchants **Nilson2020d<empty citation>** shows a larger reduction in the acceptance gap.

[19] The semi-elasticity linearizes the acceptance cutoff at near-universal acceptance. The OptBlue fee reduction is large enough that the cutoff responds nonlinearly, so the per-basis-point acceptance change differs from the linearized estimate.

[20] Although merchants' types γ are fixed, their incentives to accept cards can change because the effect of card acceptance on total sales depends on γ and the share of consumers using each card.

Table 5: Counterfactual Effects on Prices and Shares

	Price Controls		Change Competition		
	Cap Fees	Uncap Debit	Monopoly	Dual Routing	CBDC
Δ Prices (bps)					
Credit Fees	-105 (0.0)	-3.3 (0.7)	36 (3.2)	-13 (3.2)	0.8 (0.1)
Credit Rewards	-110 (0.5)	-6 (0.7)	-90 (30)	-15 (3.4)	0.7 (0.2)
Debit Fees	0.0 (0.0)	25 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
Debit Rewards	1.3 (0.2)	25 (0.2)	-37 (7)	0.4 (0.0)	3.9 (1.0)
Δ Shares of Total Spending (pp.)					
Cash	6 (1.5)	-7 (1.2)	29 (0.8)	0.0 (0.1)	-1.8 (0.1)
Debit	13 (1.9)	13 (2.4)	-6 (0.9)	0.8 (0.5)	-4.4 (0.1)
Credit	-19 (3.0)	-6 (1.3)	-23 (0.9)	-0.8 (0.6)	-0.6 (0.1)
Δ Fees, Rewards ($Bn)					
Total Fees	-58 (2.0)	4.4 (1.4)	-40 (3.3)	-7 (2.2)	-1.0 (0.4)
Total Rewards	-57 (1.2)	0.5 (1.6)	-66 (9)	-8 (2.3)	0.7 (0.3)

Notes: Bootstrap standard errors are in parentheses. The "cap fees" scenario caps credit card merchant fees at 120 bp. The "uncap debit" scenario raises the cap on debit card merchant fees by 25 bps. Monopoly refers to merging all three networks. Dual routing increases the multi-homing parameter by the equivalent of 20 bps in rewards for the median consumer. CBDC introduces a zero-fee public payment option. Its own market share is the residual of the changes in the shares of the other payment methods. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

VI.A Capping Credit Card Merchant Fees

In my main counterfactual, I cap credit card merchant fees at 120 bp, roughly half their current level. Debit fees are unchanged. The cap raises total welfare by $27 billion per year. Lower merchant fees reduce credit card rewards, correcting excessive adoption driven by price coherence. The gains are progressive. Low-income consumers benefit from lower retail prices, while high-income consumers lose from reduced rewards.

VI.A.1 Effects on Prices and Shares

Capping credit card merchant fees reduces rewards and card use, illustrating the see-saw principle in **Rochet.Tirole2003<empty citation>**. Table **??** shows that networks respond to the credit fee cap by cutting credit card rewards by around 1.1 pp., largely eliminating rewards. Spending on credit falls by nearly half as consumers substitute toward debit and cash (Table **??**). Price and quantity changes together reduce total merchant fees and rewards by roughly equal amounts. I normalize total model income to the $10 trillion in 2019 consumer spending (**Nilson2020a**), so each bp corresponds to $1 billion. Hence fees and rewards fall by around $58 billion each.

Table 6: Welfare Effects of Counterfactual Policies

	Price Controls		Change Competition		
	Cap Fees	Uncap Debit	Monopoly	Dual Routing	CBDC
Δ Welfare ($Bn)					
Consumers	28 (3.5)	4.3 (1.6)	-11 (11)	3.3 (1.4)	3.7 (1.0)
Merchants	0.4 (0.2)	-0.7 (0.2)	-4.6 (1.8)	-0.1 (0.0)	0.3 (0.1)
Networks	-1.5 (0.3)	3.0 (0.2)	31 (6)	0.6 (0.0)	-2.4 (0.6)
Total	27 (3.6)	7 (1.5)	15 (7)	3.8 (1.4)	1.6 (0.5)
Δ Consumer Surplus by Income (bps consumption)					
Low	48 (2.6)	7 (2.0)	13 (7)	6 (1.9)	4.6 (1.3)
Median	37 (3.4)	5 (1.5)	0.9 (10)	4.0 (1.6)	4.0 (1.1)
High	15 (4.0)	2.4 (1.6)	-29 (14)	1.3 (1.1)	3.4 (0.9)

Notes: Bootstrap standard errors in parentheses. Cap Fees caps credit card merchant fees at 120 bp. Uncap Debit raises the debit fee cap by 25 bps. Monopoly merges all three networks. Dual Routing increases the multi-homing parameter by the equivalent of 20 bps in rewards for the median consumer. CBDC introduces a zero-fee public payment option. Low (high) income consumers have log income at -2 ($+2$) standard deviations relative to the median. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

VI.A.2 Welfare

The gains from the credit fee cap stem from correcting excessive adoption, not from reducing market power. I measure consumer welfare by compensating variation:

$$\mathrm{CS} = \int \mathbb{E}\left[\max_w \log V_i^w(y)\right] \times y \, \mathrm{d}F(y) \tag{30}$$

The inner expectation computes surplus for households at a given income as a share of baseline income. The outer integral weights by baseline income to yield aggregate welfare. Most gains accrue to consumers ($28 billion), not merchants ($0.4 billion). Networks lose only $1.5 billion—around 6% of baseline profits. The total gain is roughly double the $12 billion consumer welfare gain from the CARD Act (**Agarwal.etal2015**), highlighting the importance of regulating networks in addition to issuers.

Table **??** isolates the welfare sources by holding different margins fixed. The first row holds merchant prices and acceptance at their baseline while networks re-optimize fees and rewards and consumers adopt new payment methods. The second row additionally allows merchant retail prices to adjust, holding acceptance fixed. The third row allows acceptance to adjust as well.

Even though the credit fee cap targets merchants, merchants benefit little in equilibrium. The first row shows that merchants benefit from lower fees, but the second row shows that they dissipate these profits by lowering retail prices to compete for

Table 7: Welfare Decomposition by Channel: 120 Bps Cap

	Consumer		Merchant		Network		Total	
Network Effect	-30	(1.7)	50	(2.5)	-6	(0.7)	14	(4.3)
Price Passthrough	59	(1.9)	-49	(2.6)	0.6	(0.2)	11	(2.2)
Merchant Adoption	-1.3	(0.9)	-0.7	(0.2)	3.9	(0.7)	1.9	(1.1)
Total	28	(3.5)	0.4	(0.2)	-1.5	(0.3)	27	(3.6)

Notes: Bootstrap standard errors in parentheses. Network Effect freezes merchant retail prices and acceptance at baseline while networks re-optimize fees and rewards. Price Passthrough then allows merchants to re-optimize retail prices, holding acceptance fixed. Merchant Adoption allows merchants to adjust acceptance decisions. All values in $Bn, normalized to $10 trillion of total consumer purchases (**Nilson2020a**).

consumers.

The harm to networks is also modest. The first row shows that lower rewards reduce profits as networks lose consumers. The third row shows that networks recover a substantial share when merchants increase card acceptance in response to lower fees. Profits decline by only $1.5 billion, or around 6% of baseline profits.

Consumers gain $28 billion even though Table **??** shows fees and rewards falling by roughly equal amounts. The reason is that the credit fee cap moderates excessive adoption of credit cards. By revealed preference, the marginal credit card user is indifferent between credit and debit or cash. Rewards compensate for non-pecuniary costs of credit card use, including budget control, debt aversion, and account-management hassle. I refer to these costs as "credit aversion" (Online Appendix **??**). Lower rewards cause this marginal consumer to switch away from credit. But while lost rewards are merely transfers, the credit aversion that switchers avoid is a real social benefit. The first row of Table **??**, where payment choices shift most, shows total welfare rising by only $14 billion, around half the overall gain. That row conflates two offsetting forces: a large gain from reduced credit aversion and a loss from reduced consumption when rewards fall but retail prices remain fixed. Positive retail markups mean that reducing total consumption increases deadweight loss. Merchant price cuts in the second row reverse the consumption distortion, so the full welfare gain of reduced credit aversion materializes only across both rows.

Credit card adoption is socially excessive because of price coherence. Because merchants do not surcharge, each new cardholder raises retail prices for everyone else. Consumers collectively prefer lower retail prices and lower credit card use, but individually face incentives to defect to earn cross-subsidies. The credit fee cap corrects this prisoner's dilemma by reducing the rewards that drive excessive adoption.

VI.A.3 Distributional Effects

Fee and reward cuts redistribute consumption toward lower-income consumers. Consumer surplus rises by 48 bp of consumption for low-income consumers (log income at -2 SD), 37 bp for median-income consumers, and 15 bp for high-income consumers (log income at $+2$ SD). Lower retail prices benefit everyone, while reward reductions fall disproportionately on higher-income consumers who are more likely to use credit. All groups gain on net because reduced credit aversion is spread across the income distribution. To the extent that one believes revealed preference does not apply in payment markets, the average gain in consumer surplus can be subtracted from each group. Such a correction would still leave gains for low- and median-income consumers, but losses for high-income consumers.

VI.B Repealing the Durbin Amendment

The credit fee cap raises welfare by correcting adoption distortions, not by reducing market power. If they did, capping fees on any payment instrument should help. The Durbin Amendment provides a direct test, since it caps debit but not credit card fees. I study repealing the Durbin Amendment by raising the cap on debit card fees by 25 bps, matching the share of the 75 bp interchange decline that flowed through to rewards (**Hayashi2012**).[21]

Repealing the Durbin Amendment moderates the rewards race. If the debit fee cap were lifted by 25 bps, merchant fees would rise and networks would increase debit rewards. Consumers would switch to debit, especially reward-sensitive ones. As the marginal credit consumer becomes less reward-sensitive, networks pull back on rewards competition, triggering the see-saw principle and lowering credit merchant fees. Credit rewards fall 6 bps and credit fees fall 3.3 bps as credit networks pull back. Net of the debit increase, total merchant fees rise by \$4.4 billion but rewards rise by only \$0.5 billion, reflecting substitution toward lower-reward debit.

Repealing the Durbin Amendment generates large welfare gains. Consumers gain \$4.3 billion from lower credit aversion as higher debit rewards draw consumers away from credit cards, yielding \$7 billion in total welfare gains. These gains are progressive: low-income consumers gain 7 bps of consumption, versus 2.4 bps for high-income consumers. The current regime, capping debit but not credit, is worse than both laissez-faire and uniform caps. Uniform regulation benefits consumers (**Rochet.Tirole2011**).

[21] Durbin reduced debit interchange by roughly 75 bps, but not all of this flowed through to debit rewards. Banks also absorbed the loss through higher checking account fees (**Kay.etal2018**; **Mukharlyamov.Sarin2025**). I ignore the increase in checking account fees since they do not influence the relative costs and benefits of different payment methods.

Asymmetric regulation does not.

VI.C Welfare Effects of Reducing Network Competition

One might expect that increasing competition among networks would substitute for fee regulation. In payment markets, the opposite holds. Competition fuels the rewards arms race that amplifies distortions from price coherence. I test this by modeling a merger to monopoly, with fees and rewards set to maximize industry profits.

Without competitive pressure to fund rewards, a monopolist cuts credit card rewards by 90 bps. Because rewards are the primary reason consumers carry credit cards, spending on credit collapses. The credit share falls 23 pp. The merger reduces aggregate merchant fees by $40 billion despite a 36 bps rise in per-transaction fees, because far fewer transactions clear on high-fee credit cards. Unlike in **Armstrong2006<empty citation>**, where competition raises per-transaction merchant fees, here it raises the total fee burden by shifting payment composition toward high-fee credit cards.

The merger increases total welfare by $15 billion because eliminating rewards competition more than offsets the consumer losses. Consumers lose $11 billion (though imprecisely estimated) while networks gain $31 billion. Lower credit card use reduces retail prices for all consumers, but lower rewards mainly harm high-income cardholders. Low-income consumers gain 13 bps in consumption while high-income consumers lose 29 bps. These results do not support mergers, since the consumer losses are large and imprecisely estimated. They do show that rewards competition is so harmful that increases in market power can raise total welfare.

VI.D Consumer Multi-Homing and Dual Routing

Rather than regulating fees directly, policy can redirect the locus of network competition. Routing requirements, like those in the Credit Card Competition Act, would force banks to issue cards that are compatible on multiple networks (**Andriotis2022**). This increases effective multi-homing rates and allows merchants to route payments to cheaper networks, shifting the locus of competition from the consumer side toward merchant fees (**Teh.etal2022**). I model this by raising the χ^{22} parameter, which captures the complementarity of holding two credit cards, thereby increasing consumer multi-homing. For the median consumer, this is equivalent to a 20 bps reward increase for those bundles. Credit card multihoming then rises by 20 pp. from a baseline level of around 60%. Credit card fees fall 13 bps and rewards fall 15 bps. Consumer welfare, excluding the mechanical impact of increasing χ^{22}, still rises $3.3 billion, and total welfare rises $3.8

billion.[22]

The intuition for these effects follows from the merchant acceptance condition in Section **??**. A merchant's fee cost from accepting card j depends on all consumers who use it, but the incremental sales gain comes only from single-homers who would otherwise pay cash. When consumers single-home on one credit network, merchants must accept that network to capture those sales—what trial testimony in Ohio v. American Express called "cardholder insistence" (**Conrath2014**). When a consumer carries credit cards from multiple networks, the merchant can decline the expensive network's card without losing the sale—the consumer simply pays with the cheaper alternative. The incremental sales gain from accepting the expensive network shrinks. Networks must then compete on merchant fees rather than relying on consumer lock-in to extract rents from the merchant side.

Current proposed dual-routing legislation often requires that the secondary network not be a large established network. This exclusion is unnecessary. The power of the dual-routing mechanism stems from increasing multi-homing, not from reducing the size of the largest networks. The exclusion may even be counterproductive if merchants are unwilling to invest in accepting smaller networks.

VI.E Central Bank Digital Currencies and Public Entry

Proposals for central bank digital currencies (CBDC) and faster payments (**Shin2021**; **FederalReserve2022**; **Berg.etal2024a**) motivate the possibility that government entry could discipline network pricing. I simulate a public debit network with MC debit's demand and cost characteristics that sets merchant fees at cost and offers zero consumer rewards, earning zero profit. Because the entrant offers no consumer-side subsidies, its resulting market share is small. Consumer welfare gains are roughly \$3.7 billion, smaller than repealing Durbin caps. These results depend on modeling the entrant as a debit product that does not substitute for credit at the point of sale. The segmentation assumption (Section **??**) limits the competitive pressure a new payment method can exert on incumbent merchant fees. A public product that also displaced credit could generate larger inroads, but is beyond the scope of most of the current proposals for public options. The estimated welfare gains do not include any fixed costs of setting up the public network. Directly regulating fees or mandating multi-homing is thus likely more effective than

[22] Because dual routing directly modifies χ^{22}, it mechanically raises consumer surplus even at unchanged prices and portfolios. I net this mechanical component—around \$4.5 billion—out of the reported consumer surplus so that the \$3.3 billion figure reflects only the equilibrium response. Online Appendix **??** derives the adjustment. To the extent that changing preferences complicates the interpretation of welfare, the effects on fees and rewards are more robust as they are a consequence of how merchant acceptance depends on consumer multi-homing across networks.

introducing a public option.

VI.F Discussion

Online Appendix **??** re-estimates the model under alternative assumptions about consumer constraints, pass-through, and reward sensitivity.

Constraints vs Preferences Consumer payment choices in the model reflect preferences, not constraints. Some consumers' choice to primarily use debit over credit is evidence of "credit aversion." Survey evidence documents concrete motives, including budget control, debt aversion, and account-management hassle (Online Appendix **??**).

An alternative interpretation is that some consumers cannot obtain credit cards. The presence of exogenous constraints does not affect the estimated welfare effects. Online Appendix **??** uses credit score data from the DCPC to estimate a model in which some consumers are credit-constrained, with the extent of the constraint varying with income. The alternative specification predicts smaller credit aversion for the median consumer but greater reward sensitivity among remaining consumers. The estimated consumer and total welfare results are similar to the baseline model.

Pass-through The CES functional form implies full pass-through from merchant fees to retail prices. One reason full pass-through is a reasonable baseline for interchange is because interchange is an aggregate shock to all merchants (**Amiti.etal2019**). I also lack the required merchant-level interchange data matched to retail prices to test pass-through. Online Appendix **??** compares the full pass-through baseline against zero pass-through, in which retail prices are unresponsive to merchant fees in either direction. The alternative shifts the consumer-merchant welfare split but yields similar equilibrium fees and rewards.

Reward Sensitivity A more conservative specification targets only half the observed Durbin decline, attributing the remainder to non-price steering. Online Appendix **??** re-estimates the model against this lower-bound target. The halved target produces negative marginal costs for Visa, which conflict with accounting data on issuer, acquirer, and network costs. The credit fee cap remains welfare-improving in this specification. The monopoly counterfactual is the most sensitive to this calibration: its welfare gain reverses sign under the halved target. If non-rewards steering imposes real costs without direct consumer benefits, these estimates understate the true losses from high merchant fees.

Choice of Credit Fee Cap The 120 bp cap keeps the counterfactual close to observed fee levels, avoiding the large out-of-sample extrapolation required by more aggressive caps. Online Appendix **??** compares this cap to a 30 bps cap modeled after the tourist test of **Rochet.Tirole2011<empty citation>** and to the Ramsey planner's solution. The 120 bps cap achieves roughly 93% and 77% of the respective welfare gains with less out-of-sample extrapolation.

VII Conclusion

Rising credit card use and rising merchant costs both follow from intense network competition channeled toward the wrong side of the market. Because consumers are far more sensitive to rewards than merchants are to fees, networks compete for cardholders with generous rewards funded by merchant fees. Price coherence ensures that these fees pass through to higher retail prices borne by all consumers, including those who pay with cash and debit. The rewards draw more consumers to credit cards, increasing merchants' costs.

The counterfactual results speak to contemporary legal and policy debates in payments and platform markets. The Supreme Court's 2018 decision in *Ohio v. American Express* held that plaintiffs must show net harm to the two-sided market as a whole, requiring evidence on both sides of the platform. This paper provides a structural framework for carrying out that analysis, linking consumer rewards, merchant fees, and retail prices within a single equilibrium model of a two-sided card market. Applied to credit card merchant fees, the framework shows that a modest cap reduces both fees and rewards yet raises total welfare by $27 billion per year through more efficient payment choice.

The Durbin Amendment shows that capping the wrong fees makes things worse. Congress capped debit interchange in 2010 because it appeared supracompetitive, but those fees fund rewards that draw consumers away from high-fee credit cards. Repealing the cap would raise welfare by $7 billion per year, making the current U.S. regime of capping debit but not credit fees worse than laissez-faire.

The merger counterfactual challenges a second piece of conventional wisdom: that high merchant fees reflect too little competition. The Walmart class action against Visa and MC alleged that the networks collectively set interchange at supracompetitive levels (**Constantine2012**). A monopoly network raises per-transaction fees, which is the harm that antitrust cases have targeted. Yet aggregate merchant costs fall because the monopolist has no rival to outbid for cardholders, so it cuts rewards and credit use declines. The merger increases total weflare by $15 billion because eliminating rewards competition reduces the overuse of credit cards. Antitrust typically targets harms from low output. Under price coherence, the harm is excessive credit card adoption. Buy Now, Pay Later is the latest technology to woo consumers with generous terms funded by high merchant fees (**Berg.etal2024**). Whether these services raise welfare depends on whether they shift competition toward merchants or simply open another front in the war for cardholders.

A broader lesson from the counterfactuals is that which side of the market networks compete for matters as much as how intensely they compete. The dual routing counterfactual illustrates this: when consumer multi-homing forces networks to compete for merchants rather than cardholders, both credit card fees and rewards fall, and total welfare rises on net.

A Data Details

A.1 Issuer Payment Volumes

I build an annual panel of issuer payment volumes from the Nilson Report, supplemented with FFIEC call reports (banks) and NCUA call reports (credit unions), to estimate consumers' reward sensitivities. For credit unions, I proxy interchange income with non-interest income. **CUToday2016<empty citation>** find that interchange accounts for roughly half of non-interest income even after the Durbin Amendment. The panel spans 2007–2014.

I exclude issuers with assets below \$2 billion or above \$200 billion to remove systemically important banks subject to other regulations. I also exclude issuers whose acquisitions exceeded 50% of equity during 2006–2018, since their volume changes mainly reflect portfolio expansion rather than organic consumer choices. The excluded issuers are First Tech FCU, Firstmerit Bank, BMO Harris, Regions Bank, and Synovus/Columbus B&T. The 50% threshold removes only transformative mergers where acquired volumes dominate the issuer's pre-merger base. Coverage varies by year (Online Appendix Figure **??**) because the Nilson Report covers only the largest issuers and reporting categories shift over time. Table **??** reports summary statistics of institutions in the panel. Signature debit has the highest coverage (the top 200 issuers were reported in 2007), so I use it as the main volume measure. The analysis sample requires at least one observation of both signature debit and credit card volumes on each side of the Durbin Amendment, restricting to issuers that offered both products throughout. The Online Appendix compares event studies with and without merging issuers and finds similar results.

A.2 Aggregate Shares and Prices

The Nilson Report publishes annual consumer purchases by payment method, including network-specific volumes for Visa, MC, AmEx, and Discover (Figure **??**) and an estimate of cash spending. Its concept of consumer purchases covers most Personal Consumption Expenditures (PCE) but excludes items like imputed rent that do not trigger a payment between two parties. I restrict the model to Visa, MC, and AmEx but scale their volumes to match aggregate credit and debit card totals, using 2019 shares.

The Nilson Report provides annual merchant fees from acquirer surveys, split by V/MC Credit, AmEx, and V/MC Debit. I take 2019 Visa Credit, Visa Debit, and MC Debit fees from network financial data and recover MC Credit and AmEx fees in the estimation, treating their Nilson values as measured with error (Online Appendix **??**).

I collect total rewards expense from the annual reports of AmEx, Chase, Citi, Bank of America, Capital One, and US Bank, which together cover about 80% of US credit card volume in 2019. Dividing by credit card spending gives volume-weighted average rewards of 1.74% (non-AmEx) and 1.85% (AmEx) for 2019.

Total rewards expense overstates consumer benefits because it ignores annual fees. AmEx reports annual fees of about 38 bps of volume in 2019. For other banks, **Adams.Bord2020<empty** find average annual fees of 20 bps over 2014–2019. Annual fees roughly doubled from 2015 to 2019 (**CFPB2021**). Under a linear trend, the 20 bps mid-period average implies about 15 bps in 2015 and 30 bps in 2019. I subtract 30 bps from non-AmEx rewards to get the net consumer benefit for 2019.

A.3 Homescan

The NielsenIQ Homescan panel tracks payment decisions of over 100,000 households at consumer packaged goods stores. I use the full 2013–2023 panel for payment event studies but draw multi-homing moments from 2017–2019 households.

A.3.1 Defining Payment Preferences

I first drop households with more than 1% missing payment information, removing 28.9% of the sample (concentrated in 2013–2014). I classify a household as cash-only if their cash share exceeds a cutoff calibrated to match the DCPC share who prefer cash for non-bill payments (19.7%). Among non-cash consumers with more than 20 card trips, I define the primary card as the one with the most trips and the secondary card (if any) as the one with the second-most trips. This restriction removes 8% of non-cash consumers. Table **??** shows that trip-based and spending-based rankings are highly correlated: the top-trip card is also the top-spending card for about 96% of consumers (volume-weighted across card types).

When counting trips, I focus on stores accepting all four main card brands to separate consumer preferences from merchant acceptance. I drop store-years with fewer than 500 trips (0.7% of the remaining sample), then remove stores where any network's transaction share falls below 1.5% (15.7% of the remaining sample). Store-years below this threshold are classified as non-acceptors and dropped to avoid confounding acceptance with non-acceptance in the multi-homing moments. Figure **??** shows the resulting distribution of payment mix across merchant-years.

Although Visa and MC operate separate debit networks, almost no consumers multi-home across them (Figure **??**), so I aggregate Visa Debit and MC Debit into a single "debit" category when constructing the multi-homing moments. This aggregation does not affect estimates of credit card multi-homing. The structural model in Section **??**

retains distinct Visa Debit and MC Debit products on the fee side while using this aggregation for consumer adoption moments.

A.3.2 Payment acceptance event studies

I build event-study samples around acceptance changes at grocery stores, selecting households in zip codes where a grocer operates in year $Y-1$ whose preferred payment method I can define in that year. I use data from 24 months before and after each change, setting months with no observed transactions to zero within each household's observed span.

A.4 Diary of Consumer Payment Choice

The Diary of Consumer Payment Choice (DCPC) provides consumer demographics, stated payment preferences, and transaction-level records of method choice and merchant acceptance. I use the 2017–2019 waves, conducted by the Federal Reserve Bank of Atlanta and fielded through USC's Understanding America Study. For card transactions, acceptance is observed directly. For cash transactions, the diary asks whether the merchant would have accepted a card. Check transactions (2.1% of all transactions) lack this acceptance question. To estimate non-acceptance among check transactions, I use the survey's "why not preferred" question: among card users, 21% report writing a check because the merchant did not accept cards. I count all confirmed non-acceptance cases plus 21% of check transactions when estimating the share of transactions at non-accepting stores. I resample this fraction of check transactions when bootstrapping standard errors. The DCPC also asks respondents to report their credit score in rough categories, which I use to parameterize credit access in the credit-constrained robustness check (Online Appendix **??**).

A.5 Second-Choice Survey

I ran an online second-choice survey of English-speaking US adults under 50 via Prolific in two waves (June and August 2024), approved under IRB STU00221445. The survey elicited: primary payment method for in-person transactions, household income, primary bank and consideration set, second choice if the primary method became unavailable, and rewards sensitivity. For credit users, rewards sensitivity was whether they would switch if rewards halved; for debit users, whether they would switch if credit rewards doubled. The final sample contains 357 primary credit and 383 primary debit users; Table **??** reports summary statistics. Online Appendix **??** provides additional design details.

B Model Details

B.1 Merchant Profits and Optimal Pricing

Merchant profits equal the integral of per-unit margins times quantities across consumer types:

$$\Pi\left(\gamma, p, M\right) = \int \sum_{w \in \mathcal{W}} \tilde{\mu}_y^w \times q^w\left(\gamma, p, M, y\right) \times L_M^w\left(\gamma, p\right) \, \mathrm{d}F\left(y\right) \tag{31}$$

where $\tilde{\mu}_y^w$ is the mass of consumers at income level y with wallet w.

Per-unit margins. The average margin L_M^w weights each instrument by its share of sales:

$$L_M^w\left(p\right) = \underbrace{\frac{1 - \pi_{M,w_1}^w - \pi_{M,w_2}^w}{1 + \gamma v_M^w}}_{\text{share on cash}} \times \left(p - 1\right) + \sum_{j=1}^{2} \underbrace{\frac{\pi_{M,j}^w\left(1+\gamma\right)}{1 + \gamma v_M^w}}_{\text{share on card } w_j} \times \left(p\left(1 - \tau_{w_j}\right) - 1\right) \tag{32}$$

Collecting terms yields

$$L_M^w\left(p\right) = p\left(1 - \frac{(1+\gamma)\tau_M^w}{1 + \gamma v_M^w}\right) - 1 \tag{33}$$

where $\tau_M^w = \sum_{j=1}^{J} \pi_{M,w_j}^w \tau_j$ is the wallet-average fee.

Profits. CES preferences are homothetic, so $q^w(\gamma, p, M, y) = q^w(\gamma, p, M, 1) \cdot y$; the income distribution then integrates out of Π. Using $\tilde{\mu}^w q^w(\gamma, p, M, 1) = C\mu^w p^{-\sigma}\left(1 + \gamma v_M^w\right)$ (with C the aggregate demand constant and μ^w the demand shares from Equation **??**). Merchant profits become:

$$\Pi\left(\gamma, p, M\right) = C \times \sum_{w \in \mathcal{W}} \mu^w\left(1 + \gamma v_M^w\right) p^{-\sigma} \times \left[p\left(1 - \frac{(1+\gamma)\tau_M^w}{1 + \gamma v_M^w}\right) - 1\right] \tag{34}$$

This matches Equation **??** in the main text.

Optimal price. Setting $\partial\Pi/\partial p = 0$:

$$\sum_{w \in \mathcal{W}} \mu^w\left(1 + \gamma v_M^w\right)\left[\left(1 - \sigma\right) p\left(1 - \frac{(1+\gamma)\tau_M^w}{1 + \gamma v_M^w}\right) + \sigma\right] = 0 \tag{35}$$

Solving for p gives the optimal price in Equation **??**.

B.2 Linearizing Merchant Profits

Define quasi-profits $\overline{\Pi}$ by:

$$\overline{\Pi}\left(\gamma, M, \tau\right) \equiv C \times \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} \left\{-a_M + b_M\gamma + \frac{1}{\sigma}\right\} \tag{36}$$

where C is the aggregate demand constant from Equation **??**, and

$$a_M = \sum_{w\in\mathcal{W}} \mu^w \tau_M^w, \qquad\qquad b_M = \frac{1}{\sigma}\sum_{w\in\mathcal{W}} \mu^w \left(v_M^w - \sigma\tau_M^w\right) \tag{37}$$

recovering the coefficients in Equation **??**.

Theorem 1. *For any γ, M, and τ with $\tau^{\max} \equiv \max_j \tau_j$,*

$$\widehat{\Pi}\left(\gamma, M, \tau\right) - \overline{\Pi}\left(\gamma, M, \tau\right) = \left(1+\gamma\right) O\left(\left(\tau^{\max}\right)^2\right)$$

Proof. The optimal payment-specific prices are $p_j = \frac{\sigma}{\sigma-1}\frac{1}{1-\tau_j}$ for each payment method j. By the envelope theorem, any price within $O(\tau_j)$ of p_j delivers the same profit up to second-order terms in τ_j. Setting $\overline{p} = \frac{\sigma}{\sigma-1}$ and collecting terms recovers the linear approximation. Collecting terms in γ:

$$\Pi\left(\gamma, \overline{p}, M\right) = C \times \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} \left(-a_M + b_M\gamma + \frac{1}{\sigma}\right) = \overline{\Pi}\left(\gamma, M, \tau\right)$$

□

Online Appendix **??** confirms that $\overline{\Pi}$ is numerically accurate.

C Estimation Details

I estimate the model by matching data and simulated moments. Inference is by bootstrap. Table **??** maps each data source to the parameters it identifies.

C.1 Bootstrap Inference

Standard errors come from 100 bootstrap samples. Each sample redraws data moments independently across datasets, preserving within-block correlation at fixed sample sizes. For each draw, I re-estimate the full model and re-simulate all counterfactuals. Reported standard errors are the standard deviations of these bootstrap distributions.

C.2 Consumer Parameters

Random coefficients Σ. Two sets of survey moments pin down Σ. In the 2024 second-choice survey, I compute the share of credit card users who would switch to cash if credit cards did not exist, and the analogous share for debit users. I simulate these by dropping the relevant wallets from the choice set and computing joint first- and second-choice probabilities using the logit formula of **Berry.etal2004<empty citation>**, integrating over income and random coefficients. The second set of moments is the shares diverting to the same card type on a different network when their bank drops their primary card, also from the 2024 survey. I adjust for within-network bank moves (Online Appendix **??**). To match these, I perturb Ξ^{w_1} for each primary card and measure the market-share-weighted diversion share.

Mean reward sensitivity α_0. I target the last post-period coefficient from the Durbin event study using issuer-level Nilson data (2007–2014; Figure **??**). I simulate the equilibrium effect of a 25 bps cut to debit rewards to match it.

Income gradients β_y and α_y. For β_y: I compute mean log income by primary payment type in the DCPC (2017–2019; Table **??**). Since $\Pr(\text{wallet} = w \mid y)$ is given by $\tilde{\mu}^w_y$, Bayes' rule pins down expected income by payment type, and β_y is set to match (Table **??**).

For α_y: I target the elasticity of switching probability with respect to income among credit card holders asked whether they would switch issuers if rewards halved (Table **??**). In a conditional logit with small issuer shares ($s_{ij} \ll 1$), switching probability is proportional to α_i, so this elasticity directly reveals α_y.

Multi-homing parameters χ^{lm}, χ^{lm}_y, ω, π. From Homescan (2017–2019), I compute the conditional probability $\hat{P}(l \mid k)$ that a household's secondary card is type l given primary type k, for $(k,l) \in \{$(credit, debit), (debit, credit), (credit, credit)$\}$. I compute these separately for above- and below-median income households. The model counterparts are the analogous conditional probabilities, re-derived from the simulated adoption shares $\tilde{\mu}_y^w$ separately for each income group. The level of each conditional probability (averaged across income groups) identifies χ^{lm}; the income gradient identifies χ_y^{lm}. The weight ω is matched to the gap between Visa's share among primary and secondary credit cards. The spending share π is matched to the average primary card spending share among multi-homing households (Table **??**). These eight moments exactly identify the eight multi-homing parameters: six conditional probabilities (three card-type pairs $\times$ two income groups), the Visa primary/secondary gap, and the spending share.

Mean unobserved characteristics $\bar{\Xi}_j$. I match dollar spending shares by network from the Nilson Report (2019). In the model, $d_j / \sum_{j'} d_{j'}$ gives the simulated counterpart, where d_j is defined in Equation **??**.

C.3 Network Parameters

Visa Debit, Visa Credit, and MC Debit fees come directly from network financial data (2019). MC Credit and AmEx fees are less precisely measured, so I estimate them.

Adoption utilities A_j and network marginal costs c_j. I seek A_j and c_j satisfying two conditions: each network's first-order condition for A_j holds at c_j, and inverting Equation **??** yields implied rewards matching observed per-dollar rates $r_j = f^j / s_j$, where s_j is the spending share of a single-homer on network j. Cash and debit rewards are set to zero.

Mean merchant benefit $\bar{\gamma}$ and unobserved merchant fees. Visa Credit's first-order condition with respect to its merchant fee pins down $\bar{\gamma}$. The analogous first-order conditions for MC Credit and AmEx identify their merchant fees.

C.4 Merchant Parameters

Benefit dispersion ν_γ. I target the share of spending at card-accepting merchants (Table **??**). In the model, this share is spending at merchants above $\underline{\gamma}$, the acceptance cutoff for the grand bundle of credit cards (Equation **??**), divided by total spending (Equation **??**). Given $\bar{\gamma}$, the acceptance rate pins down ν_γ.

CES elasticity σ. I target the average of the four post-period coefficients from the grocery event study (Section **??**). The simulated counterpart is the same cutoff $\underline{\gamma}$ (Equation **??**), which depends on σ through merchants' quasi-profits (Equation **??**).

Online Appendices: Not for Publication

OA Additional Tables

Table OA.1: Summary statistics of the Homescan sample

	N	Mean	P25	Median	P75
Years per Household	118537	4.07	1.00	3.00	6.00
Transactions	118537	680.07	147.00	373.00	917.00
Average Tx Size	118537	59.02	37.01	51.60	72.01

Notes: NielsenIQ Homescan panel, 2013–2023. Each observation is a household-year.

Table OA.2: Comparing Homescan payment shares to aggregate shares

Payment Method	Homescan	Nilson
AmEx	0.04	0.10
Cash	0.24	0.20
Debit	0.37	0.36
MC	0.11	0.11
Visa	0.24	0.24

Notes: Homescan payment shares are each method's dollar volume divided by total spending. Aggregate shares are from **Nilson2020a<empty citation>**.

Table OA.3: Event study estimates for the effects of card acceptance on sales to credit-card consumers

	Trips	Spending
Credit Share at Treated Retailer, t=-4	-0.012	-0.020
	(0.034)	(0.046)
Credit Share at Treated Retailer, t=-3	-0.004	-0.044
	(0.027)	(0.035)
Credit Share at Treated Retailer, t=-2	0.000	-0.025
	(0.019)	(0.025)
Credit Share at Treated Retailer, t=0	0.085***	0.076**
	(0.022)	(0.027)
Credit Share at Treated Retailer, t=1	0.085**	0.093**
	(0.026)	(0.034)
Credit Share at Treated Retailer, t=2	0.142***	0.160***
	(0.033)	(0.041)
Credit Share at Treated Retailer, t=3	0.153***	0.184***
	(0.034)	(0.043)
N	330746	330746
Period × Retailer × Income × Zip3 FE	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Regression coefficients from the triple-difference Poisson specification described in Section **??**. The dependent variable is spending (dollars) by household h at retailer j in period t. Periods are six months. Standard errors clustered at the household level.

Table OA.4: Average share of card spending on consumers' top two cards, by primary card

Primary Card	Primary Share	Secondary Share	Top Two Total
AmEx	0.73	0.19	0.93
Discover	0.76	0.18	0.94
MC	0.74	0.20	0.94
Visa	0.78	0.17	0.95
Debit	0.81	0.14	0.96

Notes: Homescan panel, 2017–2019. The primary card has the most trips; the secondary has the second-most. Each cell reports the average share of total card spending on the primary or secondary for households with the indicated primary type.

Table OA.5: Credit card multi-homing rates: usage-based vs. ownership-based measurement

Dataset	Measure	Years	Rate	CI	N
DCPC	Ownership	2017-2019	57.7%	[57.0%, 58.4%]	20,772
Homescan	Usage	2017-2019	56.7%	[56.3%, 57.1%]	53,714

Notes: Share of primary credit card users who hold (DCPC) or use (Homescan) cards from two or more of {Visa, MC, AmEx}. DCPC counts are from the Diary of Consumer Payment Choice survey, which asks respondents directly which cards they own. Homescan rates are computed from observed primary and secondary card usage, with a consumer classified as multi-homing if both their primary and secondary card are credit cards. Both samples restrict to 2017–2019.

Table OA.6: Multihoming rate by income: data vs. model

	Data	Model
Below Median Income	53.6%	54.0%
Above Median Income	57.9%	58.2%
Gradient (High - Low)	+4.3 ppts	+4.3 ppts

Notes: Homescan panel, 2017–2019. Multihoming rate is the share of households whose primary and secondary payment methods are both credit cards. Income split at the median. The reported gradient is computed from the unrounded values; displayed cells are rounded to one decimal, so subtracting them may give a slightly different number.

Table OA.7: Data sources and parameters identified

Data Source	Years	Moments	Parameters Identified
Panel A: Identification			
Second-choice survey	2024	Substitution patterns when card types unavailable	Random coefficient distribution (Σ)
		Switching probability when rewards cut in half	Income gradient of reward sensitivity (α_y)
Durbin event study (Nilson Report)	2007–2014	Change in debit spending following fee cap	Reward sensitivity (α_0)
DCPC	2017–2019	Log income differences by primary payment method	Income gradients in base preferences (β_y)
Homescan	2017–2019	Conditional probabilities of carrying card pairs	Multi-homing complementarity (χ^{lm}, χ_y^{lm}), weights (ω, π)
Nilson Report	2019	Dollar spending shares by payment network	Mean unobserved characteristics ($\bar{\xi}$)
Merchant event study + Aggregate pricing	2013–2023 / 2019	Sales effect of credit card acceptance at grocer	Merchant benefit distribution ($\bar{\gamma}$, ν_γ)
Panel B: Validation			
MRI-Simmons	2009–2022	Payment shares across merchants; bank switching after Durbin	Merchant sorting (Appendix OE)
Yelp Reviews	2010–2018	Merchant card acceptance by sector	Acceptance rate cross-check

Table OA.8: Summary statistics of Financial Institutions in the Nilson Panel in 2010

	N	Mean	P25	P50	P75
Assets	39	28704.60	4131.11	9841.50	25562.84
Credit	39	1515.09	365.15	586.38	1500.00
Debit	39	4622.54	1188.69	2334.12	4510.50
Signature Debit	39	2944.55	762.62	1138.75	2638.25
Sig Debit Ratio	39	0.64	0.61	0.68	0.77
Treated	39	0.46	0.00	0.00	1.00

Notes: Treated indicates the institution had more than $10 billion in assets in 2010. Institution-year observations are averaged to the institution level. Assets are in millions. Credit, Debit, and Signature Debit are card volumes in millions.

Table OA.9: Correlation between top card by trip count and top card by spending share

		Top Card by Spend				
Top Card by Trips		AmEx	Debit	Discover	MC	Visa
AmEx	N	17524	150	107	324	500
	% row	94.2	0.8	0.6	1.7	2.7
Debit	N	586	213679	531	2736	3814
	% row	0.3	96.5	0.2	1.2	1.7
Discover	N	114	114	21036	457	428
	% row	0.5	0.5	95.0	2.1	1.9
MC	N	308	653	342	50153	1314
	% row	0.6	1.2	0.6	95.0	2.5
Visa	N	693	1234	587	2082	104907
	% row	0.6	1.1	0.5	1.9	95.8

Notes: Homescan panel, 2017–2019. Rows classify households by the card with the most trips. Columns classify by the card with the highest spending share. Each cell reports the share of households in that row–column combination.

Table OA.10: Second-choice survey summary statistics

Variable	Mean (SD)	N
Primary Debit	0.52	740
Primary Credit	0.48	740
Switch Rewards	0.71	347
Bank Removed Other Debit	0.76	383
Bank Removed Other Credit	0.87	357
Card Removed Debit	0.41	740
Card Removed Credit	0.24	740
Card Removed Cash	0.35	740
Female	0.51	740
Wave 1	0.50	740
Wave 2	0.50	740
Additional Cards Used	0.79 (0.93)	740
Age	33.09 (7.89)	740
Income	88,572 (64,705)	740

Notes: Credit and debit card users from the second-choice survey (Online Appendix **??**). *Primary Debit/Primary Credit*: primary payment method is debit/credit card. *Switch Rewards*: would switch from primary credit card if rewards were halved. *Bank Removed Other Debit/Credit*: respondent would switch to another bank if their own bank dropped that payment type (defined only for primary debit/credit users). *Card Removed Debit/Credit/Cash*: respondent would switch to debit, credit, or cash if their primary payment type were no longer available. *Female*: customer is female. *Wave 1/Wave 2*: survey wave. *Additional Cards Used*: number of credit or debit cards beyond the primary card. *Age*: customer's age. *Income*: household annual income, made continuous using the geometric mean of each category's bounds.

Table OA.11: Log income regression on primary payment method

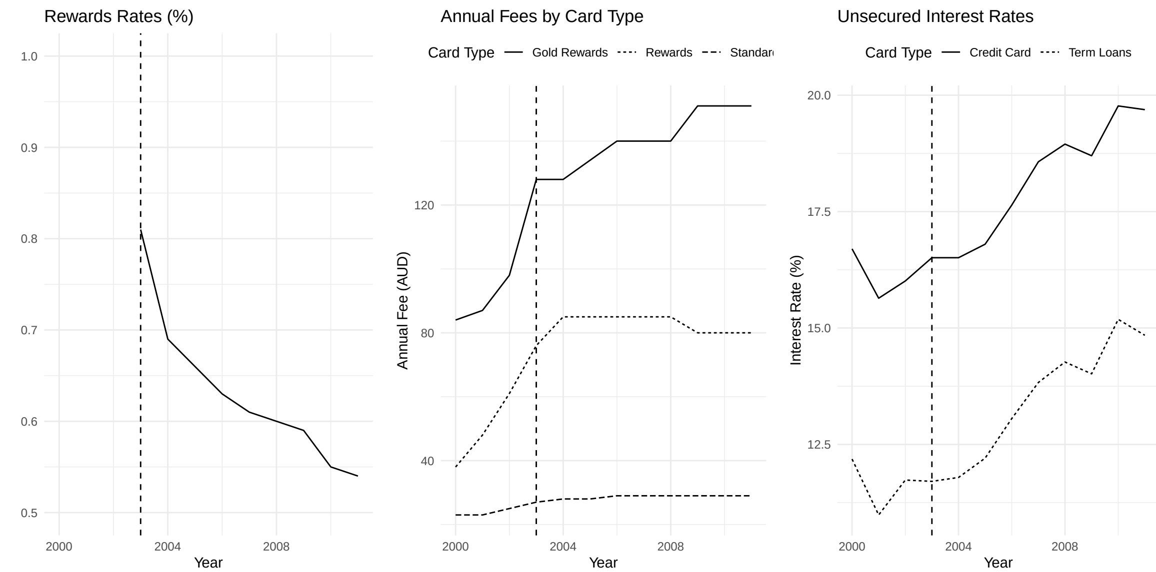
	DCPC
Prefers Debit	0.23***
	(0.04)
Prefers Credit	0.57***
	(0.04)
Num.Obs.	5986
R2 Adj.	0.090
R2 Within	0.089

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Data from the 2017–2019 waves of the DCPC. The dependent variable is log household income. The omitted category is primary cash users.

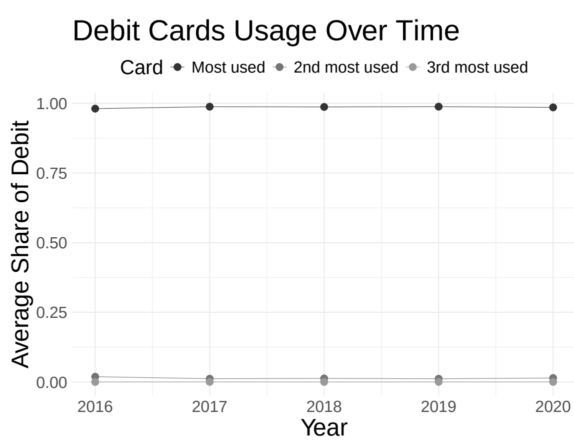
OB Descriptive Figures

Figure OB.1: Australian credit card market after interchange regulation



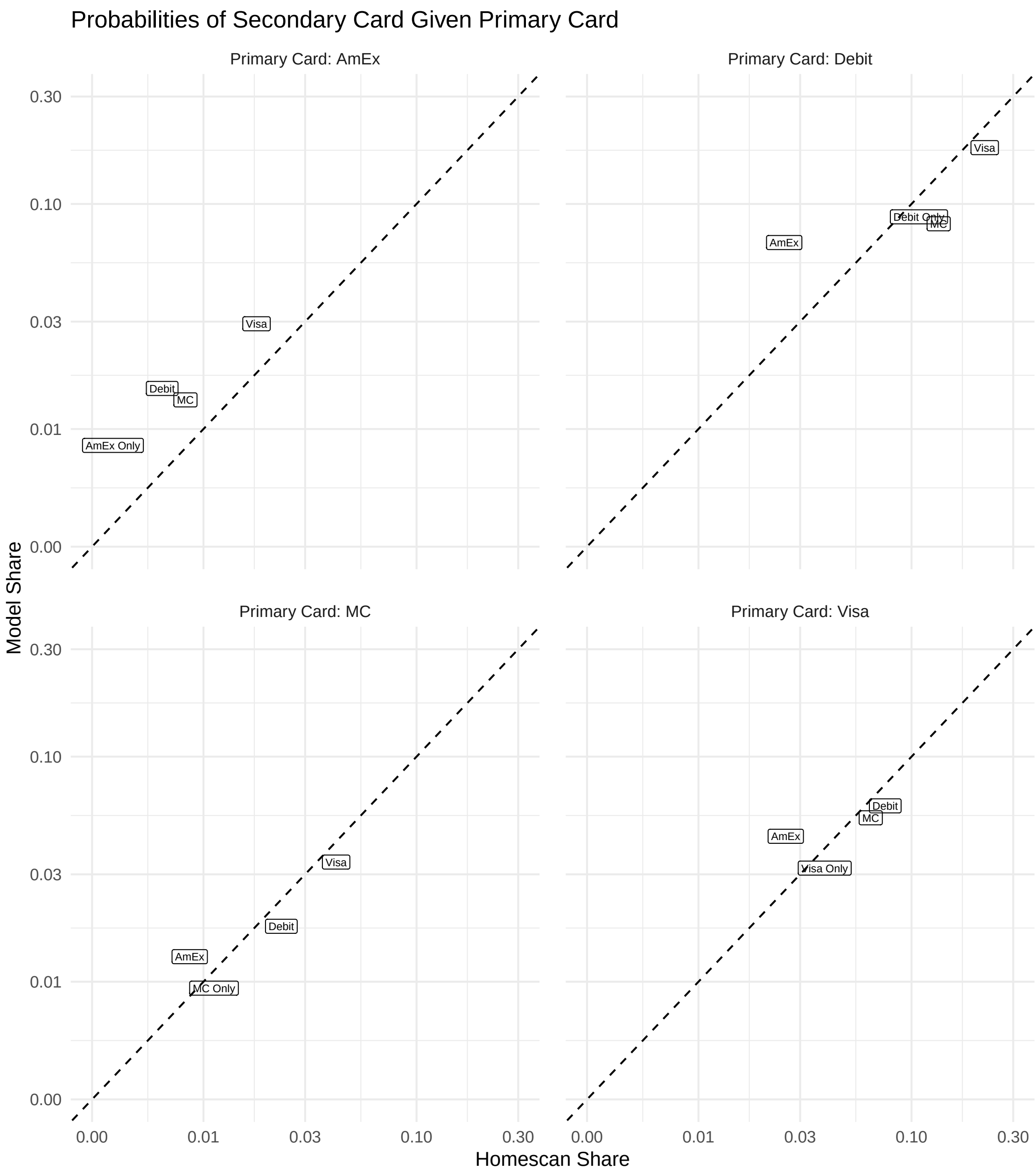
Notes: Vertical line marks 2003, when interchange regulation began. 'Gold': highest-tier rewards cards. 'Rewards': basic-tier rewards cards. 'Basic': cards without rewards. Rewards data from **Chan.etal2012<empty citation>**; annual fees from "Banking Fees in Australia" reports; interest rates from the Reserve Bank of Australia's F05 publication.

Figure OB.2: Share of debit card spending on primary debit card



Notes: DCPC data, 2017–2019. The primary debit card is the card with the plurality of a consumer's debit transactions by count.

Figure OB.3: Model fit of market shares by primary–secondary card combination



Notes: Each point plots a card combination's model share against the data share. Facets correspond to primary card choices. Points are labeled by secondary card. The dashed 45-degree line marks perfect fit.

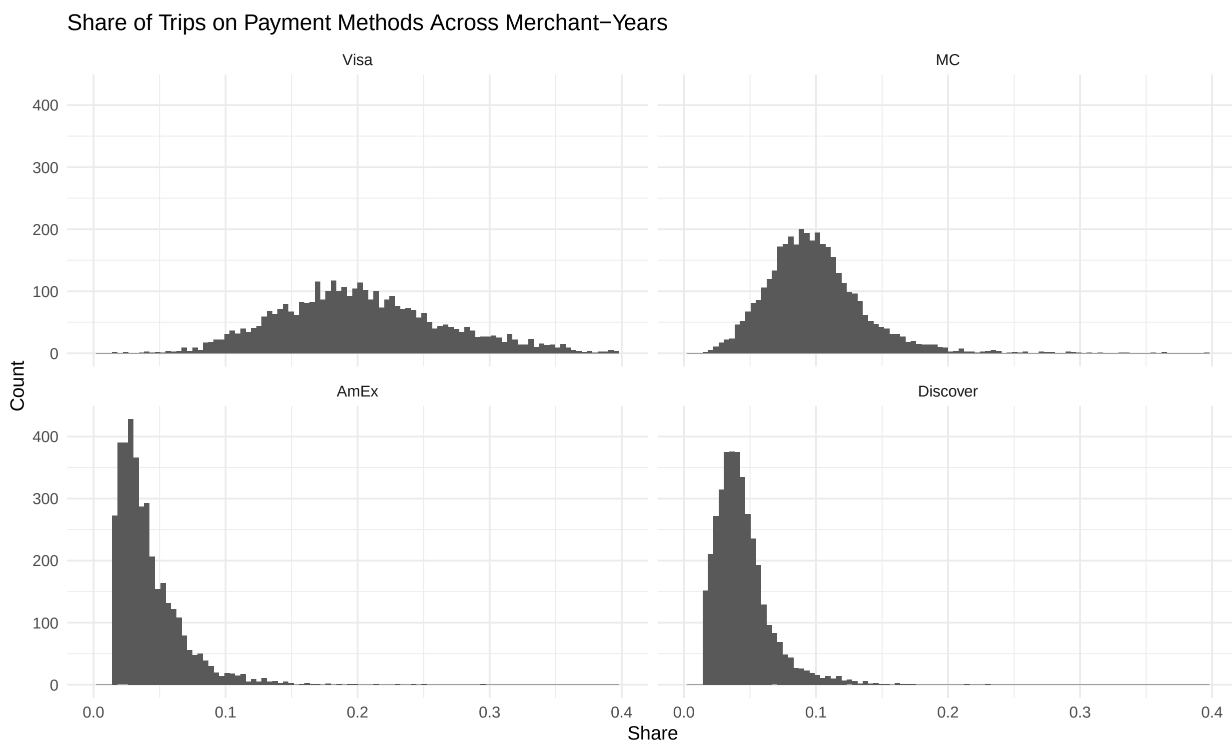


Figure OB.4: Payment mix distribution across merchants in the Homescan data

Notes: Homescan panel, 2013–2023. Each observation is a merchant-year. The figure shows the distribution of credit, debit, and cash payment shares across merchants.

OC Additional Data Details

OC.1 MRI-Simmons

MRI-Simmons USA is a nationally representative panel that merges two surveys: MRI's Survey of the American Consumer and Simmons Research's National Consumer Study. The survey covers roughly 25,000 consumers per year from 2009–2020 and 50,000 from 2021 onward. I draw on 2009–2022 data for financial institutions, payment method ownership, income, and bank switching; the 2022 wave adds shopping behavior across 214 merchant chains. These data support robustness checks throughout the online appendices but do not enter the structural estimation. Table **??** reports summary statistics.

OC.1.1 Payment method

The data report credit and debit spending by bank and network but not cash spending. The survey asks "Which of the following best describes your attitude toward using cash?" with responses including "I prefer to use cash whenever possible." I classify respondents who choose this answer as cash users. Among the rest, I assign each consumer to credit or debit based on whichever card carries higher spending.

OC.1.2 Bank type

The MRI data list each consumer's deposit institutions, which I use to sort consumers into small or large banks. Small bank customers hold deposit accounts only at community banks and credit unions. These institutions are largely exempt from the Durbin Amendment because they hold less than $10 billion in assets. Large bank customers hold deposit accounts only at the four largest banks (Chase, Bank of America, Citibank, Wells Fargo) plus U.S. Bank. Online Appendix **??** uses this split to test whether consumers switched banks or gained credit card rewards after Durbin.

OC.1.3 Share of consumers across merchants

The 2022 wave records whether each respondent shopped at 214 merchant chains: grocery stores, department stores, restaurants, and other sectors. For each merchant, I compute the survey-weighted share of customers preferring each payment method. The average customer base is 48% credit, 38% debit, and 14% cash, with standard deviations of 8–10 pp. Table **??** reports consumer-level summary statistics from the MRI survey.

OC.2 Yelp Open Dataset

I use the Yelp Open Dataset (reviews mainly from 2010–2018) to document which payment methods merchants accept and which they refuse. I identify cash-only men-

Table OC.1: Payment preferences and bank choice in the MRI survey

	2009-2021		2022	
Variable	Mean (SD)	N	Mean (SD)	N
Has Debit Card	0.6	349,546	0.75	51,697
Has Credit Card	0.58	349,546	0.75	51,697
Debit User			0.29	51,697
Credit User			0.34	51,697
Cash User			0.37	51,697
Multiple Credit Cards	0.42	349,546	0.57	51,697
Multiple Networks	0.25	349,546	0.34	51,697
Rewards	0.41	349,546	0.63	51,697
Factor Rewards	0.09	323,343	0.16	51,697
Small Banks	0.27	349,546	0.35	51,697
Big Banks	0.37	349,546	0.46	51,697
Switch Banks	0.04	323,343	0.04	51,697
Share of Debit Exp			0.4 (0.37)	43,136
Share of Credit Exp			0.6 (0.37)	43,136
Income	80,749 (60,890)	349,546	99,336 (69,037)	51,697

Notes: Has Debit Card and *Has Credit Card* indicate ownership of each card type. *Debit User*: non-cash user who either holds only debit or whose debit expenditure share exceeds credit. *Credit User*: non-cash user who either holds only credit or whose credit expenditure share exceeds debit. *Cash User*: agrees completely with "I prefer to pay cash for things I buy, whenever I can." *Multiple Networks*: owns credit cards from multiple networks. *Multiple Banks*: owns credit cards from multiple issuers. *Factor Rewards*: rated rewards as very important when choosing a bank. *Rewards*: receives credit card rewards. *Big Banks*: uses Chase, Citibank, Wells Fargo, Bank of America, or US Bank. *Small Banks*: uses community banks or credit unions. *Switch Banks*: changed banks in the last 12 months (not defined for 2009). *Share of Debit Exp* and *Share of Credit Exp*: debit and credit shares of card expenditure. *Income*: household income in dollars, made continuous using the geometric mean of each category's bounds.

tions with regular expressions ("cash only" or "only...cash"), then flag card-acceptance discussions with keywords such as "accept," "take," "took," and "bring." Among these, I search for specific payment methods (Visa, MC, AmEx, Discover, credit, debit).

I classify acceptance reviews with the ChatGPT 4o API. The prompt instructs the model to determine which payment methods each review indicates are accepted, keeping networks separate (accepting Visa does not imply MC) and keeping types separate (accepting credit does not imply debit). Table **??** shows representative review snippets for each acceptance category.

Yelp business profiles also report whether merchants accept credit cards. Table **??** compares DCPC acceptance rates to Yelp business attributes by sector. The two sources match in aggregate: 95.8% vs. 96.3%.

Table OC.2: Examples of Yelp reviews

Classification	Review (relevant snippet)
Debit, no Credit	You won't be disappointed Keep In Mind: -They only accept debit and cash -Weekends are super busy so be prepared for a crowd
Debit + Credit	Street parking, accept credit/debit cards
Visa and MC	They flat out charge you more if you don't pay with cash, now they do accept debit and consider it the same as cash so its really negligible but honestly they even label everything with two prices the cash price and non cash price. I seldom carry cash, and I seldom use debit and they see fit to penalize me for this. The cashier naturally refused to take my card as it didn't have my name on it and they always require a photo ID to make a credit purchase
Visa, no AmEx	The have complementary water and take all major credit cards except for AMEX.
Credit, no Debit	DEB'S NOW TAKES CREDIT CARDS. No Amex tho and no debit cards, but you can now pay for your delicious and cheap breakfast with Visa, Mastercard, or Discover
Only one: Visa or MC	When I went inside to pay the cashier said they do not take Mastercard. Who doesn't take mastercard. I had to pay with a credit card and with tax I was charged $45.78
AmEx, no Visa	One item that might concern some - American Express is the only credit card they accept.

OC.3 Second-Choice Survey Design Details

Appendix **??** describes the sample and summary statistics. This section provides additional design details.

The survey included an attention check; I drop 47 respondents who failed it and 12 who completed the survey outside the collection window. Household income was collected in eight brackets and made continuous with geometric means of bracket bounds.

The survey instrument first asked respondents to identify their primary payment method for in-person transactions. Respondents then answered three hypothetical questions. (i) Which method they would use if their primary payment type (credit or debit) did not exist. (ii) Which method they would use if their bank stopped offering their primary card. (iii) Whether they would switch cards if rewards changed (halving for credit users, doubling for debit users). For question (ii), the survey also asked respondents

Table OC.3: DCPC and Yelp acceptance rates by sector

Sector	DCPC Accept	N (DCPC)	Yelp (raw)	Yelp (wtd)	N (Yelp)
Grocery	97.7%	5,397	98.5%	98.2%	3,087
Shopping	98.1%	4,756	96.4%	96.8%	16,739
Fast Food	95.7%	4,214	98.1%	97.3%	11,552
Gas Station	98.9%	3,366	99.2%	99.6%	896
Restaurant	96.4%	2,165	97.4%	98.2%	23,840
General Services	82.3%	926	93.5%	95.7%	14,146
Entertainment	74.6%	789	91.0%	89.2%	3,201
Medical	94.2%	525	97.5%	98.4%	8,758
Other	91.9%	320	92.3%	96.7%	4,061
Total	95.8%	22,458	96.3%	97.6%	86,280

Notes: DCPC Accept is the share of transactions at merchants accepting cards, from the 2017–2019 Diary of Consumer Payment Choice. Because respondents record acceptance at each transaction, DCPC rates are sales-weighted across merchants visited. Yelp (raw) is the unweighted share of Yelp business profiles reporting credit card acceptance. Yelp (wtd) reweights Yelp businesses by estimated transaction volume within each sector. N columns report the number of transactions (DCPC) or businesses (Yelp) in each sector.

to name the banks they would consider switching to, which I use to construct network diversion ratios (Online Appendix **??**).

OD Additional Reduced Form Results

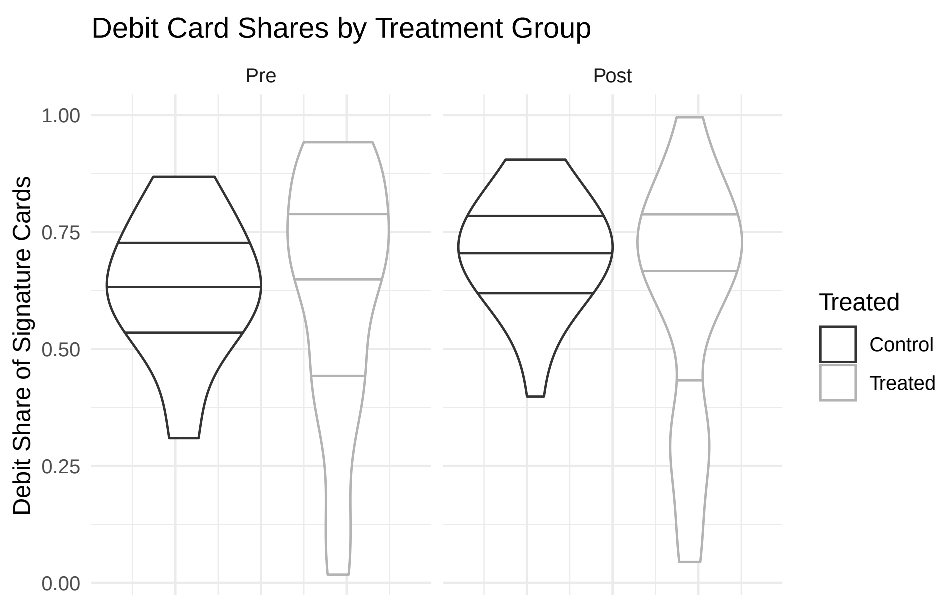
OD.1 Durbin Amendment Robustness Checks

This section checks sensitivity of the main specification and identifies the mechanism behind the debit volume decline.

OD.1.1 Sensitivity of the Main Specification

Treatment and control issuers had similar pre-policy payment mixes. The percentage decline estimate is sensitive to functional form only if treatment and control issuers have different baseline debit-versus-credit mixes. Figure **??** shows the pre-policy distributions of signature debit shares overlap closely across treatment and control banks.

Figure OD.1: Comparing debit versus credit shares at treatment and control banks



Notes: Each panel shows the distribution of signature debit as a share of total card transactions by value for control (<$10 billion in assets in 2010) and treatment banks (>$10 billion) in the pre- and post-periods. Dashed lines mark the 25th, 50th, and 75th percentiles.

Plausibility of the effect of Durbin on interchange revenue. The estimated interchange decline (Table **??**) matches Durbin's expected effect. Debit interchange at regulated issuers fell by roughly half while credit interchange was unaffected. Given that

Table OD.1: Event study estimates for the effect of the Durbin Amendment on interchange fees, signature debit payment volumes, and deposits

	Interchange	Signature Debit	Deposits	Total Debit	Credit
Treat, t=-4	-0.023	0.003	0.057	-0.114	-0.041
	(0.088)	(0.051)	(0.053)	(0.083)	(0.127)
Treat, t=-3	0.061	-0.010	-0.001	-0.005	0.027
	(0.088)	(0.039)	(0.034)	(0.064)	(0.119)
Treat, t=-2	-0.088	-0.021	0.000	0.003	0.009
	(0.073)	(0.024)	(0.018)	(0.043)	(0.070)
Treat, t=0	0.006	-0.077*	-0.012	-0.019	0.159*
	(0.056)	(0.037)	(0.018)	(0.063)	(0.069)
Treat, t=1	-0.418***	-0.105*	-0.024	-0.007	0.147
	(0.101)	(0.047)	(0.024)	(0.068)	(0.090)
Treat, t=2	-0.318*	-0.222***	-0.038	-0.090	0.258**
	(0.119)	(0.048)	(0.030)	(0.078)	(0.093)
Treat, t=3	-0.328**	-0.287***	-0.039	-0.188*	0.309**
	(0.109)	(0.063)	(0.032)	(0.083)	(0.108)
N	289	280	309	280	290
Bank FE	X	X	X	X	X
Year FE	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Difference-in-differences estimates comparing financial institutions above and below the $10 billion Durbin asset threshold. The dependent variable in each column is stated in the column header. Institutions involved in large mergers or divestitures are excluded. Standard errors clustered at the issuer level.

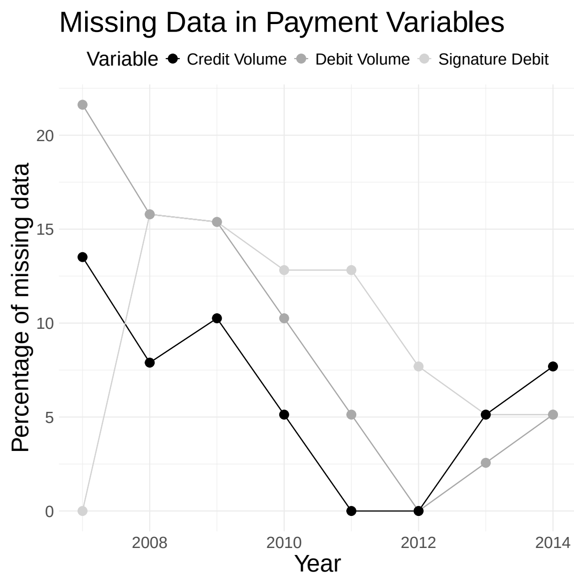
debit accounted for about two-thirds of total interchange revenue before the policy, this reduction in debit interchange is consistent with the policy (**CUToday2016**).

The estimates are somewhat sensitive to whether we use total debit or signature debit volumes. I focus on signature debit because of better data coverage in the early sample (Figure **??**), which is essential for evaluating pre-trends. Total debit volumes also fall after Durbin, albeit by a smaller amount (17% versus 25% for signature debit, Table **??**).

The estimates are not driven by outliers or the asset size cutoff. The Nilson sample contains 39 institutions (18 treated, 21 control). Figure **??** shows the treatment group distribution shifted left throughout its support, not just at a few outliers. Figure **??** shows stable estimates as asset size cutoffs vary toward the $10 billion threshold.

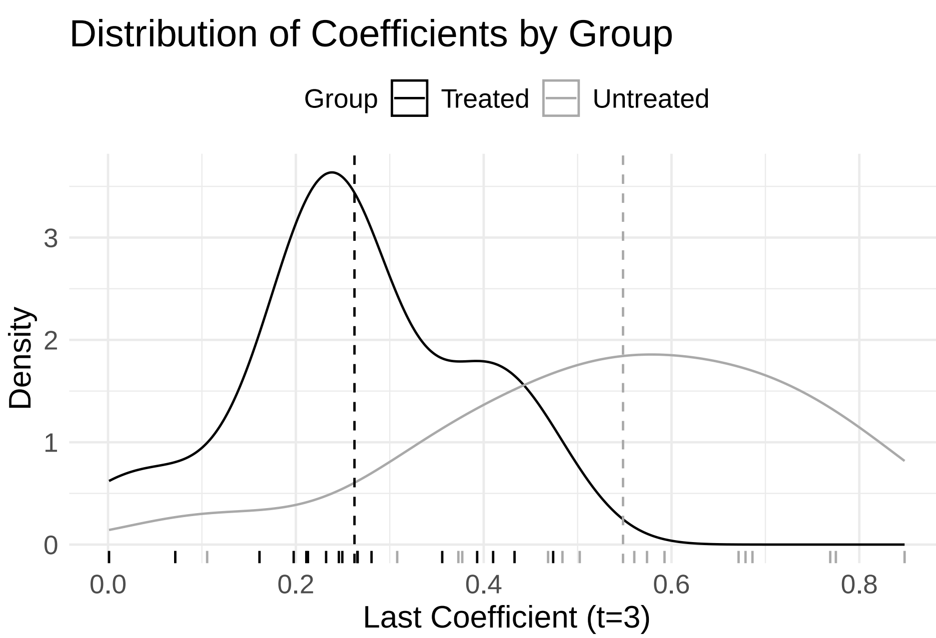
Results are robust to including mergers. The baseline excludes issuers with large acquisitions or divestitures (target assets exceeding half the acquirer's), whose volume changes reflect portfolio transactions rather than consumer payment choices. Including these issuers does not change the estimates (Figures **??** and **??**).

Figure OD.2: Share of panel observations with missing data by year and card type



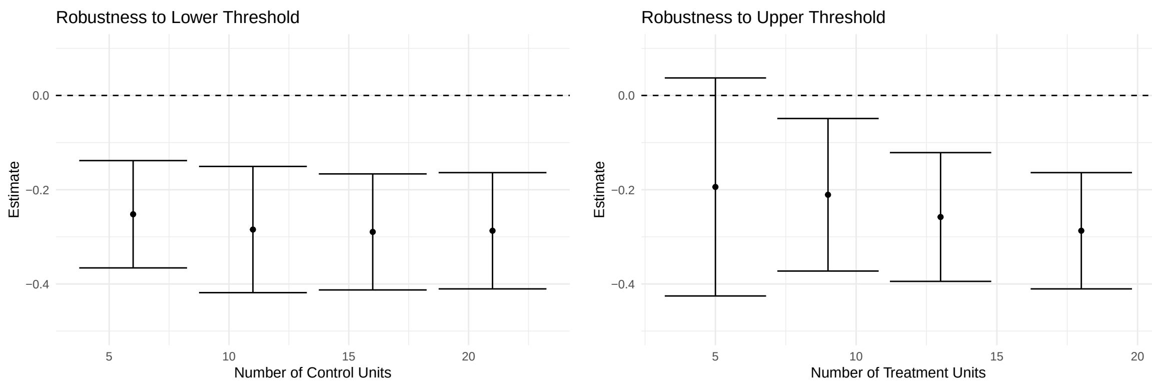
Notes: Data are from the Nilson Report. The figure plots the fraction of issuers in the analysis sample with missing volume data, by year and card type (signature debit, total debit, and credit). The Nilson Report reported the signature debit volumes for the top 200 issuers in earlier years, which improved data coverage for signature debit early on.

Figure OD.3: Distribution of institution-level estimates in the Nilson analysis



Notes: Density plots of institution-level pre-post changes in signature debit volumes for treated and control groups, from which the difference-in-differences estimate of Durbin's effect is derived. Each observation is a financial institution. The main text coefficient roughly equals the difference between the treated and control group means.

Figure OD.4: Robustness of the effect of the Durbin Amendment on debit card volumes to varying asset size cutoffs

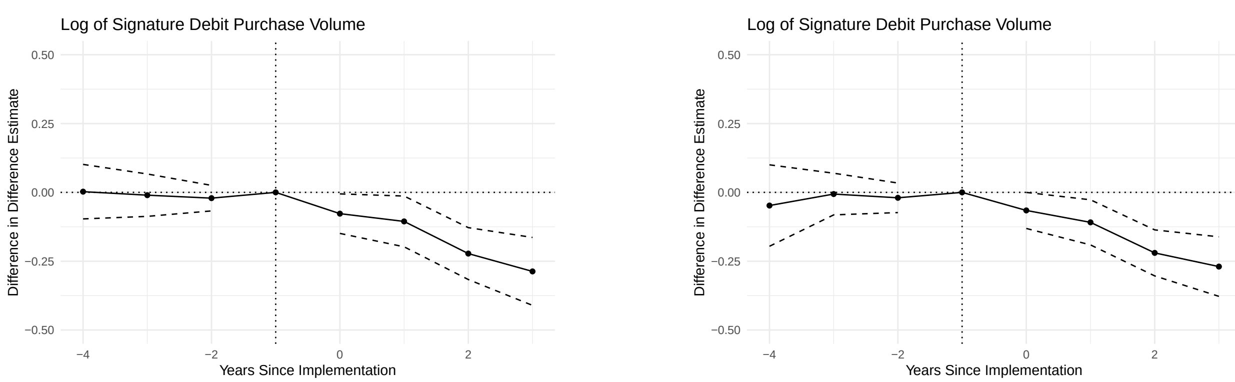


Notes: Difference-in-difference estimates of Durbin's effect on signature debit volumes as the minimum asset requirement increases toward $10 billion (control group) or the maximum decreases toward $10 billion (treatment group). Dots are point estimates. Error bars are 95% confidence intervals.

Figure OD.5: Robustness of signature debit volume estimates to including merger banks

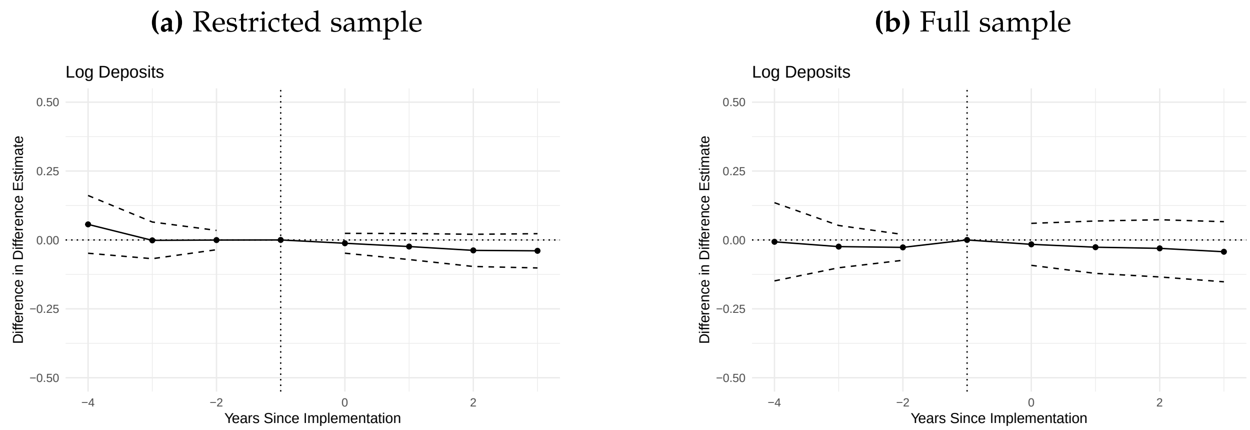
(a) Restricted sample

(b) Full sample



Notes: Same difference-in-differences design as Figure **??**. The restricted sample (left) excludes issuers with acquisitions or divestitures exceeding half the acquirer's assets. The full sample (right) includes all issuers.

Figure OD.6: Robustness of deposit estimates to including merger banks



Notes: Same design as Figure **??**, with log deposits as the dependent variable.

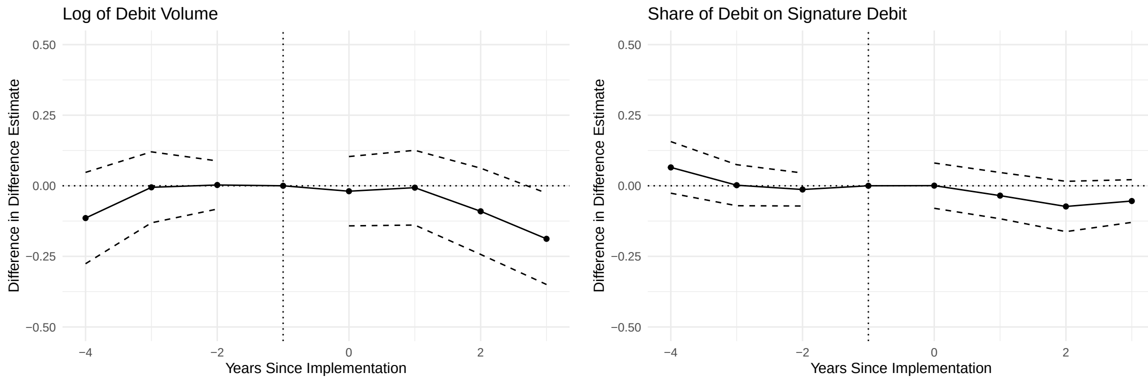
Results extend to total debit and credit card volumes. The main text focuses on signature debit because coverage in the Nilson Report is most complete. Figures **??** and **??** apply the same design to total debit volumes, credit card volumes, and signature debit's share of total debit. The panel conditions on banks with pre- and post-Durbin observations for both signature debit and credit, but not for the PIN-inclusive total, so the total-debit and compositional panels reflect a different and smaller set of reporting banks (Figure **??**). Total debit volumes decline after Durbin, although by less than signature debit and with noisier estimates. Among banks that report both signature and total debit, the compositional panel shows little shift between signature and PIN debit, suggesting the decline was broad-based rather than driven by substitution across debit types. Credit card volumes at large banks appear to rise relative to small banks after Durbin (Table **??**, Figure **??**), although pre-trends in the credit column are not cleanly zero, so this result should be interpreted with caution.

OD.1.2 Mechanism

The debit volume decline could reflect three mechanisms: bank switching by consumers fleeing Durbin-exposed institutions, a pull toward credit from improved large-bank rewards, or within-bank substitution driven by loss of debit rewards. Banks that ended debit rewards lost volumes comparable to the full treatment effect. Deposits did not shift from large to small banks, ruling out bank switching. And credit card rewards did not differentially improve at large banks, ruling out a pull toward credit at large banks.

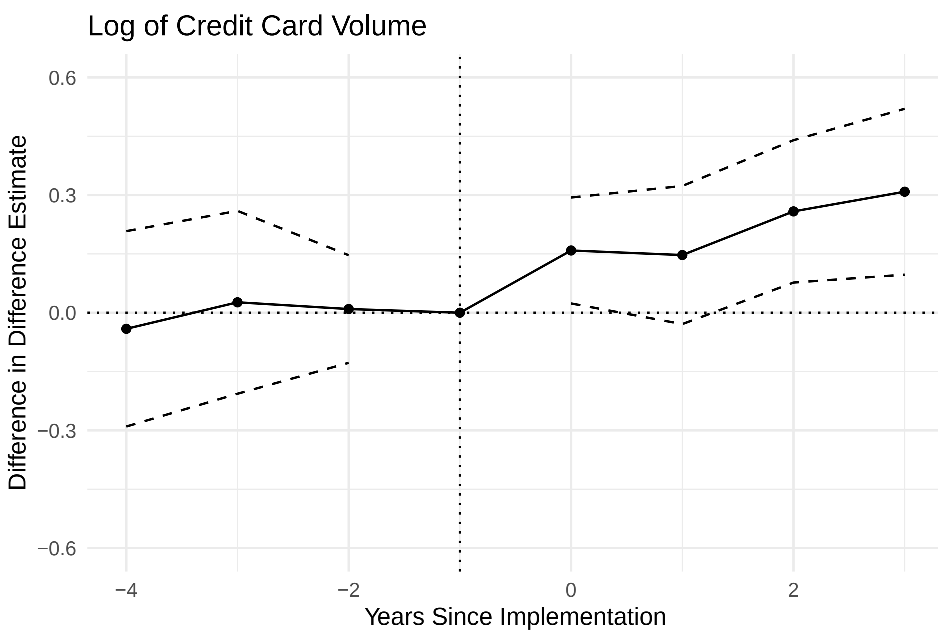
Evidence on the rewards channel. Large banks changed multiple debit card program features after Durbin, including rewards, advertising, and employee incentives. To as-

Figure OD.7: The effect of the Durbin Amendment on overall debit volumes



Notes: Data are from the Nilson Report. Both panels use the same difference-in-differences design as Figure **??**, comparing issuers above and below the $10 billion Durbin threshold. Left: log total debit volumes (signature plus PIN). Right: signature debit as a share of total debit volumes. The vertical line marks 2010, the year before the policy began to be implemented.

Figure OD.8: The effect of the Durbin Amendment on credit card volumes



Notes: Data are from the Nilson Report. The design is identical to Figure **??**, comparing issuers above and below the $10 billion Durbin threshold. The vertical line marks 2010, the year before the policy began to be implemented.

sess how much rewards contributed to the decline, I collect bank-level data on which institutions paid debit rewards before and after Durbin from archived bank websites via the Wayback Machine. Of the 39 institutions in my baseline panel, only 15 paid debit rewards before Durbin. Of these, 7 ended rewards after Durbin while 8 continued. I compare debit volume changes between these groups:

$$y_{jt} = \sum_{k \neq -1} \beta_k I\{t = k\} \times T_j + \delta_j + \delta_t + \epsilon_{jt} \tag{38}$$

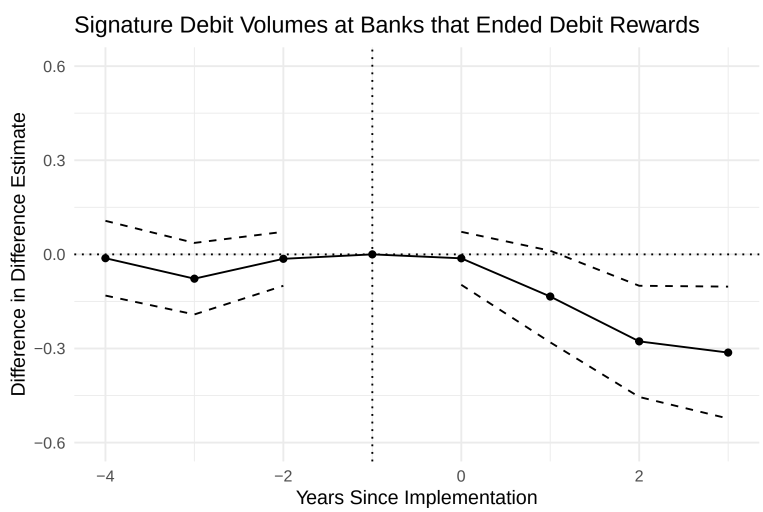
where y_{jt} is the log of debit card payment volumes, T_j is an indicator that institution j ended its rewards program after Durbin, β_k are the dynamic difference-in-difference estimates, δ_j are bank fixed effects, and δ_t are time fixed effects.

Figure **??** shows the difference-in-difference estimates. On this restricted sample, banks that ended rewards experience debit volume growth about 25% slower than banks that kept rewards, a gap that matches the aggregate estimate and motivates the calibration choice in Section **??**.

A Roy-model interpretation connects the 25% to the rewards channel specifically. Banks fund debit promotion from interchange revenue, allocating across rewards, advertising, and teller incentives. If banks differ in their productivity across these levers and each chooses the mix that maximizes consumer response per dollar, non-price steering at reward-paying banks would then be small to begin with. The within-subsample comparison then reflects mainly the rewards change.

I do not run a horse race between being above the cutoff and terminating rewards. A regression that controls for rewards and being above or below the cutoff would be misspecified. It assumes equal non-price baselines across banks, whereas the Roy model described above implies the opposite.

Figure OD.9: Direct effect of rewards on signature debit card volumes

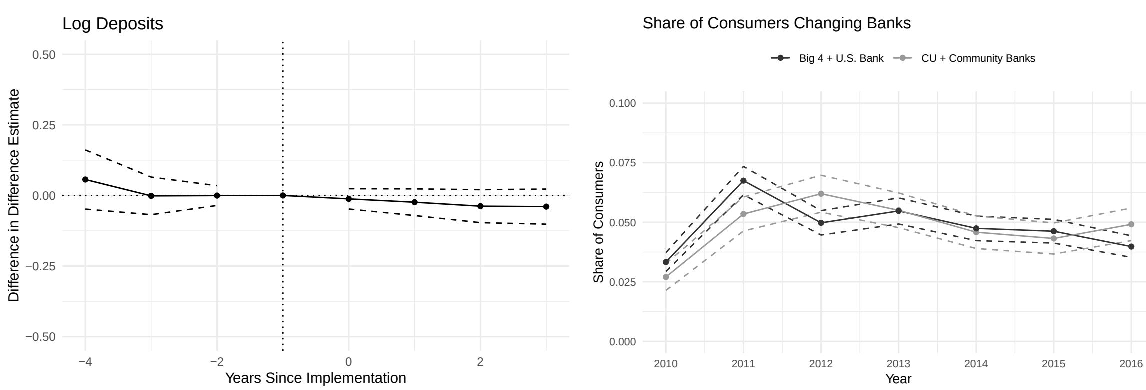


Notes: Point estimates from a regression of log debit volumes on an indicator for not paying rewards, with bank and time fixed effects. Sample: 15 banks and credit unions that paid debit rewards before Durbin. Standard errors clustered at the issuer level.

The decline reflects within-bank payment switching, not bank switching. An alternative explanation for the debit volume decline is that consumers switched from large banks to small banks after Durbin, taking their debit volumes with them. If this were the case, deposits at large banks should have fallen relative to small banks, and consumers at small banks should report higher rates of recent bank switching.

Figure **??** tests both predictions. The left panel uses the same Nilson Report panel and difference-in-differences design as the main specification, with log deposits as the outcome. Deposit growth at large banks tracked deposit growth at small banks closely throughout the post-Durbin period, with no evidence of differential outflows. The right panel uses MRI-Simmons survey data, which asks consumers whether they recently changed their primary financial institution. Consumers at small banks did not report higher bank switching rates. The evidence rules out bank switching as the driver: deposit flows and consumer behavior both stay stable across bank sizes. Debit volumes fall without customers leaving. They must be substituting toward other payment methods within their current banks. The importance of within-issuer substitution is also why I focus on issuers with both debit and credit franchises, which provide a more representative picture of aggregate switching behavior than single-product issuers.

Figure OD.10: The effect of the Durbin Amendment on deposits and bank switching behavior

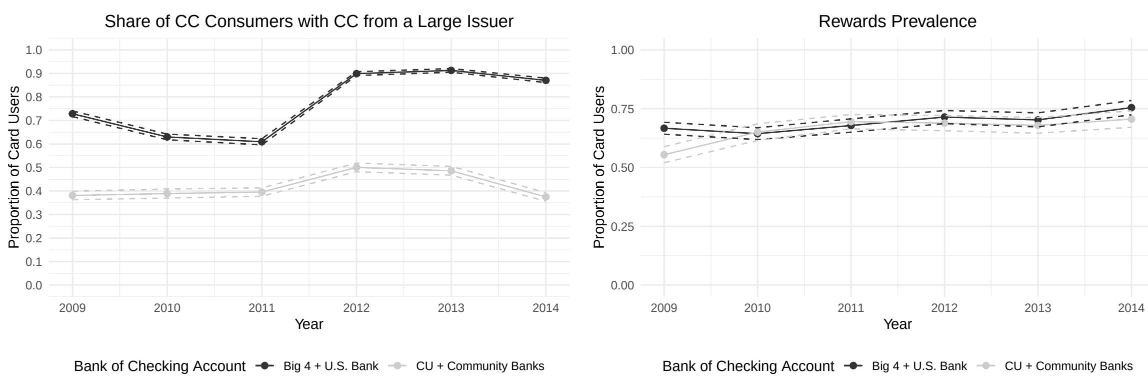


Notes: Left: estimated effect of Durbin on deposits at large banks, using the Nilson Report panel and the same difference-in-differences design as Figure **??**. The vertical line marks 2010, the year before implementation. Right: MRI-Simmons data (2009–2022) showing consumers at small banks (credit unions or community banks, largely exempt from Durbin) did not report higher rates of recent bank switching compared to consumers at large banks.

Credit card rewards did not differentially change at large banks. An alternative explanation for why consumers decreased their use of debit cards at large banks is because those banks increased credit card rewards. If so, the debit volume decline could reflect consumers being pulled toward credit by improved rewards rather than being pushed away from debit by the loss of debit rewards. This story predicts that credit card reward availability or credit card holding rates should have increased differentially at large banks after Durbin.

Figure **??** is inconsistent with this alternative. Even if large banks improved credit card rewards, the effect on debit usage would be dampened because around 40% of small-bank customers also hold credit cards from large issuers (left panel), so they would benefit from any such improvement too. More directly, credit card reward receipt rates did not differentially increase at large banks after Durbin (right panel). The debit volume decline at large banks thus reflects the loss of debit promotion, not a pull toward improved credit rewards.

Figure OD.11: Credit card rewards for consumers who use large and small deposit institutions after the Durbin Amendment



Notes: Left: probability of holding a credit card from a large issuer, by deposit institution size. Right: probability of receiving credit card rewards, by deposit institution size. Small deposit institutions: credit unions or community banks only. Large deposit institutions: Chase, Bank of America, Citibank, Wells Fargo, or U.S. Bank. Large credit card issuers: Chase, Bank of America, Citibank, Wells Fargo, Capital One, Discover, and AmEx.

OD.2 Evidence on Whether Rewards Shift Shopping Behavior

Discover's 5% Cashback Bonus program provides a natural experiment for whether rewards affect store choice. The rewards apply at grocery stores but not discount or warehouse stores. Consumers do not change where they shop during reward quarters, but they do shift payments toward Discover. Most of the Discover gain comes from other credit cards. The remaining gain comes from debit and cash in equal amounts (-0.7 pp each; Table **??**), so debit and cash appear equally good substitutes for credit at the point of sale (Online Appendix **??**). Rewards thus affect payment choice but not shopping location, even when consumers are aware of the rewards.

Table OD.2: Discover 5% Cashback Bonus Categories (2017-2022)

Year	Quarter	Reward category
2017	Q1	Gas stations, ground transportation, wholesale clubs.
	Q2	Home improvement stores, wholesale clubs.
	Q3	Restaurants.
	Q4	[redacted], [redacted].
2018	Q1	Gas stations and wholesale clubs.
	Q2	**Grocery stores**.
	Q3	Restaurants.
	Q4	[redacted] and wholesale clubs.
2019	Q1	**Grocery stores**.
	Q2	Gas stations; Uber and Lyft.
	Q3	Restaurants; PayPal.
	Q4	[redacted]; [redacted] (in-store and online); [redacted].
2020	Q1	**Grocery stores**; [redacted] and [redacted].
	Q2	Gas stations; Uber and Lyft; wholesale clubs; [redacted].
	Q3	Restaurants; PayPal.
	Q4	[redacted], [redacted] & [redacted].
2021	Q1	**Grocery stores**, [redacted] & [redacted].
	Q2	Gas stations, wholesale clubs & select streaming services.
	Q3	Restaurants & PayPal.
	Q4	[redacted], [redacted] & [redacted].
2022	Q1	**Grocery stores**; fitness clubs; gym memberships.
	Q2	Gas stations; [redacted].
	Q3	Restaurants; PayPal.
	Q4	[redacted]; digital wallets.

Note: Retailer names have been redacted. General retailer types include a major e-commerce platform, a large online store, a large discount store, two large drugstores, and a home improvement store.

The Discover program rotates reward categories quarterly. Grocery stores have appeared once a year since 2018 (Table **??**). I compare shopping and payment behavior at grocery versus discount and warehouse stores during quarters with and without Discover grocery cashback. I focus on these store categories because they are substitutes for consumers (**Ellickson.etal2020**).

For the graphical evidence, I restrict to households whose primary or secondary card is Discover.[23] The triple-difference regressions below use the full panel of Homescan households, with Discover cardholders as the treated group, so that the household-by-time fixed effects are identified. I exclude merchants not accepting Discover and household-years without trips to both store categories.

[23] Primary and secondary cards are defined by trip counts, with spending as a tiebreaker. See Appendix **??**.

OD.2.1 Graphical Evidence

Figure **??** shows the effects on payment and shopping behavior. The top panel shows grocery spending shares are flat for Discover cardholders during reward quarters (grey shaded areas). The bottom left panel shows Discover cardholders shift their spending share toward Discover during reward quarters, roughly a 7.6 pp. increase (Table **??**). Consumers respond to rewards without changing where they shop. The bottom right panel shows no change in payment composition at discount and warehouse stores, so time-varying confounds are unlikely to explain the pattern.

OD.2.2 Regression Evidence

I supplement the graphical evidence with regression estimates. I first confirm that consumers do not shift between grocery and discount/warehouse stores during reward quarters. Column 1 of Table **??** reports a difference-in-differences specification:

$$y_{it} = \alpha \cdot \mathbf{1}\{\text{Reward Qtr}\}_t \times \mathbf{1}\{\text{Discover HH}\}_i + \delta_i + \delta_t + \epsilon_{it} \tag{39}$$

where y_{it} is the grocery spending share for household i in month t, and δ_i and δ_t are household and time fixed effects. The grocery spending share coefficient is noisy and economically marginal, confirming that consumers do not systematically change where they shop.

I then show that these same consumers do respond to rewards on the payment margin, ruling out inattention. Using a triple-difference design, I estimate for each payment method m (columns 2–5):

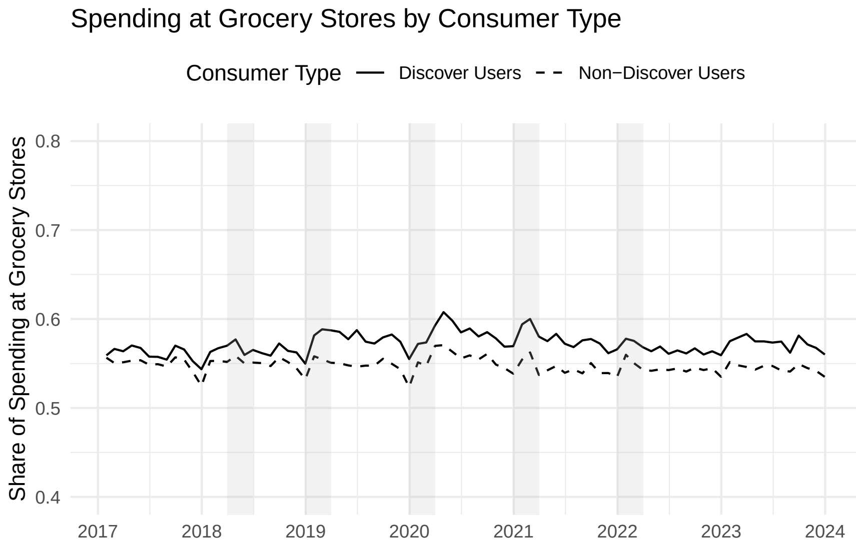
$$y^m_{igt} = \beta \cdot \mathbf{1}\{\text{Reward Qtr}\}_t \times \mathbf{1}\{\text{Discover HH}\}_i \times \mathbf{1}\{\text{Grocery}\}_g + \delta_{ig} + \delta_{it} + \delta_{gt} + \epsilon_{igt} \tag{40}$$

where y^m_{igt} is the spending share of payment method m for household i at store type g (grocery or discount/warehouse) in month t. The specification includes three sets of two-way fixed effects. Household-by-store-type (δ_{ig}) absorbs time-invariant differences in a household's payment mix across store types. Household-by-time (δ_{it}) absorbs household-level trends in payment behavior common to both store types. Store-type-by-time (δ_{gt}) absorbs aggregate shocks to payment composition at grocery versus discount/warehouse stores. Standard errors are clustered at the household level throughout.

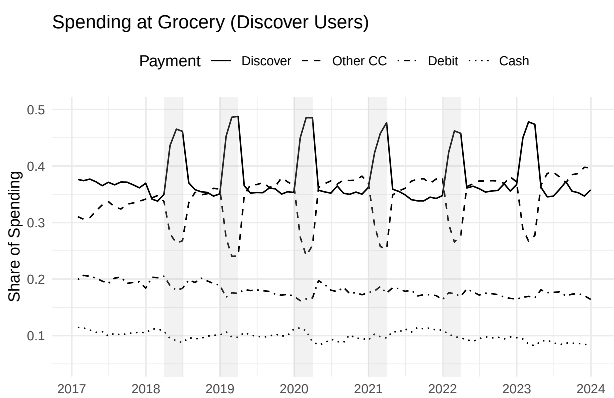
The Discover spending share at grocery stores rises by 7.6 pp. during reward quarters ($p < 0.001$), offset by declines in other credit (-6.0 pp), debit (-0.7 pp), and cash (-0.7

Figure OD.12: Effects of Discover's quarterly reward program at grocery stores on consumer spending on different payment methods across retailer types

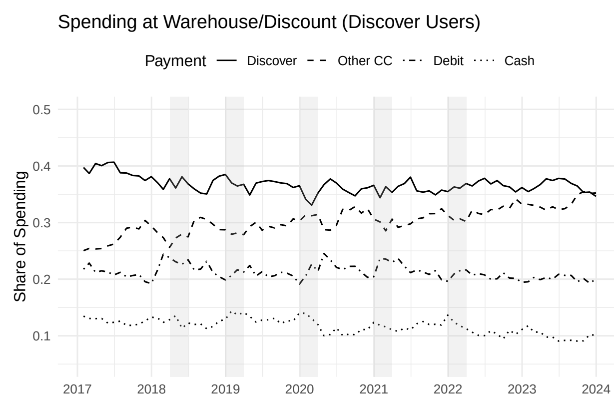
(a) Share of dollars spent at grocery stores



(b) Payment mix at grocery stores



(c) Payment mix at warehouse/discount stores



Notes: Grey areas mark quarters when Discover offers 5% cash back at grocery stores but not discount, warehouse, or other retailers. The first panel shows shopping decisions across retailer types. The second and third panels show payment composition at grocery stores (treated) and discount/warehouse stores (control). All samples condition on Discover cardholders.

pp). Consumers are clearly aware of reward timing and actively shift which card they use, yet they do not change where they shop.

Table OD.3: Effect of Discover cashback rewards on payment composition

	Grocery	Cash	Debit	Other Credit	Discover
Discover HH × Grocery × Reward Month		-0.007***	-0.007***	-0.060***	0.076***
		(0.002)	(0.001)	(0.003)	(0.003)
Discover HH × Reward Month	0.002+				
	(0.001)				
Observations	5152886	8580301	8580301	8580301	8580301
R^2	0.586	0.863	0.911	0.913	0.902
HH × Grocery FE		Yes	Yes	Yes	Yes
HH × Time FE		Yes	Yes	Yes	Yes
Grocery × Time FE		Yes	Yes	Yes	Yes
Household FE	Yes				
Time FE	Yes				
Clustered SE	HH	HH	HH	HH	HH

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Column 1: grocery spending share (difference-in-differences with household and time FE). Columns 2–5: spending share by payment method at grocery versus discount/warehouse stores (triple-difference with household × store-type, household × time, and store-type × time FE). Standard errors clustered at the household level.

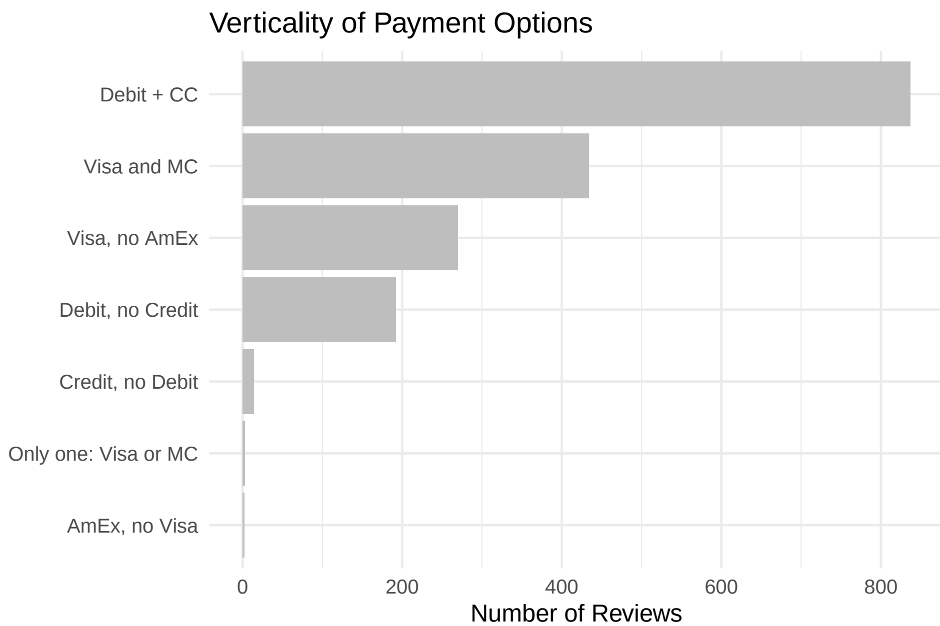
OD.3 Hierarchical Card Acceptance

In the main text, I show that the number of merchants that accepts AmEx is similar to the number of merchants that accepts Visa. However, aggregate acceptance counts alone cannot prove that merchants multi-home. This section uses Yelp reviews to show that when AmEx and Visa charged different fees, acceptance patterns were hierarchical. Merchants add payment methods sequentially—cash, then debit, then Visa/MC, then AmEx—rather than specializing in one network at the expense of another. This provides evidence of the multi-homing fact.

I collect roughly 3,000 Yelp reviews that mention at least two payment methods by name (see Online Appendix **??** for details on the sample construction). The reviews come primarily from the period 2010–2018, before AmEx's acceptance gap with Visa had fully closed, so they capture merchant acceptance decisions during a period of meaningful variation across networks.

Figure **??** shows a clear hierarchy in which payment methods co-occur in reviews. The most common combinations are debit and credit together, Visa with MC, Visa with-

Figure OD.13: Hierarchical card acceptance patterns in Yelp reviews



Notes: Each bar shows the frequency of a particular combination of payment methods mentioned in Yelp reviews. The sample contains roughly 3,000 reviews that mention at least two payment methods by name, primarily from 2010–2018. See Online Appendix **??** for sample construction details.

out AmEx, and debit without credit. Almost no reviews mention credit without debit, AmEx without Visa, or only one of Visa or MC. This pattern indicates that merchants add payment methods sequentially—cash, then debit, then Visa/MC, then AmEx—rather than specializing in one network at the expense of another.

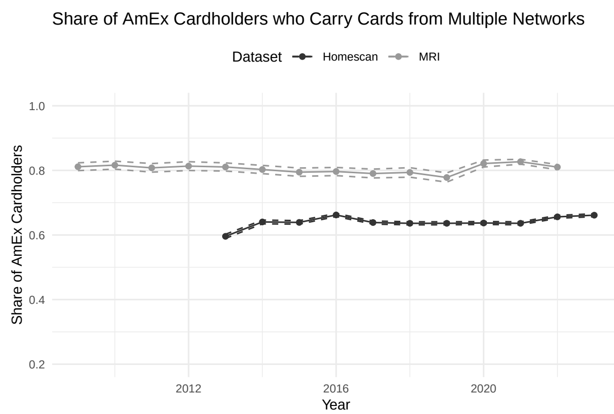
The pattern contrasts sharply with platform markets such as food delivery, where merchants often operate on one platform but not another (**Sullivan2025**). In payment networks, multi-homing is nearly universal among merchants who accept cards at all.

OD.4 Consumer Multi-homing Behavior

Consumer multi-homing across credit card networks reflects differences in rewards and card features, not insurance against acceptance gaps. This section tests whether AmEx cardholders reduced multi-homing as AmEx's acceptance gap with Visa closed. AmEx provides the sharpest test because it historically had the largest gap among major networks, so acceptance-driven multi-homing should be most visible there. If consumers held multiple credit cards mainly as insurance against acceptance gaps, the share of AmEx cardholders who also hold cards from other networks should have fallen as AmEx became accepted nearly everywhere.

Figure **??** plots the share of AmEx cardholders who also hold a credit card from another network, using data from MRI-Simmons (2009–2022) and NielsenIQ Homescan. Despite the substantial closing of the AmEx–Visa acceptance gap documented in Figure **??**, the AmEx multi-homing rate has been flat throughout the sample period. Consumers multi-home because cards differ in rewards and features, not to hedge acceptance risk. This finding supports the model's treatment of consumer card choice as driven by card attributes rather than acceptance coverage.

Figure OD.14: AmEx Consumer Multi-homing Over Time



Notes: Data from MRI-Simmons (2009–2022) and NielsenIQ Homescan. The figure plots the share of AmEx cardholders who also hold a credit card from another network over time.

OD.5 Consumer Substitution Patterns from a Second Choice Survey

A second-choice survey confirms that credit and debit are distinct product categories and that higher-income consumers are more reward-sensitive.

OD.5.1 Distinct Product Categories

Debit cards are closer substitutes to cash than credit cards are, but credit and debit remain distinct product categories. Table **??** shows the results. When asked what they would use if credit cards did not exist, most credit users would switch to debit. Only 15% would switch to cash. Among primary debit users, 53% would switch to cash—consistent with cash and debit being closer substitutes. When asked how they would pay if their bank stopped offering their primary card type, 87% of credit users and 76% of debit users would switch to another bank's card of the same type. These high same-type diversion rates support modeling credit and debit as distinct categories.

Table OD.4: Substitution patterns from a second choice survey

Current Card	Choice Set	Share
Credit	{Cash, DC}	P(Cash) = 0.15
Debit	{Cash, CC}	P(Cash) = 0.53
Credit	{Cash, DC, Other Banks' CC}	P(Other Banks' CC) = 0.87
Debit	{Cash, Other Banks' DC, CC}	P(Other Banks' DC) = 0.76

Notes: How consumers would pay under counterfactual choice sets. The first column is the consumer's current primary payment method, the second is a hypothetical choice set, and the third is the probability of adopting a given method. The sample contains 357 primary credit and 383 primary debit users. See Online Appendix **??** for survey design details.

Table OD.5: Income heterogeneity in rewards sensitivity

	P(Switch)	P(Rewards Important)
Log Income	0.14***	0.02***
	(0.04)	(0.00)
DV Mean:	0.71	0.11
N	347	375040
Dataset:	2nd Choice	MRI
Year FE:		X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Model 1 uses second-choice survey data. The dependent variable indicates willingness to switch if current credit card rewards were halved. Model 2 uses MRI-Simmons data. The dependent variable indicates the consumer cites rewards as important for financial institution choice. Both regressions use log household income as the independent variable.

OD.5.2 Heterogeneity in Reward Sensitivity by Income

High-income individuals are more reward-sensitive: a 1% income increase raises the propensity to switch if rewards are halved by 0.19% (Table **??**). MRI data from 2009–2022 show the same pattern: higher-income individuals are more likely to cite rewards as important for financial services choice.

OD.5.3 Technical Details of Constructing Network Diversion

The survey asks how consumers would switch if their bank stopped offering their preferred payment type. The model requires diversion between networks (e.g., Visa to MC), but the survey measures diversion between banks (e.g., Chase to Bank of America). I designed the question this way because consumers often do not know which cards are Visa or MC. The survey therefore overstates true network switching: a Visa consumer switching from Chase to Bank of America might stay on Visa if both banks offer it.

To correct this, I adjust responses by removing the expected "within-network" bank switches:

1. From the Nilson Report, I collect the share of Visa and MC cards at each major bank.
2. For each respondent, I define a choice set of alternative banks: (i) banks where they already hold cards, plus (ii) banks they researched when last shopping for a card.
3. For each alternative bank, I calculate the probability a switch keeps the consumer on the same network, using network composition at the current and alternative banks.
4. I reduce the reported diversion ratio by subtracting the expected fraction of bank switches that keep consumers on the same network.

Formally, let $r_i \in \{\mathrm{CC}, \mathrm{DC}\}$ denote the primary card of survey respondent i. Let $N_c = \sum_i I\{r_i = \mathrm{CC}\}$ be the number of primary credit card consumers, $N_d = \sum_i I\{r_i = \mathrm{DC}\}$ be the number of debit card consumers, and $N = N_d + N_c$. Primary cash consumers have already been dropped from the sample.

Let $S_{i,1}$ be an indicator for switching to cash if credit cards did not exist, $S_{i,2}$ an indicator for switching to cash if debit cards did not exist, and $S_{i,3}$ an indicator for switching to the same card type if their bank stopped offering their primary payment type.

For survey respondent i, let the imputed choice set of alternative banks be $\mathcal{C}_i$, the current bank be b_i, and the customer's card type be t_i (credit or debit). For each bank j and network $k \in \{\mathrm{Visa}, \mathrm{MC}\}$, let $n_j^{k,t}$ denote the share of type-t cards (credit/debit) on network k offered by bank j.

The adjustment term ι is the expected number of respondents who report switching to the same card type but also switch networks:

$$\iota = \sum_{i=1}^{N} S_{i,3} \times \left(1 - \left(\sum_{k \in \{\mathrm{Visa,MC}\}} n_{b_i}^{k,t_i} \times \left(\frac{1}{|\mathcal{C}_i|} \sum_{j \in \mathcal{C}_i} n_j^{k,t_i}\right)\right)\right)$$

The data moments for the second-choice survey are then:

$$\hat{g}_1 = \begin{pmatrix} N_c^{-1} \sum_{i, r_i = \mathrm{CC}} S_{i,1} \\ N_d^{-1} \sum_{i, r_i = \mathrm{DC}} S_{i,2} \\ \frac{\sum_i S_{i,3} - \iota}{N - \iota} \end{pmatrix} \tag{41}$$

Table OD.6: Share switching to the same card type on a different network, under alternative assumptions about whether a bank switch (e.g., Chase to Bank of America) also changes the network (Visa to MC)

Assumption on Diversion	Credit Card	Debit Card
Consideration	0.81	0.63
All Divert	0.87	0.76
Half	0.76	0.62

In practice, diversion to credit cards does not depend on how within-network moves are modeled. Table **??** shows diversion to the same type under varying assumptions about the share of between-bank moves that also involve network switching.

OD.6 Survey Evidence on Credit Aversion

SCPC and marketing surveys show that many consumers actively prefer debit over credit for substantive reasons, not because they are unaware of credit card rewards. Debit-preferring consumers cite budget control and debt aversion as their primary motivations. The model captures these motivations as higher non-pecuniary utility for debit.

Table OD.7: Most important characteristic of preferred payment instrument, by payment preference

	Cash	Debit	Credit
Budget control	0.14	0.06	0.01
Convenience	0.46	0.61	0.34
Rewards	0.00	0.00	0.39

Notes: Three consumer groups defined by preferred non-bill payment instrument: cash, debit, or credit. Each cell reports the share of consumers in that group who cite the row feature as the "most important characteristic" of their preferred payment instrument. Source: SCPC 2017–2019. All statistics use individual-level sampling weights.

Table **??** shows 14% and 6% of cash and debit users in the SCPC cite budget control as most important, versus 1% of credit users. Debit consumers are about 27 pp. more likely to cite convenience (Table **??**), perhaps because debit cards come bundled with checking accounts and require no monthly bill. By contrast, 39% of credit card consumers say rewards are the most important reason they pay with credit. Among card owners who hold both products, 79% nonetheless make debit their primary instrument, showing that budget control and debt aversion keep many consumers on debit even when credit would pay them more.

These results are consistent with findings from marketing surveys. About a quarter of consumers feel "impulsive," "anxious," or "overwhelmed" with credit cards, twice the rate for debit (**Issa2017**). Among consumers without credit cards, 37% cite debt aversion ("prefer not to carry any debt"), versus only 26% who do not qualify (**Boehm2018**).

Behavioral research finds some consumers prefer immediate payment to avoid carrying balances (**Prelec.Loewenstein1998**). **Zinman2004<empty citation>** finds that around 30% of debit card users show spending-control motives. Australia banned unsolicited credit limit increases in 2018 after finding they caused "significant consumer detriment" (**RetailBankerInternational2018**).

OE Additional Model Details

OE.1 Consumer Payment Probabilities

This section derives the payment probabilities $\pi^w_{M,j}$ from Section **??** for a model with two credit cards and one debit card, with results collected in Table **??**.

Single-homers are straightforward: a consumer uses her card if the merchant accepts it and pays cash otherwise. For multi-homers carrying two credit cards, substitution is also simple. Both cards serve the same function, so if only one is accepted the consumer switches to it with probability 1 (middle panel of Table **??**).

For a multi-homer carrying one credit and one debit card, credit and debit are segmented: each card is used only for its intended transaction type. When both are accepted, the consumer uses credit with probability π and debit with probability $1-\pi$. When only one card is accepted, the consumer uses it for the share of transactions assigned to that type and pays cash for the rest. A consumer whose credit card is accepted but whose debit card is not still uses credit with probability π—she does not redirect debit transactions to the credit card.

Table OE.1: Consumer Payment Probabilities π^w_M (Baseline)

Primary Card	**Secondary Card**	**Merchant Accepts**	π^w_{M,w_1}	π^w_{M,w_2}
Any Card	None	Primary card accepted	1	0
Any Card	None	Primary card not accepted	0	0
Visa	MC	Both Visa and MC	π	$1-\pi$
Visa	MC	Only Visa	1	0
Visa	MC	Only MC	0	1
Visa	MC	Neither	0	0
Credit	Debit	Both Credit and Debit	π	$1-\pi$
Credit	Debit	Only Credit	π	0
Credit	Debit	Only Debit	0	$1-\pi$
Credit	Debit	Neither	0	0

OE.2 Acceptance Thresholds

Every acceptance decision hinges on the same two forces: consumer multi-homing patterns and the fee spread across networks. Consider two cards with fees $\tau_1 \leq \tau_2$. Every payment method generates the same incremental sales per dollar. A merchant already accepting card 1 decides whether to also accept card 2. The incremental quasi-profit (Theorem **??**) depends on two consumer groups:

- **New transactions** (N): single-homers on card 2 who would otherwise pay cash, generating incremental fees and sales.
- **Diverted transactions** (D): multi-homers who switch from card 1 to card 2, changing the fee without generating new card transactions.

Both N and D are weighted sums of wallet shares $\mu^{(\cdot)}$, with weights depending on π and card holdings. The incremental quasi-profit coefficients are:

$$\Delta a = N\tau_2 + D(\tau_2 - \tau_1) \tag{42}$$

$$\Delta b = \tfrac{1}{\sigma}\big[N(1-\sigma\tau_2) - D\sigma(\tau_2 - \tau_1)\big] \tag{43}$$

The merchant accepts card 2 if and only if $\gamma \geq \underline{\gamma}_2$, where

$$\underline{\gamma}_2 = \frac{\sigma\big[N\tau_2 + D(\tau_2 - \tau_1)\big]}{N(1-\sigma\tau_2) - D\sigma(\tau_2 - \tau_1)} \tag{44}$$

This threshold rule applies whenever $\Delta b > 0$, i.e., the incremental sales gain from accepting card 2 is positive; in the empirical setting $\Delta b > 0$ holds for all observed networks at baseline parameters. When $\Delta b \leq 0$, accepting the card destroys value for every type and the merchant declines unconditionally.

OE.2.1 Two Credit Cards

Multi-homing across credit cards affects acceptance, and relative fees matter. Since both are credit cards, multi-homers switch to whichever is accepted (Table **??**, middle panel), so all divert: $D = (1-\pi)\mu^{(1,2)} + \pi\mu^{(2,1)}$. The new-transaction mass is $N = \mu^{(2,0)}$ (single-homers on card 2). Substituting into (**??**):

$$\underline{\gamma}_2 = \frac{\sigma\mu^{(2,0)}\tau_2 + \sigma\left[(1-\pi)\mu^{(1,2)} + \pi\mu^{(2,1)}\right](\tau_2 - \tau_1)}{\mu^{(2,0)}(1-\sigma\tau_2) - \sigma\left[(1-\pi)\mu^{(1,2)} + \pi\mu^{(2,1)}\right](\tau_2 - \tau_1)} \tag{45}$$

By inspection, cutting τ_1 raises $\underline{\gamma}_2$ by widening $\tau_2 - \tau_1$, which increases diversion drag without changing incremental sales. Three special cases follow by inspection: $D = 0$ (no multi-homers) gives $\underline{\gamma}_2 = \sigma\tau_2/(1-\sigma\tau_2)$; $N \to 0$ (no single-homers) sends $\underline{\gamma}_2 \to \infty$; and $\tau_2 = \tau_1$ collapses the threshold to $\underline{\gamma}_1$. This is consistent with Figures **??** and **??**: AmEx acceptance responded sharply to its fee relative to Visa.

OE.2.2 Credit and Debit (Baseline)

Multi-homing across credit and debit does not affect credit acceptance, and relative fees do not matter. Credit and debit are segmented at the point of sale (Table **??**, bottom panel): adding credit does not divert any spending from debit, so $D = 0$. The threshold reduces to

$$\underline{\gamma}_c = \frac{\sigma\tau_c}{1 - \sigma\tau_c}$$

independent of the debit fee. Segmentation ensures Δa and Δb share the same wallet-share weights, so the threshold depends only on credit's own fee. This is consistent with the Durbin evidence: after debit interchange fees fell by roughly half, credit interchange fees did not change (Figure **??**) and yet credit card acceptance was unchanged (Figure **??**).

OE.3 CES Price Index and Multi-Homing Pecuniary Utility

A more natural pecuniary utility for multi-homers would use the CES price index directly:

$$\log U^w = \pi f^{w_1} + (1 - \pi) f^{w_2} - \log P^{(w_1, w_2)}.$$

I do not use the CES-index definition because it counterfactually predicts that Visa-AmEx multi-homing rates fall as AmEx acceptance rises (Appendix **??**).

When w_1 and w_2 are different types (credit and debit), the card-payment probability decomposes linearly: $v_M^w = \pi \cdot \mathbf{1}[w_1 \in M] + (1 - \pi) \cdot \mathbf{1}[w_2 \in M]$, and $(P^w)^{1-\sigma} = \pi (P^{(w_1,0)})^{1-\sigma} + (1 - \pi)(P^{(w_2,0)})^{1-\sigma}$. The only remaining gap is a Jensen correction in the log price index, second-order in the difference between credit and debit price indices.

For same-type multi-homers (e.g., two credit cards), there is an additional first-order gap from union acceptance: the consumer pays by card whenever *either* card is accepted, so $\pi \log P^{(w_1,0)} + (1 - \pi) \log P^{(w_2,0)} \geq \log P^{(w_1,w_2)}$, with equality when w_1 and w_2 have identical acceptance sets. Since credit card acceptance is symmetric across networks in the estimated model and all counterfactuals, both cases yield identical welfare numbers.

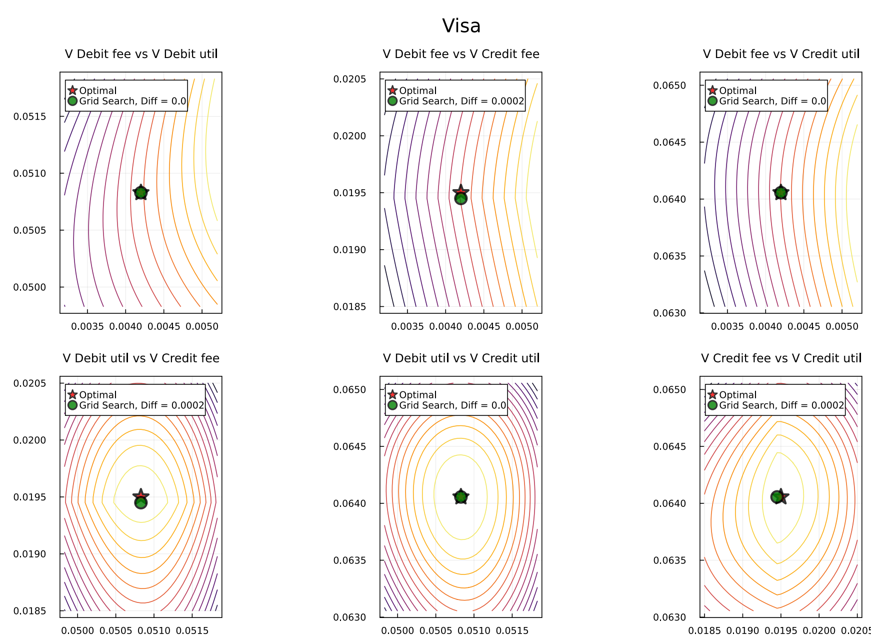
OE.4 Details on the Conduct Assumption

Why networks compete in gains A_j rather than utilities or rewards. **Weyl2010<empty citation>** shows that guaranteeing utility gains captures penetration pricing when merchant acceptance is low, giving consumers a dominant adoption strategy and pinning down a unique subgame equilibrium.

Non-differentiability with multiple networks. Network profits are non-differentiable in merchant fees when more than two networks compete. A network raising its fee competes with acceptance of all other networks, while a network cutting its fee competes with cash. These regions have different marginal revenues, creating non-differentiability at the symmetric equilibrium. **Rochet.Tirole2003<empty citation>**'s two-network model avoids this issue. **Teh.etal2022<empty citation>** show the problem arises with three or more networks.

Trembles for equilibrium selection. To select a unique equilibrium, I introduce small trembles ν_x in networks' fee choices, smoothing the profit function and ensuring equal profit loss from marginal fee increases or decreases (following **Bilal.Lhuillier2021<empty citation>**). Convolving with smooth trembles makes Ψ differentiable, and as the perturbation vanishes, the smoothed objective converges uniformly to the original. To assess approximation error, Figure **??** plots pairwise heatmaps of Visa's profit (without trembles) across all control-variable pairs, comparing the perturbed-FOC solution with a grid search over the original profit function. The match is almost exact.

Figure OE.1: Comparing the maximum from solving the FOCs of the perturbed profit function with the maximum of the original profit function



OE.5 Model Solution Algorithm

The algorithm has two layers. An inner fixed point solves for the allocation (consumer wallet shares and merchant acceptance) given network fees A_j and τ_j. An outer loop then jointly solves networks' first-order conditions over fees, taking the inner equilibrium as given.

OE.5.1 Fixed Point Iteration

The inner fixed point iterates on wallet-specific price indices P^w. Given a conjecture for these price indices, the algorithm recovers the implied uniform reward rates by inversion and proceeds through the following steps:

1. Compute consumer wallet shares $\tilde{\mu}_y^w$ for each income level.
2. Solve merchant acceptance: each merchant type γ maximizes quasi-profit to determine the optimal card bundle $M^*(\gamma)$.
3. Update price indices P^w by integrating the price-index integrand over the merchant type distribution.
4. Check convergence: if $\max_{w,s} |P^{w,\text{new}} - P^{w,\text{old}}| < \epsilon$, stop. Otherwise, apply damping and return to step 1.

Profits for each network follow from Equations **??** and **??**. Demand integration uses 2,000 Monte Carlo draws jointly over unobserved preference heterogeneity and log-normal income.

The outer loop solves networks' first-order conditions jointly, respecting the ownership matrix that accounts for Visa's co-ownership of credit and debit products. Capped fees (e.g., under Durbin or counterfactual caps) bypass the first-order condition and are instead confirmed binding via Lagrange multiplier checks. For the normal expectations in the trembles, N-dimensional Gauss-Hermite quadrature with 2 points per dimension ensures equal profit loss from raising or lowering fees. Second-order conditions for local maxima are verified numerically at every solution.

OE.6 Numerical Validation of the Quasi-Profit Approximation

The quasi-profit functions approximate true profits closely. Figure **??** compares exact and approximate profits for all 32 potential payment bundles in the baseline equilibrium. The fit holds across all values of γ. Figure **??** plots the upper envelopes of true and approximate profit functions. Slope changes in the envelope mark where the merchant's optimal acceptance strategy shifts. The close fit confirms that the approximation accurately captures how optimal acceptance varies with merchant type.

OE.7 Mechanical Welfare Adjustment for Dual Routing

The dual routing counterfactual in Section **??** differs from the other counterfactuals because it directly perturbs preferences rather than fees, ownership, or the set of products. I raise χ^{22}, the credit–credit complementarity parameter in the consumer's portfolio

Figure OE.2: Comparison of exact profits (which factor in price changes) and quasi-profits (which do not) in the baseline equilibrium for every possible subset of accepted cards.

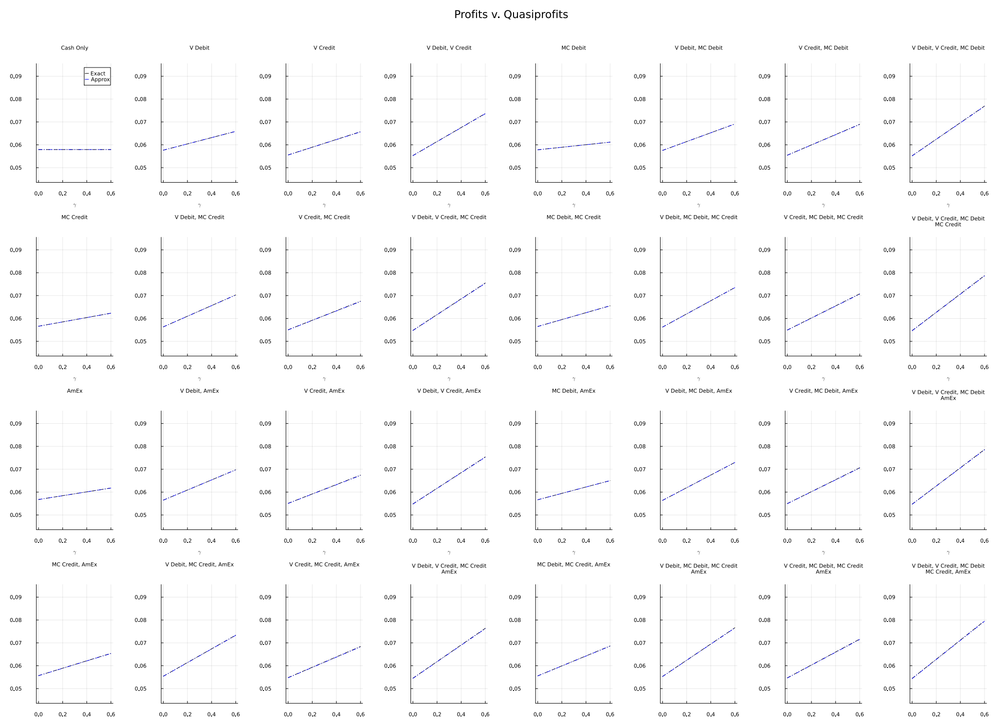
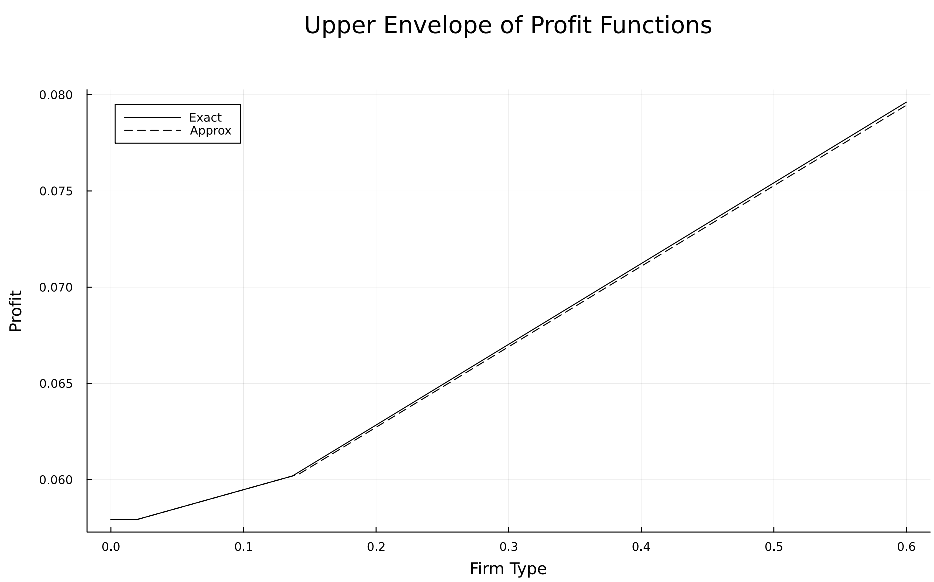


Figure OE.3: Comparison of the upper envelope of profits computed with the exact profit formula versus the quasi-profit approximation.



choice, by $\Delta\chi = 0.0020 \cdot \alpha_0$, equivalent to a 20 bp reward increase on credit–credit bundles for the median consumer. A direct preference perturbation mechanically raises the consumer surplus in Equation **??** through the logit log-sum, even if prices, portfolios, and merchant acceptance were held fixed. To keep the dual routing results comparable with the other counterfactuals in the welfare tables, I subtract this mechanical component so that the reported consumer surplus reflects only the equilibrium response.

Let $\mathcal{W}_{CC} = \{w = (w_1, w_2) \in \mathcal{W} : w_1, w_2 \text{ are both credit cards}\}$ denote the set of credit–credit wallets. For a consumer of type i at baseline income y with reward-sensitivity α_i, a small utility boost $\Delta\chi$ applied to the credit–credit wallets $\mathcal{W}_{CC}$ in the portfolio problem raises consumer surplus, to first order, by

$$\frac{\Delta\chi}{\alpha_i} \cdot y \cdot P(\mathcal{W}_{CC} \mid i),$$

where $P(\mathcal{W}_{CC} \mid i) = \sum_{w \in \mathcal{W}_{CC}} \tilde{\mu}_i^w$ is i's probability of holding a wallet in $\mathcal{W}_{CC}$, i.e., two credit cards. Dividing by α_i converts utils to equivalent consumption, and scaling by y expresses the gain in dollars. The aggregate mechanical gain integrates over the baseline income distribution F, mirroring Equation **??**:

$$W^{\text{mech}}(\tilde{\mu}) = \int \mathbb{E}\left[\frac{\Delta\chi}{\alpha_i} \cdot P(\mathcal{W}_{CC} \mid i; \tilde{\mu})\right] \cdot y \, dF(y). \tag{46}$$

Equation **??** evaluates $P(\mathcal{W}_{CC} \mid i)$ at a single set of portfolio shares, but the preference boost itself raises credit–credit holding. Using only the baseline shares understates the mechanical component; using only the counterfactual shares overstates it. I therefore average the two using the trapezoidal rule:

$$W^{\text{mech}}_{\text{trap}} = \tfrac{1}{2}\big(W^{\text{mech}}(\tilde{\mu}^{\text{base}}) + W^{\text{mech}}(\tilde{\mu}^{\text{cf}})\big), \tag{47}$$

where $\tilde{\mu}^{\text{base}}$ and $\tilde{\mu}^{\text{cf}}$ are the credit–credit portfolio shares at the baseline and dual routing equilibria. The counterfactual shares $\tilde{\mu}^{\text{cf}}$ incorporate the full equilibrium response (new fees, rewards, and acceptance), not just the preference change. Both integrals are computed by Monte Carlo over 500 joint draws of income and preference shocks per bootstrap sample. The welfare tables subtract $W^{\text{mech}}_{\text{trap}}$ from the dual routing consumer surplus before reporting, so the \$3.3 billion consumer-surplus figure in Section **??** is net of this adjustment. The mechanical component is roughly \$4.5 billion in the baseline specification.

OF Credit-Debit Segmentation

The baseline model assumes consumers do not substitute between credit and debit at the point of sale. While incorporating substitution would make the model more realistic along some dimensions, it also generates predictions at odds with the empirical evidence, so this appendix presents the case for segmentation and two alternatives that relax it.

Section **??** weighs the direct evidence. Antitrust testimony and consumer surveys describe credit and debit as distinct products at the point of sale, while the Discover rewards experiment shows some substitution at the margin.

Section **??** generalizes the model to allow substitution and derives two equilibrium predictions: credit acceptance depending on credit-debit multi-homing rates, and debit fee caps disciplining credit fees. *Ohio v. AmEx* testimony contradicts the first, and Durbin's null on credit interchange contradicts the second.

Section **??** estimates a specification that allows point-of-sale substitution and incremental debit sales relative to cash. This matches the consumer-side substitution evidence most closely at the cost of contradicting the merchant-side patterns from *Ohio v. AmEx* and Durbin. Section **??** estimates a specification that treats debit as generating no incremental sales relative to cash. This gives up some fidelity to the consumer-side substitution evidence but matches the merchant-side acceptance pattern in *Ohio v. AmEx*, and is silent on Durbin pricing because debit fees are zero by construction. The welfare effect of repealing the Durbin Amendment flips sign across the specifications that can evaluate it. Total welfare effects are similar across all three for the credit fee cap ($23–28B) and the merger ($8–15B).

Section **??** closes with a microfoundation in which payment methods substitute at adoption but segment at usage.

OF.1 Empirical Evidence on Credit-Debit Substitution at the Point of Sale

Antitrust testimony and consumer surveys describe credit and debit as distinct products on many transactions. But the Discover rewards experiment finds substitution between credit and debit in response to rewards at the point of sale. Segmentation captures the dominant pattern but understates the substitution in response to rewards.

Testimony and surveys consistent with segmentation. Antitrust testimony in *Ohio v. American Express* identifies the credit facility as the primary reason credit and debit are not close substitutes at the transaction level (**Conrath2014**). Credit cards let consumers defer payment and carry balances. Debit payments are drawn immediately from a checking account. Large-ticket merchants rely on this distinction. Sears, Southwest Airlines, and Home Depot each testified that their customers expect or require access to a line of credit. Hotels and car rental companies use the credit card as a security deposit. Hilton described debit at check-in as "atrocious" because holds can take six weeks to clear. Alaska Airlines, Crate & Barrel, and Best Buy called credit and debit "totally different products that cannot substitute for each other" and never considered debit-only acceptance.

Consumer-side evidence describes similar compartmentalization. AmEx's own analysis finds customers using debit for everyday purchases below a personal threshold of \$20–\$200, with Official Payments putting the switching point closer to \$300. An internal AmEx note describes credit and debit as "not substitutes in the consumer's (or Durbin's) mind."

Reward experiments consistent with some substitution. The Discover quarterly rewards program (Online Appendix **??**) shifts grocery spending onto Discover from both cash and debit when Discover pays higher rewards, not from cash alone. Under the segmentation assumption, only the use of cash and other credit cards should decline. The fact that debit card volumes also decline suggests that credit and debit are at least partial substitutes for consumers.

Summarizing the evidence. The direct evidence does not cleanly resolve the question. Most antitrust testimony and survey responses describe segmentation, but the Discover experiment shows some limited substitution in routine grocery purchases. Below, I explore how relaxing the no substitution assumption changes the model.

OF.2 Generalizing Credit-Debit Substitution

The alternative specifications below differ in how much consumers substitute between credit and debit at the point of sale. The model therefore treats segmentation as a continuous parameter. The parameter $\zeta \in [0, 1]$ governs how often a consumer holding both card types reaches for the wrong type when a merchant accepts only one, and $\kappa_d \in [0, 1]$ governs the demand boost merchants get from debit relative to credit.

Generalized payment probabilities. When a multi-homer carrying one credit and one debit card visits a merchant that accepts only one type, she redirects a fraction ζ of the transactions intended for the missing type. When only the primary card is accepted, the consumer uses it with probability $\pi + \zeta(1-\pi)$: π from intended usage, plus a fraction ζ of what would have gone to the secondary type. Table **??** reports the general probabilities. Setting $\zeta = 0$ recovers Table **??**.

Table OF.1: Consumer Payment Probabilities π_M^w (General)

Primary Card	**Secondary Card**	**Merchant Accepts**	π_{M,w_1}^w	π_{M,w_2}^w
Credit	Debit	Both Credit and Debit	π	$1-\pi$
Credit	Debit	Only Credit	$\pi + \zeta(1-\pi)$	0
Credit	Debit	Only Debit	0	$(1-\pi) + \zeta\pi$
Credit	Debit	Neither	0	0

Generalized payment surplus. When debit generates fewer incremental sales than credit, the model needs parameters κ_j for each payment method that scale the sales increase relative to credit cards. The baseline sets $\kappa_j = 1$ for all payment methods. With heterogeneous sales benefits $\kappa_j \leq 1$, payment surplus generalizes to

$$v_M^w = \sum_{j=1}^{J} \pi_{M,j}^w \kappa_j \tag{48}$$

where $\kappa_j \in [0,1]$ governs the demand boost from accepting card j.

Merchant problem with heterogeneous κ_j. Taking the generalized payment surplus v_M^w from Equation (**??**), the merchant's per-unit margin becomes

$$L_M^w(p) = \underbrace{\frac{1 - \pi_{M,w_1}^w - \pi_{M,w_2}^w}{1+\gamma v_M^w}}_{\text{share on cash}} \times (p-1) + \sum_{j=1}^{2} \underbrace{\frac{\pi_{M,w_j}^w \left(1 + \gamma \kappa_{w_j}\right)}{1+\gamma v_M^w}}_{\text{share on card } w_j} \times \left(p\left(1 - \tau_{w_j}\right) - 1\right) \tag{49}$$

Collecting terms yields

$$L_M^w(p) = p\left(1 - \frac{\tau_M^w + \gamma \tilde{\tau}_M^w}{1+\gamma v_M^w}\right) - 1 \tag{50}$$

where the wallet-average fees are $\tau_M^w = \sum_j \pi_{M,w_j}^w \tau_{w_j}$ and $\tilde{\tau}_M^w = \sum_j \pi_{M,w_j}^w \tau_{w_j} \kappa_{w_j}$. The κ-weighted fee $\tilde{\tau}_M^w$ equals τ_M^w in the baseline.

Profits become

$$\Pi(\gamma, p, M) = C \times \sum_{w \in \mathcal{W}} \mu^w \left(1 + \gamma v_M^w\right) p^{-\sigma} \times \left[p \left(1 - \frac{\tau_M^w + \gamma \tilde{\tau}_M^w}{1 + \gamma v_M^w} \right) - 1 \right] \tag{51}$$

and the optimal price is $p^* = \frac{\sigma}{\sigma - 1} \times \frac{1}{1 - \hat{\tau}}$, with effective fee $\hat{\tau} = \frac{\sum_w \mu^w (\tau_M^w + \gamma \tilde{\tau}_M^w)}{\sum_w \mu^w (1 + \gamma v_M^w)}$. Dollar volume on card j generalizes to

$$d_j = \sum_{w \in \mathcal{W}} \tilde{\mu}^w \times \int \frac{(1 + \gamma \kappa_j)\, \pi^w_{M^*(\gamma), j}}{1 + \gamma v_M^w} q^{w*}(\gamma)\, p^*(\gamma)\, \mathrm{d}G(\gamma) \tag{52}$$

where the numerator $(1 + \gamma \kappa_j)$ reflects the demand boost specific to card type j. In the baseline ($\kappa_j = 1$), the numerator reduces to $(1 + \gamma)$, matching the main-text expression.

The quasi-profit coefficients (Appendix **??**) generalize to

$$a_M = \sum_{w \in \mathcal{W}} \mu^w \tau_M^w, \qquad b_M = \frac{1}{\sigma} \sum_{w \in \mathcal{W}} \mu^w \left(v_M^w - \sigma \tilde{\tau}_M^w\right) \tag{53}$$

In the baseline, $\tilde{\tau}_M^w = \tau_M^w$, recovering the coefficients in Equation **??**.

Two-card credit-debit acceptance threshold. With point-of-sale substitution, merchants' credit acceptance decisions depend on the gap between credit and debit fees and on credit-debit multi-homing rates. A two-card example makes this dependence visible. Keep the convention from Online Appendix **??** that card 1 is debit (fee τ_d, sales parameter $\kappa_d \leq 1$) and card 2 is credit (fee τ_c, sales parameter $\kappa_c = 1$). Let $M \equiv (1 - \pi)\mu^{(1,2)} + \pi \mu^{(2,1)}$ denote the multi-homer mass. The N/D decomposition from Online Appendix **??** generalizes to

$$N = \mu^{(2,0)} + (1 - \zeta) M$$
$$D = \zeta M$$

N captures new card transactions: single-homers on credit plus the share of multi-homer spending that stays cash under partial substitution. D captures actual diversion: multi-homer spending redirected from debit to credit. When $\zeta = 0$ (baseline), $D = 0$. When $\zeta = 1$, $D = M$, the same expression as for two credit cards (Online Appendix **??**).

The $\Delta a / \Delta b$ formulas from (**??**)–(**??**) extend to this setting with one modification. Each diverted transaction was generating debit surplus $\kappa_d(1 - \sigma \tau_d)$ and now generates credit surplus $(1-\sigma\tau_c)$, so the D component of Δb captures the net sales gain from switching:

$$\begin{aligned}\Delta a &= N\tau_c + D(\tau_c - \tau_d) \\ \Delta b &= \tfrac{1}{\sigma}\big[N(1-\sigma\tau_c) + D\big((1-\sigma\tau_c) - \kappa_d(1-\sigma\tau_d)\big)\big]\end{aligned} \tag{54}$$

The threshold follows immediately:

$$\underline{\gamma}_c = \frac{\sigma\big[N\tau_c + D\,(\tau_c - \tau_d)\big]}{N(1-\sigma\tau_c) + D\big[(1-\sigma\tau_c) - \kappa_d(1-\sigma\tau_d)\big]} \tag{55}$$

Setting $\zeta = 0$ gives $D = 0$, and the threshold reduces to $\sigma\tau_c/(1-\sigma\tau_c)$.

By inspection, the acceptance threshold depends now on the gap between credit and debit fees $\tau_2 - \tau_1$ and the share of transactions that substitute between credit and debit (D). The threshold reduces to the standard credit card acceptance threshold from the model in the main text $(\sigma\tau_c/(1-\sigma\tau_c))$ under two situations:

- $\zeta = 0$: $D = 0$, so both channels are off.
- $\kappa_d = 0$ and $\tau_d = 0$: the sales channel is off ($\kappa_d = 0$); the fee channel becomes $D\tau_c$, so $(N + D)$ factors out of numerator and denominator.

When $\zeta > 0$ and $\kappa_d > 0$ (intermediate case), both channels are active and the threshold depends on multi-homing shares. Table **??** summarizes the three specifications.

Table OF.2: Credit-Debit Acceptance Threshold Across Specifications

Specification	**Assumption**	**Dep. on credit-debit multi-homing?**	**Location**
Segmented baseline	ζ=0, κ_d=1	No	Main text
Incremental debit sales	$\zeta \in (0,1)$, $\kappa_d \in (0,1)$	Yes	Section **??**
Debit as cash	ζ=1, κ_d=0, τ_d=0	No	Section **??**

Notes: "Dep. on credit-debit multihoming" indicates whether the credit acceptance threshold depends on the rate of credit-debit multihoming. When it does, credit acceptance is linked to debit pricing, a prediction the Durbin episode does not support (Section **??**).

The generalized threshold yields two testable predictions.

Prediction 1: Credit acceptance should depend on credit-debit multi-homing. The threshold (**??**) contains $D = \zeta M$. When $\zeta > 0$, the credit acceptance threshold depends on credit-debit multi-homing rates.

Testimony in *Ohio v. American Express* is hard to reconcile with this prediction. Expert witnesses testified that merchants discipline credit fees by threatening to divert transactions to rival credit cards, not to debit (**Conrath2014**). Section **??** documents merchants treating credit and debit as distinct products.

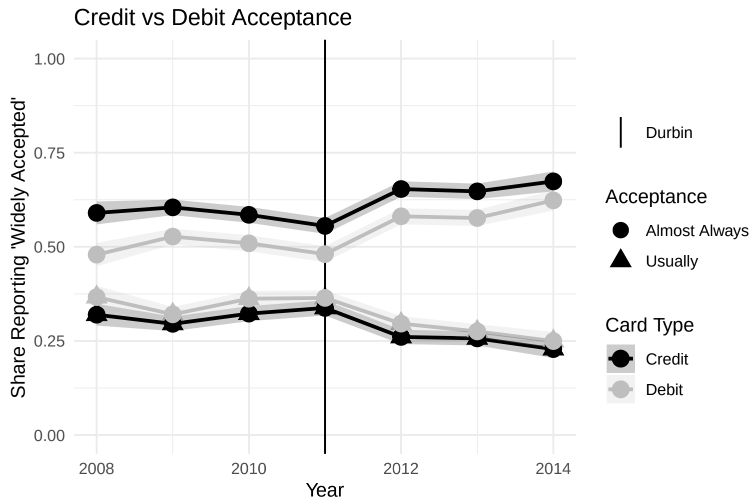
Prediction 2: Debit fee caps should discipline credit fees. The $(\tau_c - \tau_d)$ term in the threshold implies equilibrium fee co-movement: when credit and debit are substitutes, a lower debit fee allows less aggressive credit pricing before acceptance breaks down.

The Durbin Amendment cut regulated debit interchange fees by roughly half, yet credit interchange fees did not respond (Figure **??**). "Very little, if anything, happened with credit card prices after Durbin was implemented" (Katz, in **Conrath2014<empty citation>**). AmEx and Discover both told merchants that they do not price credit off debit, benchmarking instead against rival credit products. Alaska Airlines, IKEA, Best Buy, and Hilton all testified to seeing no decrease in credit pricing after Durbin. Section **??** quantifies the implied co-movement by estimating a specification with incremental debit sales and confirms the implied gap is large.

The acceptance threshold also predicts that if credit merchants fees don't respond to the debit fee cap, credit card acceptance should decline. Survey data show acceptance ratings for both credit and debit did not decline through the Durbin period (Figure **??**), so a decline in credit acceptance cannot rationalize networks holding credit fees fixed. Bundling cannot explain either pattern. The 2003 settlement had already ended bundling (**Constantine2012**). AmEx, which has no bundling with debit, also faced no competitive pressure from cheaper debit despite about a fifth of AmEx consumers also using debit (Figure **??**).

Takeaways from the model with credit-debit substitution. The goal of the model is to study how regulatory changes to interchange fees shape network competition, consumer rewards, and welfare. Every counterfactual in the paper revisits margins similar to Durbin's debit fee cap, so a credible model must reproduce Durbin's actual response. The two substitution predictions fail this test. The *Ohio v. AmEx* record shows merchants disciplining credit fees by threatening to route transactions to rival credit cards rather than debit, and Durbin's halving of debit interchange left credit interchange unchanged. The Discover evidence on reward-elastic substitution cuts the other way, and the segmented baseline misses it. Sections **??** and **??** estimate two alternatives that navigate this trade-off differently, and the credit-focused counterfactuals (the credit fee cap and the merger) deliver similar welfare across all three specifications.

Figure OF.1: Credit and Debit Card Acceptance over Time



Notes: Data from the Survey of Consumer Payment Choice (SCPC). The SCPC asks: "Please rate how likely each payment method is to be accepted for payment by stores, companies, online merchants, and other people or organizations," with separate responses for Credit card and Debit card. The figure plots the distribution of acceptance ratings for each card type over time.

OF.3 Alternative Specifications

This subsection estimates two alternative specifications that navigate the trade-off between consumer-side substitution evidence and merchant-side acceptance and pricing evidence differently.

OF.3.1 Estimating with Incremental Debit Sales

I estimate a specification that allows point-of-sale substitution between credit and debit and incremental debit sales relative to cash, with $\kappa_d = 0.5$ and $\zeta = 0.5$. This is the specification that matches the consumer-side substitution evidence from the Discover rewards shock most closely. Because debit still generates incremental sales, credit acceptance depends on the debit fee through the $(\tau_c - \tau_d)$ term in the threshold (Equation (**??**)). The estimated model implies credit fees would rise 33 bps under Durbin reversal, at odds with the flat credit fee response when Durbin actually halved debit fees.

Model Modification. Consumers holding both card types substitute between them at the point of sale half the time ($\zeta = 0.5$) rather than never (baseline) or always (debit as cash). Debit acceptance generates half the demand boost of credit ($\kappa_d = 0.5$; Eq. **??**) rather than the full boost ($\kappa_d = 1$) of the baseline or none ($\kappa_d = 0$) of debit as cash.

Two additional adjustments are required. First, AmEx is dropped, leaving four products (Visa Debit, Visa Credit, MC Debit, MC Credit). With debit-credit substitution active, each network's debit and credit arms interact in the merchant acceptance decision. A merchant considering Visa Credit must account for diversion from Visa Debit holders who can now substitute across card types. This linkage exacerbates the non-differentiability problem discussed in Online Appendix **??** and is moderated by dropping AmEx.

Second, a fixed cost of card acceptance F is added to the merchant's acceptance problem. In the baseline, acceptance solves $\text{argmax}_M -a_M + b_M\gamma$ (Equation **??**), which is linear in γ with no fixed cost. The fixed cost modifies this to $\text{argmax}_M -a_M - F\mathbf{1}[|M| > 0] + b_M\gamma$: a merchant that accepts any card pays f_1, raising the threshold at which the marginal merchant begins accepting cards. The fixed cost is identified by requiring that the first merchant that is willing to add Visa Credit is indifferent between accepting both debit networks + MC credit and the empty bundle. Without the fixed cost parameter, merchants should accept MC Credit before Visa Credit because more consumers multi-home across Visa Credit and debit than across MC Credit and debit. Accepting Visa Credit therefore diverts more transactions from cheap debit to expensive credit, making it costlier at the margin. This acceptance ordering is counterfactual. The fixed cost eliminates it by making the extensive margin (accepting any card) dominate the ordering among credit networks.

Table **??** reports estimated parameters under incremental debit sales.

Parameter Estimates. Because debit generates incremental sales and consumers substitute between credit and debit at the point of sale, cheap debit becomes a close alternative to expensive credit in the merchant's acceptance decision, so the estimator must explain why merchants keep accepting expensive credit anyway. The resolution runs through merchant margins. Higher margins weight the incremental-sales channel more heavily than the fee channel, which requires lowering the CES elasticity to $\sigma = 4.98$ from the baseline 5.61 (Table **??**). Network costs remain positive and plausible for all networks (30–60 bps).

Durbin Test of Prediction 2. Uncapping debit fees in the estimated model raises credit fees by 33 bps (Table **??**). The reduced-form Durbin evidence in Section **??** rules out a fee co-movement of this magnitude. Durbin halved debit interchange and credit interchange did not move, and merchants and networks both testified to no link between debit fees and credit pricing.

Table OF.3: Estimated parameters: Incremental debit sales

Panel A: Consumer Preference Parameters

Parameter	Est	SE
Log Price Sensitivity ($\log \alpha_0$)	7.07	0.30
Card-Credit Complement (χ^{12})	5.10	2.24
Credit-Card Complement (χ^{21})	3.40	1.05
Credit-Credit Complement (χ^{22})	-3.70	1.58
Visa Debit Utility (Ξ)	-2.80	1.23
Visa Credit Utility (Ξ)	-6.80	1.92
MC Debit Utility (Ξ)	-2.90	1.23
MC Credit Utility (Ξ)	-7.00	2.19
Primary Weight (ω)	0.61	0.14
P(Use Primary) (π)	0.83	0.00

Panel B: Consumer Heterogeneity Parameters

Parameter	Est	SE
Card Volatility	1.00	0.34
Credit Volatility	2.40	0.93
Correlation of β	-0.84	0.34
Income Volatility (ν_y)	0.75	0.01
Income Elasticity (α_y)	0.20	0.06
Card Income Effect (β_y^C)	-1.00	0.29
Credit Income Effect (β_y^L)	-0.50	0.53
Card-Credit Income (χ_y^{12})	0.70	0.34
Credit-Card Income (χ_y^{21})	0.40	0.15
Credit-Credit Income (χ_y^{22})	-0.30	0.12

Panel C: Merchant Parameters

Parameter	Est	SE
Merchant CES	4.98	1.15
Average γ	0.29	0.08
S.D. of γ	0.13	0.04
Fixed Cost F (pp)	0.40	0.05

Panel D: Network Cost Parameters (bps)

Parameter	Est	SE
Cash	30	10
Visa Debit	50	14
MC Debit	60	8
Visa Credit	40	30
MC Credit	50	10

Notes: S.D. refers to the standard deviation, and R.C. refers to the random coefficients. The Ξ are the unobserved characteristics, and the χ^{lm} is the complementarity parameter. In Panel B, volatility parameters are the standard deviations of the random coefficients. Merchant types γ are distributed according to a Gamma distribution. This specification drops AmEx, leaving four products (Visa Debit, Visa Credit, MC Debit, MC Credit). Standard errors computed by bootstrap.

Counterfactual Scenarios. For counterfactuals that primarily affect credit, the incremental-debit-sales specification still yields welfare estimates close to the baseline. The credit fee cap raises total welfare by \$28B (SE \$4.2B), nearly identical to the \$27B baseline gain (Table **??**). Credit's spending share drops 22 pp. (versus 19 at baseline), with more substitution toward debit because consumers can now switch at the point of sale (Table **??**).

Under substitution, credit and debit merchant fees move in the same direction, so this specification reverses the welfare effects of uncapping debit fees. Total welfare falls by \$10B (SE \$2.8B) versus a \$7B gain at baseline. Uncapping debit removes the competitive constraint on credit fees, so credit networks raise both fees (by 33 bps) and rewards (by 32 bps), an escalation on both sides of the credit market that reduces total welfare. The baseline instead raises debit rewards with minimal effects on credit pricing.

Credit-debit substitution amplifies dual routing: welfare gains rise to \$14B (SE \$3.9B) versus \$3.8B at baseline.[24] Point-of-sale substitution makes multi-homing more valuable: when consumers can freely switch cards, more multi-homing pushes the market closer to a world where merchants accept only the cheapest card. The resulting price effects are larger.

Table OF.4: Counterfactual effects on prices and shares: Incremental debit sales

	Price Controls		Change Competition	
	Cap Fees	Uncap Debit	Monopoly	Dual Routing
Δ Prices (bps)				
Credit Fees	-105 (0.2)	33 (9)	33 (9)	-30 (6)
Credit Rewards	-116 (5)	32 (6)	-74 (29)	-43 (8)
Debit Fees	0.0 (0.0)	25 (0.0)	0.0 (0.0)	0.0 (0.0)
Debit Rewards	-0.6 (0.6)	24 (0.4)	-15 (11)	-0.8 (0.3)
Δ Shares of Total Spending (pp.)				
Cash	6 (1.9)	-1.8 (1.1)	22 (1.7)	1.5 (0.9)
Debit	16 (3.6)	2.4 (1.5)	-0.3 (3.6)	7 (1.8)
Credit	-22 (3.5)	-0.6 (1.0)	-22 (2.4)	-8 (2.5)
Δ Fees, Rewards (\$Bn)				
Total Fees	-59 (2.0)	24 (4.9)	-34 (7)	-24 (5)
Total Rewards	-59 (1.4)	23 (3.5)	-55 (9)	-28 (6)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. This specification drops AmEx, leaving four products (Visa Debit, Visa Credit, MC Debit, MC Credit). Dollar values are computed by normalizing to \$10 trillion of total consumer purchases (**Nilson2020a**).

[24]Both figures are reported net of the mechanical χ^{22} log-sum adjustment (Online Appendix **??**), so the comparison isolates the equilibrium response. The gap between \$14B and \$3.8B reflects the stronger fee-spillover channel under incremental debit sales, not a larger mechanical variety contribution.

Table OF.5: Counterfactual welfare effects: Incremental debit sales

	Price Controls				Change Competition			
	Cap Fees		Uncap Debit		Monopoly		Dual Routing	
Δ Welfare ($Bn)								
Consumers	30	(4.5)	-11	(4.5)	-5	(13)	11	(3.6)
Merchants	0.1	(0.2)	-0.3	(0.6)	-4.0	(1.8)	-0.8	(0.2)
Networks	-1.4	(1.4)	1.3	(3.8)	24	(6)	3.4	(0.4)
Total	28	(4.2)	-10	(2.8)	15	(9)	14	(3.9)
Δ Consumer Surplus by Income (bps consumption)								
Low	49	(2.6)	-14	(5)	13	(12)	19	(4.9)
Median	39	(4.5)	-12	(4.8)	4.1	(12)	15	(4.2)
High	16	(4.9)	-10	(4.0)	-20	(14)	4.8	(2.8)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. This specification drops AmEx, leaving four products (Visa Debit, Visa Credit, MC Debit, MC Credit). Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

OF.3.2 Debit as Cash

I estimate a specification that treats debit as generating no incremental sales relative to cash, with $(\zeta, \kappa_d, \tau_d) = (1, 0, 0)$. This admits full point-of-sale substitution between credit and debit while shutting off the multi-homing channel behind Prediction 1, so credit acceptance depends only on the credit fee and multi-homing across credit cards (Table **??**). Because fees are measured relative to cash (Section **??**), treating debit as cash raises the implied cost of cash to match debit, so $\tau_d = 0$ and targeted credit fees fall by the same amount. The specification is therefore silent on Durbin pricing because debit fees are zero by construction.

The $\kappa_d = 0$ assumption is itself an imperfect description of the data. Pre-Durbin Visa Debit merchant fees were roughly 160 bps (Figure **??**), well above the cost of cash from **Felt.etal2020<empty citation>**. Merchants accepted debit voluntarily at those fees, so revealed preference says debit must generate some incremental sales relative to cash. This is why the appendix imposes universal debit acceptance below: with $\kappa_d = 0$, merchants would not voluntarily accept debit at fees above cash costs, and the imposition is required to match observed acceptance rather than derived from primitives.

Welfare results are quantitatively similar to the baseline.

Implementation. The payment surplus v_M^w (Eq. **??**) changes through $\kappa_d = 0$ and $\zeta = 1$ but not through the fee renormalization, so the merchant acceptance equation (Eq. **??**) is unchanged in form and only b_M moves. Because b_M no longer rewards debit acceptance, I impose universal debit acceptance, which is without loss since merchants are indifferent between debit and cash at equal costs.

Parameter Estimates and Counterfactuals. Treating debit as cash lowers every credit card's net cost of acceptance because fees are measured relative to cash and cash's implied cost rises to equal debit's. Rationalizing observed acceptance at a lower net cost requires lower merchant margins, which the estimator delivers by raising the CES elasticity to $\sigma = 7.16$ from the baseline 5.61 (Table **??**).

Tables **??** and **??** report counterfactual prices and welfare under the debit-as-cash specification.

Credit fee cap welfare is \$23B under debit as cash, close to the \$27B baseline result (Table **??**). The robustness follows because credit fee caps primarily change competition between credit card networks, where the credit-debit substitution assumption has little bite. Credit's spending share drops 20 pp. under the credit fee cap, close to the 19

Table OF.6: Estimated parameters: Debit as cash

Panel A: Consumer Preference Parameters

Parameter	Est	SE
Log Price Sensitivity ($\log \alpha_0$)	6.87	0.29
Card-Credit Complement (χ^{12})	3.20	0.42
Credit-Card Complement (χ^{21})	1.40	0.24
Credit-Credit Complement (χ^{22})	-1.70	0.64
Visa Debit Utility (Ξ)	-0.80	0.32
Visa Credit Utility (Ξ)	-5.10	0.36
MC Debit Utility (Ξ)	-1.00	0.39
MC Credit Utility (Ξ)	-5.30	0.43
Amex Utility (Ξ)	-5.40	0.43
Primary Weight (ω)	0.62	0.01
P(Use Primary) (π)	0.83	0.00

Panel B: Consumer Heterogeneity Parameters

Parameter	Est	SE
Card Volatility	0.90	0.34
Credit Volatility	2.30	0.85
Correlation of β	-0.78	0.03
Income Volatility (ν_y)	0.75	0.01
Income Elasticity (α_y)	0.20	0.06
Card Income Effect (β_y^C)	0.00	0.07
Credit Income Effect (β_y^L)	-0.60	0.62
Card-Credit Income (χ_y^{12})	0.30	0.15
Credit-Card Income (χ_y^{21})	0.10	0.08
Credit-Credit Income (χ_y^{22})	0.10	0.06

Panel C: Merchant Parameters

Parameter	Est	SE
Merchant CES	7.16	1.63
Average γ	0.26	0.07
S.D. of γ	0.11	0.03

Panel D: Network Cost Parameters (bps)

Parameter	Est	SE
Cash	30	10
Visa Debit	40	14
MC Debit	50	8
Visa Credit	50	13
MC Credit	60	8
Amex	60	8

Notes: S.D. refers to the standard deviation, and R.C. refers to the random coefficients. The Ξ are the unobserved characteristics, and the χ^{lm} is the complementarity parameter. In Panel B, volatility parameters are the standard deviations of the random coefficients. Merchant types γ are distributed according to a Gamma distribution. Standard errors computed by bootstrap.

pp. baseline decline, but the substitution shifts more toward cash (9 pp. versus 6) than toward debit (11 pp. versus 13).

The welfare results for other counterfactuals are more sensitive, but do not change signs. The monopoly counterfactual yields a $8B total welfare gain under debit as cash versus $15B at baseline. Network competition still does not improve welfare. The dual routing counterfactual delivers similar fee, reward, and welfare predictions. Debit-specific counterfactuals (Uncap Debit, Public Debit) are omitted because the debit as cash setup requires that debit fees are fixed at the cost of cash.

OF.3.3 Robustness Across Specifications

Welfare estimates are similar across the baseline and both alternatives for counterfactuals that primarily change competition between credit networks. The credit fee cap raises total welfare by $23–28B and the merger raises total welfare by $8–15B (Tables **??**

Table OF.7: Counterfactual effects on prices and shares: Debit as cash

	Price Controls	Change Competition	
	Cap Fees	Monopoly	Dual Routing
Δ Prices (bps)			
Credit Fees	-105 (0.0)	27 (2.7)	-10 (2.4)
Credit Rewards	-110 (0.5)	-112 (33)	-12 (2.7)
Debit Fees	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
Debit Rewards	0.9 (0.3)	-29 (8)	0.3 (0.0)
Δ Shares of Total Spending (pp.)			
Cash	9 (1.8)	29 (1.0)	0.4 (0.2)
Debit	11 (1.3)	-3.4 (1.6)	0.0 (0.2)
Credit	-20 (2.7)	-26 (1.7)	-0.4 (0.4)
Δ Fees, Rewards ($Bn)			
Total Fees	-56 (1.3)	-34 (3.0)	-5 (1.6)
Total Rewards	-56 (1.1)	-68 (8)	-6 (1.6)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

Table OF.8: Counterfactual welfare effects: Debit as cash

	Price Controls	Change Competition	
	Cap Fees	Monopoly	Dual Routing
Δ Welfare ($Bn)			
Consumers	25 (3.1)	-15 (11)	2.3 (0.9)
Merchants	0.1 (0.0)	-4.8 (1.6)	-0.1 (0.0)
Networks	-2.1 (0.4)	28 (6)	0.4 (0.0)
Total	23 (3.1)	8 (6)	2.6 (0.9)
Δ Consumer Surplus by Income (bps consumption)			
Low	45 (2.1)	10 (6)	4.0 (1.3)
Median	34 (3.2)	-4.2 (9)	2.8 (1.1)
High	11 (3.7)	-33 (13)	0.4 (0.6)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Uncap Debit and CBDC counterfactuals are omitted because these scenarios modify debit-specific regulation that interacts with the point-of-sale substitution assumption. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

and **??**). Dual routing reduces credit merchant fees and rewards under all three.

Welfare estimates diverge for counterfactuals that touch debit pricing directly. Repealing the Durbin Amendment raises total welfare by $7B under the baseline but lowers it by $10B under the specification with incremental debit sales, because uncapping debit removes the competitive constraint on credit fees. The specification that treats debit as generating no incremental sales relative to cash cannot evaluate this counterfactual because debit fees are zero by construction.

The pattern matches the underlying economics. The substitution assumption affects how debit competes with credit, so it matters most for counterfactuals that change debit pricing. The credit-focused conclusions of the paper, capping credit fees and the credit network merger, do not depend on the substitution assumption.

OF.4 A Microfoundation for Segmentation Between Credit and Debit at the Point of Sale

Payment methods can be substitutes at adoption yet poor substitutes at the point of sale. A microfoundation building on **Koulayev.etal2016<empty citation>** and **Huynh.etal2022<empty** separates usage from adoption to show that the two stages are independent. It also gives a natural origin for the complementarity parameters χ^{lm} in the main text: carrying two different card types generates option value, while carrying two cards of the same type generates diminishing returns. Rewards shift consumers between debit and credit at adoption even when the two are segmented at usage.

OF.4.1 Utility at the Point of Sale

At the point of sale, the consumer picks one card from her wallet. Let $\mathcal{J} = \{0, 1, 2, \ldots, J\}$ denote the set of payment methods, where $j = 0$ is cash, and let W denote the cards in consumer i's wallet. Let $X_j \in \mathbb{R}^k$ be the characteristics of method j (e.g., whether j is a credit card), and f_j the reward for using j. For transaction t, the utility of method j is:

$$
\begin{aligned}
u_{ijt} &= \alpha f_j + \gamma_{it}' X_j + \epsilon_{ijt} \\
\gamma_{it} &\sim \mathcal{N}(\gamma_i, \Sigma) \\
\epsilon_{ijt} &\sim \text{T1EV}
\end{aligned}
$$

The random shocks $\gamma_{it} \in \mathbb{R}^k$ capture transaction-specific preferences over card types: for a given transaction, a consumer may strongly prefer credit over debit, or cards over cash. The mean $\gamma_i \in \mathbb{R}^k$ is consumer-specific, so some consumers prefer credit cards in general. The idiosyncratic shocks ϵ_{ijt} capture why different cards of the same type (e.g., different credit cards) vary in value across transactions.

These assumptions yield the expected utility from wallet W_i:

$$
V_i(W_i) = \int \log\left(\sum_{j \in W_i} \exp\left(\alpha f_j + \gamma_{it}' X_j\right) \right) dH(\gamma_{it})
$$

where H captures the distribution of γ_{it}.

OF.4.2 Utility at Adoption

At adoption, consumers choose a wallet to maximize expected point-of-sale utility less adoption costs:

$$
\begin{aligned}
W_i^* &= \underset{W \subseteq \mathcal{J}}{\operatorname{argmax}}\, V_i(W) - \sum_{j \in W} c' X_j + \phi a_i^W \\
a_i^W &\sim \mathcal{N}(0, \nu^2)
\end{aligned}
$$

where $c \in \mathbb{R}^k$ captures adoption costs by payment-method characteristics.

OF.4.3 Mapping to the Model in the Main Text

The microfounded model clarifies two points: how segmentation at the point of sale differs from segmentation at adoption, and how payment methods can complement or substitute for each other in a wallet.

Usage-stage vs. adoption-stage segmentation. The main text assumes consumers use credit and debit for distinct transaction types, yet those carrying two credit cards freely substitute between them. This is consistent with strong transaction-level preferences over credit versus debit but low heterogeneity in adoption costs across cards. Strong usage-stage preferences arise when Σ is large: a consumer wanting a credit feature for a given transaction gets little utility from a debit card. This corresponds to certain transactions (e.g., large purchases) being better suited to credit than debit.

Strong adoption-stage substitution is compatible with $\nu^2 \to 0$. Large Σ does not induce segmentation at adoption because the consumer values the expected utility V_i, which integrates over the transaction-specific shocks γ_{it}. If adoption costs are similar across instruments, consumers remain sensitive to rewards in choosing what to adopt.

As an extreme example, suppose a consumer has 50 large-ticket and 50 small-ticket transactions, strongly preferring credit for the former and debit for the latter. At adoption, credit and debit offer similar expected utilities because each covers half of transactions. Her adoption choice is therefore sensitive to rewards even though her transaction-level preferences are sharply bifurcated.

Complements and substitutes at adoption. The microfounded model also explains why debit holders may want to adopt credit cards. At adoption, consumers know their mean preferences γ_i but have not yet drawn the transaction shocks γ_{it}; those shocks are realized only at the point of sale. A consumer with high Σ faces uncertainty over which card type she will prefer for each future transaction. Adopting both types lets her use credit when $\gamma_{it}' X_j$ favors credit and debit when it favors debit, raising V_i. This option value is positive even though the consumer knows her transaction-level preferences will be sharply bifurcated.

The model also captures incremental adoption costs. If all cards of a given type pay the same rewards f_j, then V_i is concave in the number of same-type cards, but costs are linear. This explains why consumers do not adopt every card on the market, captured in the main text by $\chi^{22} < 0$ for wallets with two credit cards from different networks.

A consumer who highly values credit cards may still want a credit card alongside a debit card to capture more transaction-specific shocks γ_{it}, which occurs when $\gamma_i' X_j$ is large for credit. The main text captures this through $\chi^{12} > 0$ and $\chi^{21} > 0$ for wallets that combine credit and debit.

OG Alternative Specifications and Additional Counterfactuals

Sections **??**–**??** re-estimate under three alternative specifications (no merchant fee pass-through, credit constraints, lower reward sensitivity); Online Appendix **??** addresses credit-debit substitution; Section **??** compares the 120 bps cap to a tighter cost-of-cash cap and the constrained social optimum.

OG.1 No Merchant Fee Pass-Through

The main counterfactuals assume merchants pass merchant-fee changes into retail prices one-for-one. Without pass-through, fee savings accrue to merchants instead of consumers. Total welfare gains keep their sign under every policy. Small counterfactuals are nearly unchanged, while credit fee cap and monopoly gains shrink by a third or more. The credit fee cap's distributional story also reverses, with merchants gaining at consumers' expense.

Model modification. This specification modifies the merchant pricing equation (Eq. **??**). Merchants set the standard CES markup $\hat{p} = \frac{\sigma}{\sigma-1}$ without the $\frac{1}{1-\hat{\tau}}$ pass-through term, so retail prices no longer respond to merchant fees. This applies symmetrically to fee increases and decreases: under the credit fee cap, all merchant-fee savings accrue to merchants rather than passing to consumers through lower retail prices. The optimal acceptance decision is unchanged because first-order price changes have second-order profit effects. All other equations (consumer demand, merchant acceptance, network conduct) are unchanged.

Table **??** shows that estimated parameters change little from the baseline. This is expected as none of the merchant acceptance decisions depend on the pricing strategy. Tables **??** and **??** report counterfactual prices and welfare under zero pass-through.

Counterfactuals with small fee and reward changes are robust: uncapping debit raises total welfare by \$8B (versus \$7B at baseline), dual routing raises \$2.9B (versus \$3.8B, both net of the mechanical χ^{22} adjustment described in Section **??**), and CBDC raises \$1.4B (versus \$1.6B).

Counterfactuals with large net fee changes are the exception. The credit fee cap welfare gain falls to \$17B from \$27B, and the monopoly gain falls to \$8B from \$15B. This gap reflects deadweight loss from under-consumption. Under full pass-through, positive retail markups make baseline consumption inefficiently low, but merchant price cuts after a credit fee cap reverse this distortion. Under zero pass-through, the credit fee cap and monopoly both reduce consumption (through lower rewards) without any offsetting price reductions, so deadweight loss rises and total welfare gains shrink.

Table OG.1: Estimated parameters: No merchant fee pass-through

Panel A: Consumer Preference Parameters

Parameter	Est	SE
Log Price Sensitivity ($\log \alpha_0$)	6.76	0.34
Card-Credit Complement (χ^{12})	4.60	0.58
Credit-Card Complement (χ^{21})	3.90	0.47
Credit-Credit Complement (χ^{22})	-4.80	0.79
Visa Debit Utility (Ξ)	-4.00	0.58
Visa Credit Utility (Ξ)	-5.90	0.52
MC Debit Utility (Ξ)	-4.10	0.61
MC Credit Utility (Ξ)	-6.20	0.55
Amex Utility (Ξ)	-6.20	0.56
Primary Weight (ω)	0.62	0.02
P(Use Primary) (π)	0.83	0.00

Panel B: Consumer Heterogeneity Parameters

Parameter	Est	SE
Card Volatility	0.90	0.35
Credit Volatility	2.40	0.88
Correlation of β	-0.78	0.08
Income Volatility (ν_y)	0.75	0.01
Income Elasticity (α_y)	0.20	0.06
Card Income Effect (β_y^C)	-1.50	0.46
Credit Income Effect (β_y^L)	0.60	0.42
Card-Credit Income (χ_y^{12})	0.60	0.22
Credit-Card Income (χ_y^{21})	0.50	0.19
Credit-Credit Income (χ_y^{22})	-0.40	0.18

Panel C: Merchant Parameters

Parameter	Est	SE
Merchant CES	5.61	1.30
Average γ	0.26	0.07
S.D. of γ	0.11	0.03

Panel D: Network Cost Parameters (bps)

Parameter	Est	SE
Cash	30	10
Visa Debit	40	18
MC Debit	60	9
Visa Credit	40	16
MC Credit	50	9
Amex	50	8

Notes: S.D. refers to the standard deviation, and R.C. refers to the random coefficients. The Ξ are the unobserved characteristics, and the χ^{lm} is the complementarity parameter. In Panel B, volatility parameters are the standard deviations of the random coefficients. Merchant types γ are distributed according to a Gamma distribution. Standard errors computed by bootstrap.

The surplus distribution also shifts under the credit fee cap: merchants retain the fee savings rather than passing them to consumers, so merchant surplus rises by \$49B while consumer surplus falls by \$31B (versus gains for both groups under full pass-through).

Table OG.2: Counterfactual effects on prices and shares: No merchant fee pass-through

	Price Controls		Change Competition		
	Cap Fees	Uncap Debit	Monopoly	Dual Routing	CBDC
Δ Prices (bps)					
Credit Fees	-105 (0.0)	-3.6 (0.7)	39 (3.1)	-14 (3.4)	0.7 (0.1)
Credit Rewards	-110 (0.5)	-6 (0.7)	-83 (30)	-16 (3.6)	0.7 (0.1)
Debit Fees	0.0 (0.0)	25 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
Debit Rewards	1.5 (0.2)	25 (0.2)	-37 (7)	0.5 (0.1)	3.9 (1.0)
Δ Shares of Total Spending (pp.)					
Cash	7 (1.5)	-7 (1.1)	29 (0.8)	0.1 (0.2)	-1.8 (0.2)
Debit	13 (1.7)	13 (2.4)	-7 (0.8)	0.9 (0.5)	-4.3 (0.1)
Credit	-20 (2.8)	-6 (1.3)	-22 (0.9)	-1.0 (0.7)	-0.6 (0.1)
Δ Fees, Rewards ($Bn)					
Total Fees	-59 (2.0)	4.0 (1.3)	-38 (3.3)	-8 (2.5)	-1.0 (0.4)
Total Rewards	-57 (1.2)	0.2 (1.6)	-64 (9)	-8 (2.5)	0.7 (0.3)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

Table OG.3: Counterfactual welfare effects: No merchant fee pass-through

	Price Controls		Change Competition		
	Cap Fees	Uncap Debit	Monopoly	Dual Routing	CBDC
Δ Welfare ($Bn)					
Consumers	-31 (1.7)	9 (0.2)	-50 (11)	-3.9 (0.9)	2.7 (0.7)
Merchants	49 (2.7)	-4.0 (1.2)	27 (4.0)	6 (2.0)	1.1 (0.4)
Networks	-1.6 (0.3)	3.0 (0.2)	31 (6)	0.6 (0.0)	-2.4 (0.6)
Total	17 (3.9)	8 (1.2)	8 (7)	2.9 (1.1)	1.4 (0.5)
Δ Consumer Surplus by Income (bps consumption)					
Low	-10 (1.6)	11 (0.9)	-26 (7)	-1.3 (0.4)	3.6 (1.0)
Median	-22 (1.7)	10 (0.4)	-38 (9)	-3.0 (0.7)	2.9 (0.8)
High	-44 (2.5)	7 (0.4)	-67 (14)	-6 (1.3)	2.4 (0.6)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**). Totals may not sum due to independent rounding.

OG.2 Credit Constraints

The baseline model attributes low-income credit avoidance to preferences. Some of that gap may instead reflect access, since subprime consumers cannot obtain a credit card at all. Both specifications must match the same estimated Durbin Amendment effect, so reclassifying some credit avoidance from preference to access just reallocates reward sensitivity across consumers. The main counterfactual conclusions hold, with credit fee cap and monopoly welfare gains similar to baseline, if anything slightly larger.

Model modification. This specification modifies the consumer adoption stage. Before evaluating choice probabilities (Eq. **??**), some consumers are blocked from holding credit cards. Constrained customers cannot have a credit card as a primary or secondary card. All other equations are unchanged.

Credit access follows a logistic function of demeaned log income: $P(\text{access} \mid y) = \Lambda(\gamma_0 + \gamma_y \log \tilde{y})$, adding two parameters (γ_0, γ_y) estimated from DCPC credit score data and held fixed in the structural step. Table **??** reports the logistic regression. Combining these estimates with the structural income distribution ($\nu_y = 0.75$), the access probability is 82% at median income, 92% at the 90th percentile, and 62% at the 10th percentile.

Table OG.4: Logistic regression: Credit card access on income

	Credit Score Above 650
Demeaned Log Income	1.04***
	(0.07)
Intercept	1.49***
	(0.05)
Num.Obs.	3052
Log.Lik.	-1346.172

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Dependent variable is an indicator for credit score above 650. Income is log-transformed and demeaned. Standard errors in parentheses.

Constraints now partly explain low-income consumers' credit avoidance, so the income gradient in credit preference weakens: β_y^L falls from 0.60 to 0.20 (Table **??**). The median consumer's credit aversion also falls: indifference between Visa debit and credit requires 0.9% in rewards (SE 0.2%), versus 1.2% at baseline.

Adding the access constraint raises credit fee cap welfare gains to \$31B (SE \$3.9B) from \$27B at baseline, and monopoly gains to \$22B from \$15B (Table **??**). Constrained consumers are inert, so the moment-matching constraint forces unconstrained consumers to be more reward-sensitive than in the baseline. The credit fee cap then acts on fewer

Table OG.5: Estimated parameters: Credit constraints

Panel A: Consumer Preference Parameters

Parameter	Est	SE
Log Price Sensitivity ($\log \alpha_0$)	6.75	0.34
Card-Credit Complement (χ^{12})	4.80	0.61
Credit-Card Complement (χ^{21})	4.00	0.49
Credit-Credit Complement (χ^{22})	-5.00	0.88
Visa Debit Utility (Ξ)	-4.20	0.64
Visa Credit Utility (Ξ)	-5.70	0.50
MC Debit Utility (Ξ)	-4.40	0.67
MC Credit Utility (Ξ)	-6.00	0.51
Amex Utility (Ξ)	-6.00	0.52
Primary Weight (ω)	0.62	0.02
P(Use Primary) (π)	0.83	0.00

Panel B: Consumer Heterogeneity Parameters

Parameter	Est	SE
Card Volatility	0.90	0.35
Credit Volatility	2.40	0.88
Correlation of β	-0.79	0.08
Income Volatility (ν_y)	0.75	0.01
Income Elasticity (α_y)	0.20	0.06
Card Income Effect (β_y^C)	-1.60	0.47
Credit Income Effect (β_y^L)	0.20	0.33
Card-Credit Income (χ_y^{12})	0.50	0.22
Credit-Card Income (χ_y^{21})	0.50	0.21
Credit-Credit Income (χ_y^{22})	-0.30	0.20
Constraint Intercept	1.51	0.09
Constraint Slope	1.32	0.12

Panel C: Merchant Parameters

Parameter	Est	SE
Merchant CES	5.61	1.30
Average γ	0.26	0.07
S.D. of γ	0.11	0.03

Panel D: Network Cost Parameters (bps)

Parameter	Est	SE
Cash	30	10
Visa Debit	40	18
MC Debit	50	9
Visa Credit	40	16
MC Credit	50	9
Amex	50	9

Notes: S.D. refers to the standard deviation, and R.C. refers to the random coefficients. The Ξ are the unobserved characteristics, and the χ^{lm} is the complementarity parameter. In Panel B, volatility parameters are the standard deviations of the random coefficients. Constraint intercept and slope are the logistic parameters γ_0 and γ_y governing credit card access probability. Merchant types γ are distributed according to a Gamma distribution. Standard errors computed by bootstrap.

consumers but provokes a larger response from each. The same logic amplifies the monopoly counterfactual, since more reward-sensitive consumers face bigger distortions from the competitive rewards arms race, so eliminating competition delivers larger gains. Price and share responses stay close to baseline, because the cap forces similar fee cuts regardless of why consumers avoid credit (Table **??**). Other counterfactuals are close to baseline.

Table OG.6: Counterfactual effects on prices and shares: Credit constraints

	Price Controls		Change Competition		
	Cap Fees	Uncap Debit	Monopoly	Dual Routing	CBDC
Δ Prices (bps)					
Credit Fees	-105 (0.0)	-2.8 (0.5)	35 (2.7)	-13 (3.4)	0.8 (0.1)
Credit Rewards	-110 (0.5)	-6 (0.6)	-87 (33)	-15 (3.6)	0.8 (0.1)
Debit Fees	0.0 (0.0)	25 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
Debit Rewards	1.4 (0.2)	25 (0.2)	-33 (9)	0.4 (0.0)	3.9 (1.2)
Δ Shares of Total Spending (pp.)					
Cash	7 (1.8)	-7 (1.3)	29 (1.1)	0.1 (0.2)	-1.9 (0.2)
Debit	12 (1.7)	13 (2.4)	-6 (0.7)	0.7 (0.4)	-4.4 (0.1)
Credit	-19 (3.1)	-6 (1.3)	-23 (0.8)	-0.8 (0.6)	-0.5 (0.1)
Δ Fees, Rewards ($Bn)					
Total Fees	-59 (2.4)	5.0 (1.3)	-40 (3.0)	-7 (2.3)	-0.8 (0.5)
Total Rewards	-50 (1.5)	1.5 (1.2)	-57 (9)	-7 (2.1)	0.7 (0.3)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

Table OG.7: Counterfactual welfare effects: Credit constraints

	Price Controls		Change Competition		
	Cap Fees	Uncap Debit	Monopoly	Dual Routing	CBDC
Δ Welfare ($Bn)					
Consumers	32 (3.8)	3.3 (1.6)	-2.3 (12)	3.9 (1.5)	3.4 (1.2)
Merchants	0.5 (0.2)	-0.7 (0.2)	-4.2 (1.8)	-0.1 (0.0)	0.3 (0.1)
Networks	-1.9 (0.4)	3.0 (0.3)	29 (7)	0.6 (0.0)	-2.4 (0.7)
Total	31 (3.9)	6 (1.4)	22 (7)	4.4 (1.5)	1.3 (0.6)
Δ Consumer Surplus by Income (bps consumption)					
Low	42 (3.8)	8 (1.6)	3.3 (11)	5.0 (1.8)	5 (1.8)
Median	35 (4.0)	4.7 (1.4)	0.3 (12)	3.9 (1.6)	3.9 (1.4)
High	19 (5)	2.5 (1.7)	-19 (16)	1.8 (1.2)	3.3 (1.1)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

OG.3 Lower Reward Sensitivity

The Durbin Amendment event-study coefficient pins down reward sensitivity in the estimation. If that coefficient overstates how aggressively consumers switch when rewards change, every counterfactual inherits the error. Halving the target moment does change the bottom-line conclusions for the credit fee cap and merger policies. The monopoly counterfactual flips sign, credit fee cap welfare gains fall by roughly half, and only dual routing stays close to baseline. This specification is not our preferred one, since it also drives Visa Debit and Visa Credit marginal costs into implausibly negative territory.

Model modification. No model equations change. Instead, the Durbin regression coefficient that the model targets is halved.

Halving the target lowers estimated reward sensitivity from 6.74 to 5.92 and raises mean unobserved disutility to compensate (Table **??**). The estimated network costs are less plausible: Visa Debit and Visa Credit both have negative marginal costs (-10 bps), well below the baseline estimates that match accounting data (**Lowe2005**; **Mukharlyamov.Sarin2025**; **NACHA2017**; **Visa2020**).

The monopoly counterfactual reverses sign, from +\$15B at baseline to $-$\$19B (SE \$10B) here (Table **??**). At baseline, a monopolist raises welfare by eliminating the competitive rewards arms race. With less reward-responsive consumers, that arms race is smaller, so there is less distortion for the monopolist to correct. At the same time, less responsive consumers do not switch away from a merged network, leaving its market power unchecked. The net effect is a large welfare loss.

Fee cap gains are attenuated (\$15B versus \$27B) for the opposite reason: less responsive consumers face less distortion from the current fee structure, so there is less to correct (Table **??**). Dual routing still raises welfare by \$1.1B (SE \$0.2B), versus the \$3.8B baseline gain (both net of the mechanical χ^{22} adjustment), because it operates through merchant-side competition rather than consumer switching.

Table OG.8: Estimated parameters: Lower reward sensitivity

Panel A: Consumer Preference Parameters

Parameter	Est	SE
Log Price Sensitivity ($\log \alpha_0$)	5.92	0.39
Card-Credit Complement (χ^{12})	6.10	0.80
Credit-Card Complement (χ^{21})	5.10	0.64
Credit-Credit Complement (χ^{22})	-7.40	1.19
Visa Debit Utility (Ξ)	-5.40	0.84
Visa Credit Utility (Ξ)	-7.00	0.73
MC Debit Utility (Ξ)	-5.80	0.83
MC Credit Utility (Ξ)	-7.60	0.78
Amex Utility (Ξ)	-7.70	0.80
Primary Weight (ω)	0.61	0.03
P(Use Primary) (π)	0.83	0.00

Panel B: Consumer Heterogeneity Parameters

Parameter	Est	SE
Card Volatility	2.20	0.61
Credit Volatility	5.60	1.47
Correlation of β	-0.81	0.09
Income Volatility (ν_y)	0.75	0.01
Income Elasticity (α_y)	0.20	0.06
Card Income Effect (β_y^C)	-1.60	0.52
Credit Income Effect (β_y^L)	2.10	0.71
Card-Credit Income (χ_y^{12})	0.40	0.26
Credit-Card Income (χ_y^{21})	0.40	0.22
Credit-Credit Income (χ_y^{22})	-0.30	0.23

Panel C: Merchant Parameters

Parameter	Est	SE
Merchant CES	5.61	1.30
Average γ	0.28	0.07
S.D. of γ	0.12	0.04

Panel D: Network Cost Parameters (bps)

Parameter	Est	SE
Cash	30	10
Visa Debit	-10	30
MC Debit	30	13
Visa Credit	-10	30
MC Credit	20	19
Amex	20	17

Notes: S.D. refers to the standard deviation, and R.C. refers to the random coefficients. The Ξ are the unobserved characteristics, and the χ^{lm} is the complementarity parameter. In Panel B, volatility parameters are the standard deviations of the random coefficients. Merchant types γ are distributed according to a Gamma distribution. Standard errors computed by bootstrap.

Table OG.9: Counterfactual effects on prices and shares: Lower reward sensitivity

	Price Controls	Change Competition	
	Cap Fees	Monopoly	Dual Routing
Δ Prices (bps)			
Credit Fees	-105 (0.0)	40 (2.9)	-6 (0.9)
Credit Rewards	-109 (0.5)	-249 (45)	-7 (1.0)
Debit Fees	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
Debit Rewards	1.7 (0.2)	-74 (11)	0.3 (0.0)
Δ Shares of Total Spending (pp.)			
Cash	2.1 (0.6)	29 (0.9)	-0.1 (0.0)
Debit	8 (0.9)	-5 (0.6)	0.0 (0.0)
Credit	-10 (1.4)	-24 (0.8)	0.1 (0.0)
Δ Fees, Rewards ($Bn)			
Total Fees	-52 (0.8)	-41 (3.2)	-2.2 (0.4)
Total Rewards	-52 (0.7)	-111 (12)	-2.9 (0.5)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

Table OG.10: Counterfactual welfare effects: Lower reward sensitivity

	Price Controls	Change Competition	
	Cap Fees	Monopoly	Dual Routing
Δ Welfare ($Bn)			
Consumers	16 (1.5)	-71 (16)	0.7 (0.2)
Merchants	0.0 (0.1)	-12 (3.4)	-0.1 (0.0)
Networks	-1.1 (0.3)	65 (9)	0.5 (0.0)
Total	15 (1.6)	-19 (10)	1.1 (0.2)
Δ Consumer Surplus by Income (bps consumption)			
Low	41 (1.6)	-13 (10)	1.7 (0.4)
Median	25 (1.6)	-48 (13)	0.8 (0.3)
High	1.5 (2.2)	-107 (21)	-0.5 (0.2)

Notes: Bootstrap standard errors in parentheses. Counterfactual scenarios are defined in Table **??**. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**). Totals may not sum due to independent rounding.

OG.4 Comparison to Other Credit Fee Caps and the Social Optimum

The main text caps credit card merchant fees at 120 bp. This section compares that cap to a tighter cap at the cost of cash and to the constrained social optimum.

The cost-of-cash cap sets fees at 30 bp, matching the Rochet-Tirole tourist test (**Rochet.Tirole2006**) and the benchmark behind European interchange regulation (**EuropeanCommission2015**). At this level, the cap also replicates perfect merchant surcharging (**Zenger2011**). It delivers \$29 billion in total welfare gains versus \$27 billion under the 120 bps cap, roughly 7% more (Table **??**). The extra gains come from deeper credit card reward cuts. The cost-of-cash cap also requires far greater out-of-sample extrapolation, eliminating roughly 87% of credit card merchant fees versus about half under the 120 bps cap.

The constrained social optimum provides an upper bound. The planner chooses merchant fees τ^l and consumer rewards f^l for each card type l to maximize total welfare subject to non-negative network profits. A break-even constraint is necessary because, under elastic labor supply, an unconstrained planner would subsidize merchants indefinitely to reduce the CES markup distortion. The planner's main lever is eliminating network markups on merchant fees. It also sets negative rewards, though less negative than under the tourist test. The planner jointly optimizes fees and rewards, whereas the tourist test leaves reward-setting to profit-maximizing networks (Table **??**). Total welfare under the social optimum is \$35B, versus \$27B under the 120 bps cap (Table **??**). The 120 bps cap captures roughly 77% of the planner's gains without directly regulating rewards, because the dominant distortion is over-adoption of credit cards rather than network market power.

Table OG.11: Effects of Fee Caps on Prices and Shares

	Cap Credit (120 bps)		Tourist Test		Ramsey Planner	
Δ **Prices (bps)**						
Credit Fees	-105	(0.0)	-195	(10)	-198	(16)
Credit Rewards	-110	(0.5)	-199	(9)	-163	(18)
Debit Fees	0.0	(0.0)	-42	(10)	-44	(12)
Debit Rewards	1.3	(0.2)	-38	(9)	-11	(18)
Δ **Shares of Total Spending (pp.)**						
Cash	6	(1.8)	35	(10)	18	(12)
Debit	13	(2.3)	-5	(6)	8	(8)
Credit	-19	(3.7)	-30	(4.9)	-26	(5)
Δ **Fees, Rewards ($Bn)**						
Total Fees	-58	(2.5)	-101	(8)	-102	(10)
Total Rewards	-57	(1.6)	-86	(5.0)	-75	(11)

Notes: Bootstrap standard errors in parentheses. Cap Credit caps credit card merchant fees at 120 bps. Tourist Test caps both credit and debit fees at the estimated cost of cash (30 bps). Ramsey Planner maximizes total welfare over merchant fees and consumer rewards, subject to equilibrium conditions and the network break-even constraint.

Table OG.12: Welfare Effects of Fee Caps

	Cap Credit (120 bps)		Tourist Test		Ramsey Planner	
Δ **Welfare ($Bn)**						
Consumers	28	(4.4)	38	(8)	54	(6)
Merchants	0.4	(0.2)	2.8	(1.6)	5	(2.3)
Networks	-1.5	(0.3)	-11	(2.2)	-24	(10)
Total	27	(4.5)	29	(8)	35	(5)
Δ **Consumer Surplus by Income (bps consumption)**						
Low	48	(2.9)	70	(6)	82	(8)
Median	37	(4.3)	54	(8)	67	(6)
High	15	(5)	13	(8)	35	(7)

Notes: Bootstrap standard errors in parentheses. Columns correspond to the same scenarios as Table **??**. Low (high) income consumers have log income at -2 ($+2$) standard deviations relative to the median. Dollar values are computed by normalizing to $10 trillion of total consumer purchases (**Nilson2020a**).

OH Price Coherence

This appendix explains why merchants almost never surcharge despite legal ability to do so. It covers surcharging law, empirical evidence on differential pricing, and a theoretical framework for price coherence.

I extend the baseline model to allow merchants to set different prices for card and cash consumers. In this extension, I show that surcharges cannot steer consumers from card to cash. Card-preferring consumers will not switch to cash unless the surcharge exceeds the merchant fee. Second, since surcharges do not shift payment behavior, the envelope theorem makes the profit loss from uniform pricing second-order in fee dispersion. Third, returns to surcharging across card networks are even smaller, since fee differences across networks are an order of magnitude below the cash-card gap.

Cash discounts have been legal since 1981 yet remain rare, providing a revealed-preference bound. Menu costs suffice to deter surcharging even at the cash-card margin. Since card-vs-card fee gaps are far smaller, those same menu costs overwhelm any incentive to price differently across networks.

OH.1 Legal Background

Cash discounts have long been legal in the U.S., but card surcharges became permitted only gradually.[25] The Cash Discount Act of 1981 guarantees merchants' right to offer cash discounts (**Chakravorti.Shah2001**; **Levitin2005**; **Prager.etal2009**). The Durbin Amendment in 2010 granted merchants the right to offer debit discounts (**Schuh.etal2011**; **Briglevics.Shy2014**). A 2013 settlement between Visa, MC, and the DOJ removed network-level no-surcharge rules, allowing surcharging in 40 states (**Blakeley.Fagan2015**).[26] As of 2023, only Massachusetts and Connecticut ban surcharging (**CardX2023**), though disclosure rules in New York and Maine make it impractical.[27]

OH.2 Empirical Evidence

Surcharging is rare regardless of legality, confirming that menu costs rather than legal bans explain price coherence. I restrict the sample to cash, check, debit, and credit transactions at retail merchants. In the full DCPC sample (all transaction types, before

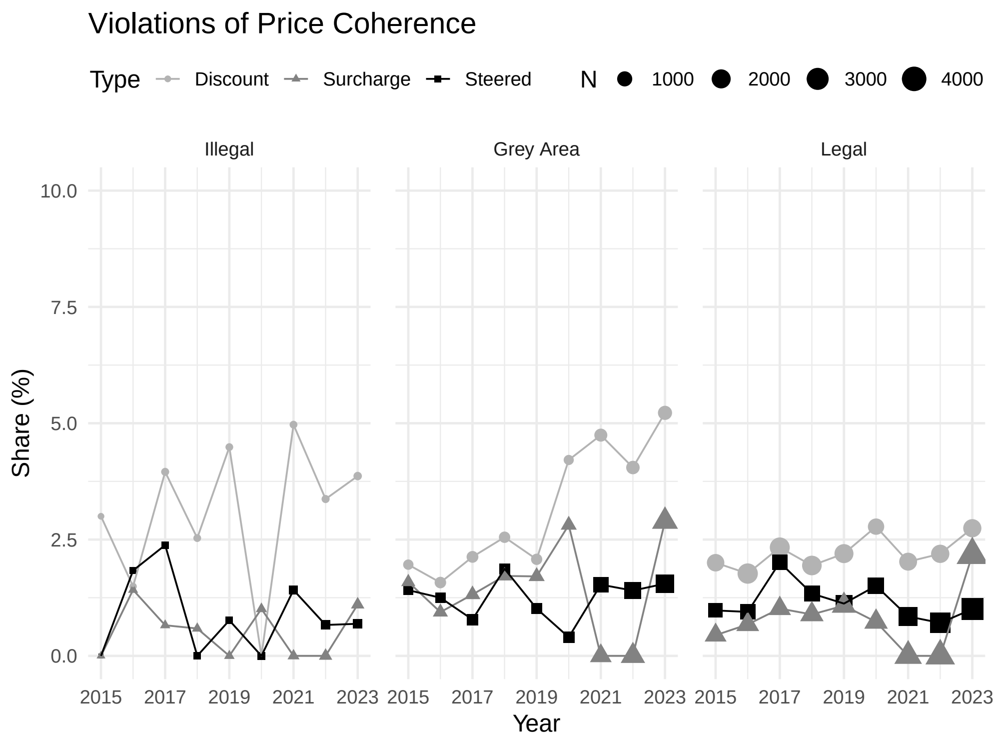
[25]Under full information, discounts and surcharges are the same. But if shrouded, cash discounts are giveaways while surcharges are add-on prices (**Bourguignon.etal2019**).

[26]Visa allowed multi-state merchants to surcharge in states that permitted it (**Visa2013**). AmEx and Discover relaxed their no-surcharge rules to match (**Merchant2016**).

[27]In New York and Maine, retailers must disclose both the credit card price and cash price in dollar amounts. In New York, this rule ensures consumers "should not have to do math to figure out whether they are paying the surcharge." (**Westchester2019**).

this restriction), about 5% of transactions involve card surcharges or cash discounts, supporting the price coherence assumption.

I flag violations of price coherence using DCPC variables on discounts, surcharges, and payment steering. The figure below plots three time series. "Discount" is the share of cash/check transactions with a discount. "Surcharge" is the share of credit card transactions with a fee. "Steered" is cash transactions by credit-preferring consumers who switched due to a discount, as a share of credit transactions plus such switched transactions. States are grouped by surcharging legality: "Legal" (never banned), "Illegal" (still banned as of 2020), and "Grey Area" (repealed bans during 2013–2020).



Rates are low across all groups, and legality does not explain the rarity. Cash discounts, always legal, are still rare. Surcharge rates do not vary with state legal status, ruling out legal bans as the main driver of price coherence. This pattern shows that menu costs deter differential pricing even at the cash-card margin, where fee gaps are largest.

OH.3 Private Incentives to Surcharge

OH.3.1 No Steering Between Cash and Card

I show merchants cannot steer consumers from card to cash in a natural extension of the baseline model that permits surcharging. The key assumption is that card convenience benefits are constant across purchases. A consumer who values cards at merchant type γ gets the same proportional benefit regardless of the item, ruling out within-trip substitution on payment-specific prices.

Consumers choose effective consumption $q(\omega)$ of each variety, a linear aggregate of card consumption $c(\omega)$ and cash consumption $a(\omega)$. Merchants charge card consumers a price $1+s(\omega)$ higher. Consumers solve

$$U = \max_{c,a} \left(\int_0^1 q(\omega)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}\omega \right)^{\frac{\sigma}{\sigma-1}} \tag{56}$$

$$\text{s.t. } q(\omega) = \left(1+\gamma(\omega) v^w_{M(\omega)}\right)^{\frac{1}{\sigma-1}} c(\omega) + a(\omega) \tag{57}$$

$$y \geq \int_0^1 \left(c(\omega)(1+s(\omega)) + a(\omega)\right) p(\omega) \, \mathrm{d}\omega \tag{58}$$

Linear aggregation makes card goods higher quality and perfect substitutes for cash goods. The baseline model corresponds to corner solutions where consumers use only their preferred payment type.

Lemma 1. *At a merchant of type γ that accepts cards, a card consumer will use cash only if $s > (1+\gamma)^{\frac{1}{\sigma-1}} - 1$*

Proof. The first-order conditions for card spending c and cash spending a yield

$$\frac{\partial \mathcal{L}}{\partial c} = I^{\frac{1}{\sigma-1}} \times q^{-\frac{1}{\sigma}} \times (1+\gamma)^{\frac{1}{\sigma-1}} - \lambda(1+s)p$$
$$\frac{\partial \mathcal{L}}{\partial a} = I^{\frac{1}{\sigma-1}} \times q^{-\frac{1}{\sigma}} - \lambda p$$

Both payment methods are used if $(1+\gamma)^{\frac{1}{\sigma-1}} = 1+s$. Since the aggregator is linear, consumers use cash exclusively if $s > (1+\gamma)^{\frac{1}{\sigma-1}} - 1$.

□

Theorem 2. *In a market with a monopoly credit card network that charges a merchant fee of τ, no merchant that accepts the credit card in the baseline model can steer consumers by setting $s = \tau$.*

Proof. The lowest type accepting credit cards satisfies $\gamma^* = \frac{\sigma\tau}{1-\sigma\tau}$. For general $\gamma > 0, \sigma > 1$, the inequality $(1+\gamma)^{\frac{1}{\sigma-1}} \geq 1 + \frac{\gamma}{\gamma+1}\frac{1}{\sigma-1}$ holds. By Lemma **??**, the required surcharge to induce switching exceeds $s^* \geq \tau\frac{\sigma}{\sigma-1} > \tau$, so setting $s = \tau$ does not induce steering.

□

The analysis assumes consumers use only their primary card type: credit users do not switch to debit, and vice versa. This is consistent with empirical evidence (**Conrath2014**) and antitrust analysis (**Jones2001**).

OH.3.2 Second-Order Losses from Uniform Pricing

Since surcharges do not change the payment method, uniform pricing generates only second-order losses by the envelope theorem. I quantify these with a second-order Taylor approximation.

Suppose merchants charge wallet-specific prices p^w (stacked into a vector). After normalization, total profits are proportional to

$$\widehat{\Pi} \propto \sum_{w \in \mathcal{W}} \mu^w \pi^w$$
$$\pi^w = \left(1 + \gamma v_M^w\right) \left(p^w\right)^{-\sigma} \left(p^w \left(1 - \tau^w\right) - 1\right)$$

Let p^* denote optimal payment-specific prices and $\hat{p}$ denote the optimal uniform price.

Theorem 3. *The percentage loss from uniform pricing is:*

$$\log \widehat{\Pi}\left(p^*\right) - \log \widehat{\Pi}\left(\hat{p}\right) = \sum_w \frac{\mu^w \left(1 + \gamma v_M^w\right)}{\sum_l \mu^l \left(1 + \gamma v_M^l\right)} \times \frac{\sigma\left(\sigma - 1\right)}{2} \left(\tau^w - \hat{\tau}\right)^2 + O\left(\tau^3\right)$$

Proof. The percentage loss is a weighted sum across wallets. By the envelope theorem, the first-order term vanishes. Computing the second derivative of log profit with respect to $\log p$ and evaluating at the optimal price $p^{w*} = \frac{\sigma}{\sigma-1}\left(1 - \tau^w\right)^{-1}$ yields $\frac{\partial^2 \log \pi^w}{\partial (\log p)^2} = -\sigma\left(\sigma - 1\right)$. Since $\log p^{w*} - \log \hat{p} \approx \tau^w - \hat{\tau} + O(\tau^2)$, the second-order Taylor expansion yields the result.

□

High fees alone do not make uniform pricing costly. Fee *dispersion* among accepted payment methods does so. At the estimated σ =5.61, losses from uniform pricing are 13 bp of profit.

OH.3.3 Card-vs-Card Surcharging

Merchants also seek the right to surcharge cards differently, e.g., charge AmEx more than Visa (**Conrath2014**). The results above bound the gains: in a type-symmetric equilibrium, fee dispersion across networks of the same type is near zero, so losses from uniform pricing are negligible.

The rarity of cash discounts sharpens this point. Cash discounts have been legal since 1981, yet merchants almost never offer them. The cash-card fee gap is roughly 2–3% (the full merchant discount fee, since cash carries no merchant fee). That merchants do not surcharge even at this margin shows that menu costs exceed the gains at a 2–3% fee gap. These costs include maintaining payment-specific prices, communicating them, and absorbing the resulting customer friction (**Caddy.etal2020**).

Fee differences across card networks are an order of magnitude smaller. Even away from exact type-symmetry, AmEx-Visa fee gaps are tens of bp, well below the cash-card gap. Since merchants do not price differently at the larger margin, they will not do so at the smaller one.

OH.3.4 Limitations

The results above assume every card transaction generates incremental sales for the merchant. If some card transactions do not boost sales, merchants would face stronger incentives to surcharge those transactions. In the limit where no card transaction generates incremental sales, surcharging at the full merchant fee would be profitable.

OI Merchant Sorting

OI.1 Sorting Concern and Main Finding

Card fees may not redistribute across consumer types if consumers with different payment preferences shop at different merchants. Under perfect sorting, credit card users patronize only high-end merchants while cash/debit users patronize discount retailers, eliminating cross-consumer redistribution. This would undermine the redistributive channel central to the welfare analysis, since fee changes would affect only consumers who already use the relevant payment method.

We measure payment shares across hundreds of merchants using NielsenIQ Homescan and MRI-Simmons data. Some sorting exists, but no large merchant serves only one payment type. Even merchants at the 90th percentile of credit card use receive 15% of transactions from debit/cash.

We derive a sufficient statistic linking sorting patterns to redistribution magnitude. With no sorting (uniform payment shares across merchants), merchant fees redistribute across consumer types because all consumers face the same price increases regardless of payment choice. Empirical sorting reduces this baseline redistribution by only 4–9 pp., leaving 91–96% intact. This validates the main-text treatment of payment markets as integrated economies.

OI.2 Structure of Analysis

The analysis has three steps. First, we measure payment share distributions across stores in Homescan and MRI, computing both unweighted and revenue-weighted covariance matrices. Second, we derive a sufficient statistic linking sorting to redistribution magnitude, valid to first order under local fee changes. Third, we apply this statistic and show sorting attenuates baseline redistribution by at most 10%.

OI.3 Measuring Payment Shares

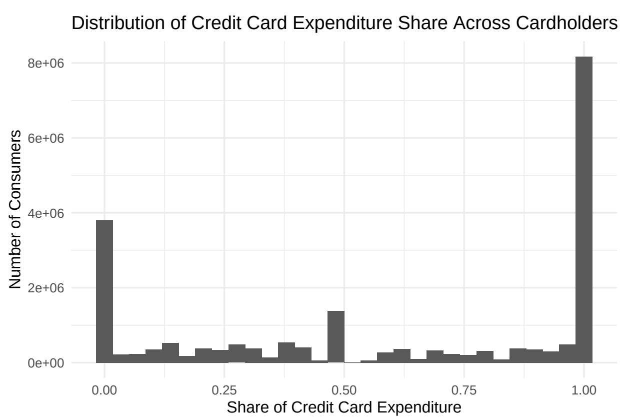
Homescan: Homescan provides transaction-level data with value-weighted shares of credit, debit, and cash spending across stores, but covers only grocery.

MRI: MRI-Simmons provides consumer-level data on payment choices and shopping across 214 large merchants: grocery, department stores, apparel, restaurants, electronics, and other retail sectors. These merchants cover a large share of U.S. consumer retail spending. The data include credit and debit spending amounts but not cash expenditures.

I classify each consumer by primary payment preference. A consumer is cash-preferring if they agree with "I prefer to pay cash for things I buy, whenever I can." Among non-cash consumers, I classify by whether they spend more on credit or debit cards. This classification captures real variation: most credit-preferring consumers put over 80% of card spending on their preferred method (Figure **??**).

To build payment shares at each merchant, I assume consumers use their preferred payment method at every store, equating a store's payment composition with the preference mix among its shoppers. This assumption fits the large national merchants in MRI, where all major payment methods are accepted. Consumers may vary methods across purchases (e.g., using credit for large transactions), but such within-consumer variation would reduce measured sorting relative to imputation. Imputation assigns each consumer a single method. Because imputation makes merchant-level payment shares more extreme, it overstates sorting and understates surviving redistribution. Any measurement error from imputation therefore biases our findings toward rejecting the integrated economy assumption, making our estimates conservative. Homescan provides a cross-check: it records actual payment choices (not imputed), and its covariance estimates are similar to MRI, suggesting imputation bias is small.

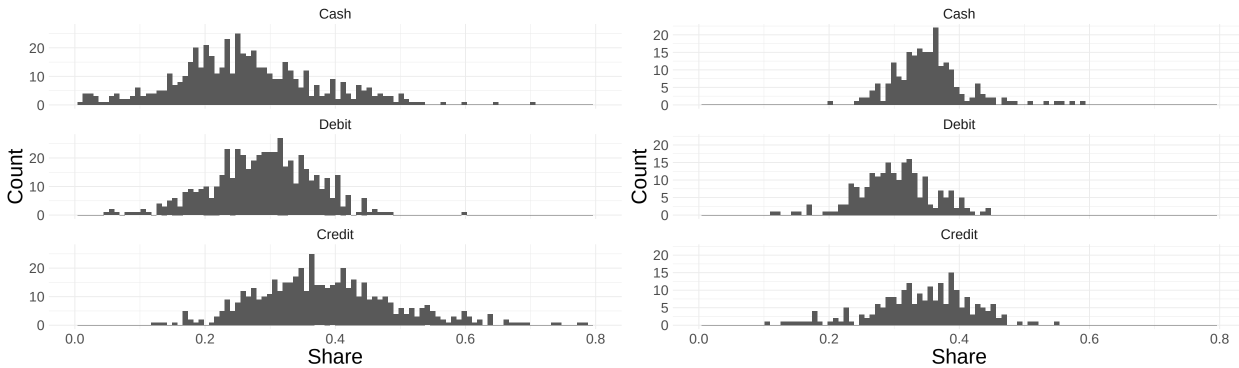
Figure OI.1: Distribution of Credit Card Expenditure Share Across Consumers with Cards



Summary of Results: Both datasets show wide dispersion in payment shares across stores, but no store has only one payment method (Figures **??** and **??**). In Homescan, credit sales average 40% of total sales but range from 10% to 80%.

Larger stores have less dispersed payment mixes (Figures **??** and **??**). This firm size relationship matters because stores with the most sorting represent a small share of the economy. Since many consumers shop at the largest stores, substantial redistribution

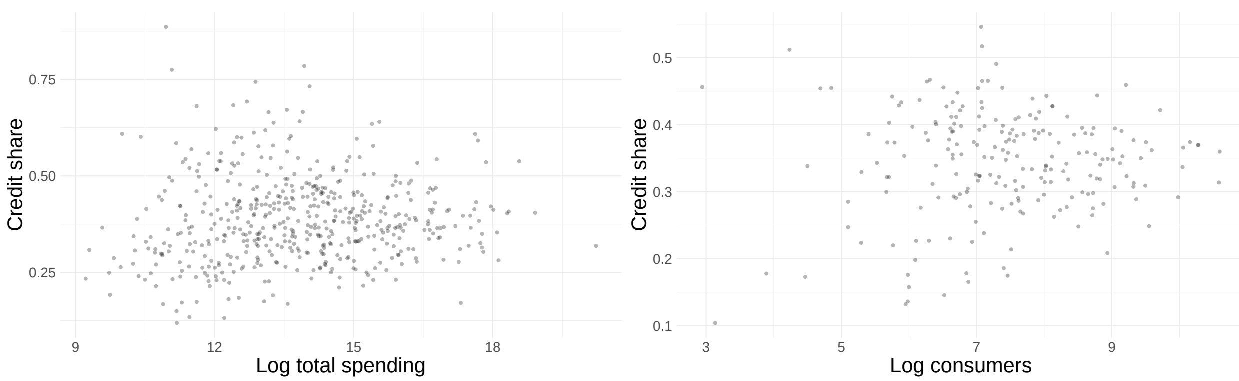
Figure OI.2: Share of Spending on Different Payment Methods Across Merchants



(a) Homescan **(b)** MRI

occurs across consumer types.

Figure OI.3: Credit card share and firm size



(a) Homescan **(b)** MRI

Table **??** shows weighted and unweighted covariances between payment shares (revenue weights in Homescan, customer weights in MRI). Cash-credit and credit-debit covariances are negative; the cash-debit covariance is positive but small, reflecting that cash and debit users shop at more similar stores than cash and credit users. Weighted variances are smaller than unweighted ones, consistent with larger stores having less dispersed payment mixes.

These negative covariances reduce redistribution from merchant fees, but the effect is small.

OI.4 Sufficient Statistics for Redistribution

I derive a sufficient statistic relating the revenue-weighted covariance matrix of payment shares to redistribution magnitude. The intuition is simple. Under uniform pric-

Table OI.1: Covariances

(a) Homescan - Unweighted

	Cash	Debit	Credit
Cash	0.0122	-0.0008	-0.0087
Debit		0.0062	-0.0031
Credit			0.0123

(b) Homescan - Weighted

	Cash	Debit	Credit
Cash	0.0056	0.0001	-0.0037
Debit		0.0045	-0.0020
Credit			0.0070

(c) MRI - Unweighted

	Cash	Debit	Credit
Cash	0.0033	-0.0003	-0.003
Debit		0.0034	-0.003
Credit			0.006

(d) MRI - Weighted

	Cash	Debit	Credit
Cash	0.0012	0.0001	-0.0013
Debit		0.0014	-0.0015
Credit			0.0028

ing, merchant fee increases hit all consumers at a given store regardless of payment method. Redistribution depends on the overlap between where different consumer types shop: if credit and debit consumers visit the same merchants, credit fee increases burden both groups. The covariance between payment shares across merchants summarizes this overlap.

Suppose prices are log-linear in fees:

$$\log p_j = \log \overline{p} + \sum_k \mu_{jk} \tau_k$$

where $\mu_{jk} = \frac{q_{jk}}{\sum_l q_{jl}}$ is the share of spending on card k at merchant j, and τ_k is the merchant fee for card k (expressed as a decimal, e.g. $0.02 = 2\%$). This log-linear form approximates the main-text CES pricing formula: with elasticity σ and ad valorem merchant fees, the exact price is $p_j = \frac{\sigma}{\sigma-1}\frac{1}{1-\hat{\tau}_j}$ where $\hat{\tau}_j = \sum_k \mu_{jk}\tau_k$, yielding the log-linear form $\log p_j \approx \log \overline{p} + \hat{\tau}_j$ for small fees.

Define $Q_j = \sum_l q_{jl}$ as total sales at merchant j. We measure welfare using compensating variation. For consumers spending on payment method l, the welfare effect of a marginal change in fee m is

$$
\begin{aligned}
\int_j q_{jl}\frac{\partial p_j}{\partial \tau_m} &= \int_j q_{jl}p_j\left(\mu_{jm}+\sum_k \frac{\partial \mu_{jk}}{\partial \tau_m}\tau_k\right)\\
&= \int_j p_j Q_j \mu_{jl}\mu_{jm} + \int_j p_j Q_j \mu_{jl}\times\left(\sum_k \frac{\partial \mu_{jk}}{\partial \tau_m}\tau_k\right)\\
&\propto \mathbb{E}_R\left[\mu_{jl}\mu_{jm}\right] + \mathbb{E}_R\left[\mu_{jl}\times\sum_k \frac{\partial \mu_{jk}}{\partial \tau_m}\tau_k\right]
\end{aligned}
$$

where $\mathbb{E}_R[\cdot]$ denotes the revenue-weighted expectation across merchants.

The welfare effect has two parts. The first is the revenue-weighted cross-moment $\mathbb{E}_R[\mu_{jl}\mu_{jm}]$ of spending shares on l and m. The second captures how payment shares respond to fees. The second term involves $\sum_k(\partial\mu_{jk}/\partial\tau_m)\tau_k$, which is $O(\tau)$ because share derivatives $\partial\mu_{jk}/\partial\tau_m$ are $O(1)$ while baseline fees τ_k are small (1–3 pp.). For a fee change $\Delta\tau_m$ that is itself $O(\tau)$, this term contributes $O(\tau^2)$, while the first term contributes $O(\tau)$. The second term is second-order for the fee changes relevant to policy analysis, so we focus on $\mathbb{E}_R\left[\mu_{jl}\mu_{jm}\right]$.

Normalize by the total spending of method-l users, which is $\int_j q_{jl}p_j = \int_j p_j Q_j \mu_{jl} \propto \mathbb{E}_R\left[\mu_{jl}\right]$. The proportionality constant (aggregate revenue $\int_j p_j Q_j$) cancels in the ratio defining w_{lm}. The complete welfare matrix is

$$
\begin{aligned}
w_{lm} &= \text{\% welfare lost by } l \text{ from fee } m\\
&= \frac{\mathbb{E}_R\left[\mu_{jl}\mu_{jm}\right]}{\mathbb{E}_R\left[\mu_{jl}\right]}\\
&= \frac{\mathrm{Cov}_R\left(\mu_{jl},\mu_{jm}\right)}{\mathbb{E}_R\left[\mu_{jl}\right]} + \mathbb{E}_R\left[\mu_{jm}\right]
\end{aligned}
$$

Here w_{lm} is the compensating variation for consumers using payment method l from a 1 pp. increase in fees for method m, expressed as a % of baseline consumption.

The formula has a clear interpretation. If all stores have identical payment mixes (zero covariance), a debit consumer's burden from a 1 pp. credit fee increase equals the economy-wide credit share $\mathbb{E}_R[\mu_{jm}]$: payment choice offers no shelter. With perfect segmentation, credit and debit consumers shop at disjoint stores ($\mu_{jm}\times\mu_{jl}=0$ for all j), yielding $w_{lm}=0$: debit consumers bear no burden from credit fee increases. The revenue-weighted covariance $\mathrm{Cov}_R(\mu_{jl},\mu_{jm})$ measures the degree of merchant segmentation. Negative covariances mean high-credit stores serve few debit consumers, reducing redistribution.

OI.5 Implementing the Sufficient Statistic

Table **??** shows the % of welfare lost for consumers of the row payment method from a 1 pp. change in fees of the column payment. Credit card consumers are more exposed to credit card fees than the average credit share would imply, because stores where credit consumers shop have high credit transaction shares and hence prices more affected by credit fees. Both datasets show similar patterns.

The derivation calls for revenue-weighted covariances because the welfare integrand $\int_j q_{jl}\,\partial p_j/\partial\tau_m$ weights each store by consumer spending. Revenue-weighting gives large stores with many shoppers appropriate weight. Unweighted covariances treat all merchants equally, giving outsized influence to small stores with idiosyncratic payment mixes.

Sorting effects are small. Table **??** shows the ratio of welfare loss under actual sorting to the homogeneous-merchant baseline. Most ratios are close to 1. Diagonal cells (row l = column m) exceed 1, meaning sorting amplifies own-fee exposure. For example, the Credit/Credit cell in the MRI weighted panel is 1.023: credit consumers face 2.3% more welfare loss from a 1 pp. credit fee increase than under homogeneous merchants, because sorting concentrates them at high-credit-share stores. Most off-diagonal cells fall below 1, meaning sorting attenuates cross-consumer redistribution. The Debit/Credit cell in the same panel is 0.986: debit consumers face 1.4% less burden from a credit fee increase, because sorting places them at stores with lower credit shares. These signs match the negative cross-covariances in Table **??**. With weighted covariances, sorting reduces redistribution from cash to credit users by only 4.1%, our preferred estimate given the theoretical case for revenue-weighting. With unweighted covariances, the reduction reaches 8.6%, an upper bound that overweights small merchants. Thus 91–96% of baseline redistribution remains intact: no large merchant serves only one payment type, so consumers cannot avoid card fee exposure through store choice. These calculations use the full MRI and Homescan samples, so standard errors are negligible given the large sample sizes.

OI.6 Implications for the Integrated Economy Assumption

The evidence validates the main text's "integrated" market assumption: consumers of all types shop at overlapping merchants, so uniform pricing passes through the average cost of every accepted method. In a "segmented economy" where credit consumers shopped only at high-credit stores and cash consumers at low-credit stores, fee increases

Table OI.2: Percentage of welfare lost from fee increases

(a) Homescan

	Debit	Credit
Cash	0.320	0.365
Debit	0.334	0.375
Credit	0.315	0.399
Mean exp share	0.320	0.381

(b) MRI

	Debit	Credit
Cash	0.307	0.341
Debit	0.311	0.340
Credit	0.303	0.353
Mean exp share	0.307	0.345

Table OI.3: Ratio of welfare lost with actual sorting to homogeneous-merchant baseline. Rows indicate the consumer type bearing the welfare loss, and columns indicate the fee increased by 1 pp. Diagonal cells (Debit/Debit, Credit/Credit) measure own-type amplification from sorting. Off-diagonal cells measure cross-consumer redistribution attenuation. Cash has no interchange fee and therefore no diagonal entry.

(a) Homescan - Unweighted

	Debit	Credit
Cash	0.988	0.914
Debit	1.075	0.973
Credit	0.973	1.082

(b) Homescan - Weighted

	Debit	Credit
Cash	1.001	0.959
Debit	1.044	0.984
Credit	0.984	1.048

(c) MRI - Unweighted

	Debit	Credit
Cash	0.997	0.975
Debit	1.037	0.971
Credit	0.971	1.051

(d) MRI - Weighted

	Debit	Credit
Cash	1.001	0.989
Debit	1.015	0.986
Credit	0.986	1.023

would affect only credit users, eliminating redistribution. The data reject this scenario: no large merchant serves only one payment type.

Sorting reduces redistribution by only 4–9 pp., leaving 91–96% intact. This directly supports the counterfactual policy analysis. The model assumes price changes from altered fees affect all consumers at a given merchant, creating cross-consumer transfers. The 91–96% figure validates this assumption: sorting does not meaningfully segment the market, so policy-induced fee changes generate the cross-consumer redistribution that the integrated economy framework predicts. Consumers cannot avoid card acceptance costs through store choice, making uniform pricing the empirically relevant regime for welfare analysis.

Dear Professor Seim,

Thank you for granting me the opportunity to re-submit my paper "Regulating Competing Payment Networks" to the American Economic Review.

I deeply appreciate your comments on how to revise the paper in order to maximize its potential. I am also thankful to the four referees, who carefully read the previous version and provided me with a clear path for this revision. I think that the paper has improved substantially thanks to the suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.

2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss how I modified the paper and extended my analysis to address it.

I also discuss all points raised by the referees and, wherever possible, address them empirically. I have attached four separate letters with my detailed answers to each referee.

Thank you again for your comments and clear guidance in revising the manuscript.

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification. This responds to concerns by R1 about the interpretation of the merchant event studies, by R1 and R2 on the Discover evidence, and your concerns about the Durbin evidence.

2. **Modeling.** A new "Key Assumptions" discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed. This responds to your guidance on data limitations, R2's comments on fixed costs, and R4's point about short-run predictions.

3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover's rewards experiment shows some substitution at the margin. Appendix **??** estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit. This responds to concerns by R1–R3 about credit-debit substitution.

4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results ($31 bn vs. $27 bn). This responds to R1's suggestion to decompose welfare effects and investigate multi-homing, R2's suggestion for a moderate fee cap, R4's suggestion on the social

optimum, your suggestion on evaluating robustness to reward sensitivity, and your request to check whether credit constraints affect the results.

5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout. This responds to your request and that of the AE to tighten the exposition.

Detailed Response to the Editor Letter

> (a) Caveat the results in Section III.A given the sensitivity to sample selection. (b) Address Referee 4 concerns that the difference-in-difference results are unable to isolate the role of rewards and that the estimated rewards sensitivity is too high. You already note that the control group aims to purge the results of some types of contemporaneous trends, such as changes in merchant acceptance behavior. Can you provide any evidence on potential strategic responses by exempt issuers to the Amendment, such as promotion of debit cards? Alternatively, can you - as a robustness check - reduce the sensitivity to rewards in model estimation?

Reply: I address these concerns with a reframing of the reduced-form evidence around multiple steering channels, new bank-level data on the rewards channel, a note on the welfare implication of non-rewards steering, and a robustness check that halves the targeted Durbin moment.

Reframing the reduced-form evidence. Section **??** now presents the Durbin evidence as showing that interchange fees steer consumer payment choices through multiple channels. The Durbin Amendment cut large issuers' interchange revenue, and those issuers responded across several margins: cutting rewards, reducing advertising, and scaling back teller incentives. Their debit volumes fell. The main text cites qualitative industry evidence supporting all three channels.

The main text also now notes that the decline is weaker once the full sample of debit card payment volumes is included.

New data on the rewards channel. I collected bank-level data from archived bank websites on which issuers paid debit rewards before and after Durbin. Among the fifteen banks that paid debit rewards pre-Durbin, those that ended rewards saw a decline in debit volumes of about 25% relative to those who did not cut rewards, matching the baseline difference-in-difference estimate. I refer to this analysis in the main text in Section **??**.

In Section **??**, I connects this 25% to the rewards channel specifically. Banks fund debit promotion from interchange revenue, allocating across rewards, advertising, and teller incentives. If banks differ in their productivity across these levers, rewards-paying banks substituted rewards for non-price promotion and did little non-price steering to begin with. The within-subsample comparison therefore reflects mainly the rewards change. Thus the baseline estimates do not overstate the role of rewards.

Robustness check. In Section **??**, I now re-estimate the model while halving the targeted Durbin moment, so the model matches only half the observed decline in debit volumes. The estimated reward sensitivity drops by around half (the log price sensitivity coefficient declines by around 0.82). Halving the moment produces negative estimated marginal costs for Visa, which are difficult to reconcile with accounting data. The qualitative welfare conclusions for the fee cap, Durbin, CBDC, and dual-routing counterfactuals are nonetheless preserved. The sign of the merger counterfactual flips, however, as networks have more market power over consumers.

In the same section, I also emphasize that if non-rewards steering (advertising, teller incentives) imposes real costs on banks without generating direct consumer benefits, the baseline welfare estimates *understate* the true losses from high merchant fees. Lowering merchant fees would also reduce steering, which would raise total welfare.

> (a) Please summarize retailer credit card acceptance rates upfront - my sense is that this grocery store must be relatively unique in not accepting credit cards. This will help motivate modeling assumptions. (b) Clarify the interpretation of the results. I agree with the R1 that one takeaway of the grocery store analysis is that debit and credit cards substitute and that the Discover evidence alone is not sufficiently conclusive to rule out this channel fully.

Reply: I now clarify early on in Section **??** that the merchants I study are highly selected: 98% of trips in Homescan occur at stores that accept credit cards. This helps motivate the estimation strategy that treats the estimated sales effects as the gains for marginal merchants who decide to accept credit cards rather than the average merchant.

I have narrowed the interpretation of my event study results in Section **??** around two main points.

1. The event study results show that incremental acceptance of credit cards increases sales for merchants, even though most consumers with credit cards own debit cards.

2. The Discover evidence suggests that consumers do not change their consumption decision between stores in response to rewards.

I agree that the Discover evidence alone is not sufficiently conclusive to rule out debit-credit substitution. I defer the discussion of the substitution assumption to the model, where I estimate two alternative specifications that navigate the trade-off between consumer-side evidence on substitution and merchant-side evidence on acceptance and pricing differently (Appendix **??**).

As an aside, my understanding of the Nielsen guidelines is that it is not possible to display results based on data from a single entity. Please ensure that the paper passes review by the Kilts Center.

Reply: The event study contains two grocery chains. The paper has passed Kilts review.

At a high level, the paper needs to be more open about the fact that the merchant data and the current state of merchant card adoption do not allow you to model merchant behavior as flexibly as one may like. A few examples:

(a) You rely on the single, possibly not representative merchant from Section III.B to pin down σ and use this single retailer to support the empirical result that merchants are fee-insensitive (presumably, to current levels of fees!). This is more like a calibration than an estimation exercise, however, that likely plays a role in the overall model predictions.

(b) The fact that most merchants accept some type of credit card today and that you have already exploited Section-III.B grocery store's adoption behavior in estimation must mean that the data do not allow you to identify fixed costs to card acceptance separately from merchant substitution and heterogeneity. Please correct me if this is wrong. If not, you should simply state this as a shortcoming of your approach and explain why you cannot be more flexible, rather than argue that fixed adoption costs are small (which R2 does not agree with). Please note that I do not expect you to incorporate fixed acceptance costs, but please add a discussion of how such costs might affect agent behavior; R2's report contains nice conjectures on this point.

(c) You say that you abstract from non-rewards characteristics because of Australia's experience of regulating merchant fees (Section IV.F.7). I assume, however, that your data does not contain information on non-rewards characteristics of consumers' credit cards and that modeling issuer choices in dimensions other than fee structures is well beyond the scope of the paper. As R4 notes, the counterfactuals are therefore best thought of as short-run predictions, with possible regulations inducing further strategic responses by issuers. R2's suggestion of representing such concerns as an artificial quality degradation in the counterfactuals is nice, but I do not necessarily expect you to implement it.

Reply: U.S. payment markets are mature, limiting the variation available for studying merchant behavior. The grocery chains that changed acceptance policies are the only natural experiments in the data. I have revised the paper to state data limitations directly wherever they constrain the analysis, rather than defending modeling choices as first-best.

The merchant-type estimation (Section **??**) is now framed as a calibration exercise anchored by the grocery-chain event studies, with an explanation of why more flexible specifications are not feasible given the data. In Section **??**, I state that fixed costs of card acceptance cannot be identified separately from heterogeneity in sales benefits, and discuss how they might affect behavior following R2's conjectures. Following R4's concerns, the counterfactuals are now described as short-run predictions that hold non-rewards characteristics fixed. The same approach extends to other limitations: credit-debit segmentation and one-dimensional merchant heterogeneity are discussed in Section **??**, and the pass-through specification in Section **??**.

> The referees continue to ask for improvements in your treatment of credit and debit cards on the consumer side. First, R1-R3 dislike that they do not substitute. They also make several suggestions for how you might introduce substitution in the model (see R1 and R2's comments here) or clarify why exactly you make this assumption (see R3's suggestion on this point). Since your consumer-side data are better, I wondered if you could make some progress on this point, but I would also be satisfied with a more detailed motivation for the need for this assumption based on modeling challenges or data shortcomings that this might introduce.

Reply: While incorporating credit-debit substitution makes the model more realistic along some dimensions, it also generates other model predictions that are at odds with the empirical evidence. As such, the main text keeps the assumption of point-of-sale segmentation between credit and debit.

The evidence on credit-debit substitution is mixed. On one hand, antitrust testimony in *Ohio v. AmEx* and consumer surveys describe credit and debit as distinct products at the point of sale (Online Appendix **??**). Alaska Airlines, Best Buy, and Crate & Barrel describe credit and debit as "totally different products that cannot substitute for each other." On the other hand, Discover's rewards shock (Online Appendix **??**) shifts consumer spending onto Discover from both cash and debit when Discover pays higher grocery rewards, consistent with some substitution at the point of sale.

The main challenge with modeling credit-debit substitution is that it generates two equilibrium predictions that are at odds with industry commentary and data. The first is that credit acceptance would depend on credit-debit multi-homing rates, but expert witnesses in *Ohio v. AmEx* instead describe merchants disciplining credit fees by threatening to route transactions to rival credit cards rather than debit. The second is that capping debit merchant fees would lower credit merchant fees, but the Durbin Amendment cut regulated debit interchange by roughly half without moving credit interchange (Figure **??**).

The segmentation assumption in the main text (Section **??**) prioritizes matching the macro facts on merchant acceptance and pricing, at the cost of imperfect fidelity to the micro evidence on consumer substitution. Online Appendix **??** estimates two specifications that navigate the trade-off between the macro and micro facts differently. The first allows point-of-sale substitution between credit and debit and incremental debit sales relative to cash (Online Appendix **??**). This matches the micro evidence most closely. The second treats debit as generating no incremental sales relative to cash (Online Appendix **??**). This gives up some fidelity to the micro evidence, but it matches the fact that merchants do not incorporate credit-debit multi-homing into their credit card acceptance decisions. It is silent on Durbin pricing because debit fees are zero by construction.

The welfare effect of repealing the Durbin Amendment flips sign across the specifications that can evaluate it, since the assumption determines how much credit networks would cut credit merchant fees in response. Total welfare effects are similar across all three for the credit fee cap ($23–28B) and the merger ($8–15B). In all cases, dual routing reduces credit card merchant fees and rewards.

> Second, R4 notes that not all consumers are able to obtain credit cards. Translating this to the model means that your choice set is mis-specified for a segment of consumers. My conjecture was that with your current setup, such constraints would lead to an understatement of the relative value of credit cards. Please clarify whether this is correct. Are you able to perform a crude robustness check by re-estimating the model under the assumption that a segment of (lower income) consumers do not have access to credit cards?

Reply: Your conjecture is correct: ignoring credit constraints understates the relative value of credit cards. However, this does not change the estimated welfare results.

Section **??** discusses this alternative interpretation in the main text, and Online Appendix **??** re-estimates the model under the assumption that credit card access varies with income, using DCPC data on credit scores to parameterize this relationship. As expected, the median consumer's credit aversion is smaller because we no longer need to rationalize the low observed credit card usage through preferences. The income gradient in credit preference also weakens, since income-dependent access now absorbs much of the correlation between income and credit card use.

The welfare results are similar ($27 bn in the baseline and $31 bn in the credit-constrained specification). Both models are estimated to match the same Durbin Amendment moment (the observed shift in payment shares after the debit fee reduction). Constrained consumers cannot contribute to this shift, so the remaining unconstrained consumers must be more reward-sensitive to rationalize the same aggregate response. The predicted change in shares from a fee cap is therefore similar, and the policy ranking is unchanged across all counterfactuals.

> Third, R2 suggests clarifying the information environment for consumers, its interaction with the consumer merchant choice, and the role of card preferences in merchant choice.

Reply: I have clarified the information environment in Section **??**. Consumers observe merchants' acceptance decisions and prices before choosing where to shop. In Section **??**, I also clarify that consumers have complete information over which merchants accept cards when they choose which cards to adopt.

> I agree with R2 that reducing merchant fees down to the cost of capital is unlikely (point 2c and point 2a, even though I do not expect you to simulate a counterfactual where credit cards are eliminated from the market, which is interesting, but maybe not as relevant in the current environment). The referee therefore suggests a more nuanced reduction sequence that would also illustrate the key features of the model better, as requested by R4.

Reply: The main fee cap counterfactual is now a more moderate 120 bps cap, matching Canadian merchant fees. By changing the fee cap, I no longer need to extrapolate as far from the current equilibrium, which helps to address concerns from R2 about fixed costs.

Online Appendix **??** compares the moderate 120 bps cap to the cost-of-cash cap and the Ramsey planner's solution. The relationship between the cap level and welfare is concave: even moderate caps generate large benefits. The cost-of-cash cap delivers only 7.4% higher welfare gains than the moderate cap that is half-way between current fee levels and the cost of cash.

> In the spirit of better illustrating model behavior, I think R1's suggestion of decomposing the fee cap effect into different channels would be very useful.

Reply: I have adopted R1's suggestion and decompose welfare effects by sequentially allowing more merchant responses. Table **??** in Section **??** reports this decomposition for the 120 bps cap. The first row (Network Effect) holds merchant prices and acceptance fixed while networks re-optimize fees and rewards. Consumers gain from reduced credit aversion but lose from lower rewards. The second row (Price Passthrough) allows merchants to adjust retail prices, which generates large consumer gains as fee reductions flow through to lower prices. The third row (Merchant Adoption) allows merchants to adjust acceptance, with a small effect. The decomposition shows that the credit-aversion costs consumers avoid are the main source of welfare gains, while changes in merchant acceptance play a comparatively smaller role.

> Allowing for substitution between payment types may increase the role of new payment methods (R2 point 2d). Depending on your ability to adopt the above suggestions, you might be able to engage more with this or at a minimum note that demand limits the impact of new payment methods.

Reply: I have added a discussion when introducing the public debit counterfactual in Section **??** of how the substitution assumption limits the effect of new payment types. The key idea is that if consumers are unwilling to substitute between the new payment method and existing ones at the point of sale, then it is unlikely to generate downward pressure on merchant fees. While this may seem mechanical, it is consistent with the fact that low fees on debit cards have not generated any downward pressure on credit card merchant fees.

> I also like R1's suggestion to investigate the importance of consumer multihoming, which would speak to the broader two- sided markets literature. That said, if you find that this takes you too far afield from presenting a coherent story with the counterfactuals, I would also be fine with leaving this point for future work

Reply: I have added a "Dual Routing" counterfactual in Section **??** that increases the share of multi-homing consumers by 20 pp, matching policy proposals such as the Credit Card Competition Act. Consistent with the theoretical predictions of **Teh.etal2022<empty citation>**, increased multihoming leads networks to compete more intensely for merchant acceptance. Credit fees fall by 13 bp and rewards decline by 15 bp. Total welfare improves by \$3.8 billion (net of the mechanical χ^{22} adjustment described in Section **??**).

> Lastly, R4 asks about unregulated optimal behavior. My guess was that this serves as the baseline for the counterfactuals already, but please correct me if I am wrong.

Reply: The baseline is the status quo equilibrium in which debit cards are regulated but credit cards are not. Each counterfactual then perturbs this baseline. For example, “Cap Fees” imposes interchange caps, while “Uncap Debit” removes the existing Durbin constraint. I have clarified this in the counterfactuals section.

> "My main concern while reading the paper relates to the exposition. The writing is reasonably well structured and polished, but I still find it hard to read. I think this is partly because the author sometimes takes for granted that a reader knows the institutions. For example, the first page refers to “poorly designed price regulations” that imply distortions but is vague on what these regulations mean. The second page then describes the finding on the impact of the basis point reduction in debit rewards after the Durbin Amendment Act, but without much context. This Act is further referred to a few times in the introduction but only explained at a late stage. In general, the author could do a better job in presenting the paper at a higher level, for a broader audience that is not necessarily familiar with the regulations."
>
> Please take a stab at further tightening the paper’s exposition with the objective of (a) clarifying what can and cannot be done with the data at hand, and (b) focusing the paper on the essential pieces of the setting that are relevant for the modeling and the interpretation of your results. I realize this was your job market paper, and it still essentially reads as one, with a lot of detail on nuances of the payments market that are unlikely to be material to your results. You may also want to shorten the footnotes and shorten the Appendix, which contains further details that – while interesting – are not central to the substance of the paper (for example, the detailed section on price coherence). To the extent that a reader would like to engage with the Appendix material, I would recommend that you include only the most important material related to the data compilation and collection, the empirical estimation approach, and the model derivation.

Reply: I have substantially streamlined the paper. The main changes are:

1. *Introduction.* I have rewritten the introduction to be accessible to readers unfamiliar with payment market institutions. The Durbin Amendment and its institutional context are now introduced in the opening paragraphs rather than being referenced early and explained late, and the framing emphasizes standard IO intuitions (market power, competitive bottlenecks, price coherence) rather than payment-specific jargon.

2. *Appendix reorganization.* The main appendices (A–C) now cover the three categories the associate editor recommended: data compilation and collection, model derivation, and the empirical estimation approach. I have moved the detailed section on price coherence, the merchant sorting analysis, and other supplementary material to a separate Online Appendix (OA–OH).

3. *Footnotes and institutional detail.* I have shortened footnotes throughout and removed extended discussions of payment market nuances that are not material to the results.

> Lastly, Frank Verboven suggests connecting with Knittel and Stango's paper "Price Ceilings as Focal Points for Tacit Collusion" (AER 2003) on price ceilings as focal points for collusion in credit card markets. Since this relates closely to price caps, it may be useful to discuss whether institutions have changed since their analysis and whether caps may create a risk of collusion in your setting (not at all to be dealt with, but to mention as a caveat)

Reply: Knittel.Stango2003<empty citation> show that state-level non-binding interest rate ceilings served as focal points for tacit collusion in credit card markets during the 1980s. I have added a caveat to the literature review noting that the mechanism is limited in the current setting. The caps I study are binding constraints, whereas the Knittel and Stango result applies to non-binding ceilings that serve as coordination devices above the competitive price.

Dear Referee,

Thank you for carefully reading the previous version of the paper and providing me with great comments and suggestions. I think that the paper has improved substantially thanks to your suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.
2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss the paper modifications and extended analysis that address it.

Sincerely,

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification.

2. **Modeling.** A new "Key Assumptions" discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed.

3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover's rewards experiment shows some substitution at the margin. Appendix **??** estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit.

4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results ($31 bn vs. $27 bn).

5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout.

Detailed Response to Referee 1

I organize my responses as follows. I begin with the four comments assessed as "no response" or "unsatisfactory" in the first round. I then address new comments from the current round, and close with follow-up clarifications on comments assessed as mostly or somewhat satisfactory. Comments assessed as fully satisfactory are not repeated here.

Previously Unaddressed and Unsatisfactory Comments

> The assumption that a merchant enjoys a fixed boost in sales γ when selling to a consumer who uses a card has some undesirable properties. When consumers carry both a credit and a debit card (as I suspect most do in practice), then the merchant has no incentive to accept credit cards; credit cards offer no incremental sales increase over debit cards.
>
> No response: I could not locate a response to this comment in the article. I struggle to imagine that consumers could carry two credit cards without carrying a debit card, which I had thought necessarily follows from having a bank account. If credit card consumers also have a debit card, and incremental consumer spending is constant across debit and credit cards, the fact that merchants adopt high-free credit cards seems to be an artifact of the somewhat arbitrary assumption that consumers may carry only up to two cards. (I may be missing something here.)

Reply: You are correct that γ is the same for credit and debit transactions in the baseline model. Factually, as shown in Table **??**, most credit card holders also own a debit card.

The reason merchants still have an incentive to accept credit in the model is that the baseline assumes credit and debit serve different transaction types (the segmentation assumption discussed in Section **??**). Under segmentation, a consumer who prefers credit for a given transaction pays with cash, not debit, if credit is unavailable, so a debit card in the same wallet does not blunt merchants' incentive to accept credit.

The two-card representation is consistent with observed behavior — around 95% of card spending is concentrated on just two networks (Online Appendix Table **??**) — but the segmentation assumption is doing the substantive work. The concentration is also informative about preferences: a consumer placing 95% of her spending on two networks is revealing that the third-best card is a poor substitute, not an equally attractive option that the model arbitrarily excludes.

The reduced-form evidence in Section **??** supports the idea that accepting credit cards in addition to debit cards increases sales. When a grocery retailer begins accepting credit cards, whether a consumer used credit at *other* stores is strongly predictive of the sales response at the newly accepting merchant. This indicates that card ownership alone is not sufficient and that usage patterns matter, but it does not rule out some degree of substitution.

The assumption that consumers do not substitute between credit and debit at the point of sale may seem strong, but antitrust testimony in *Ohio v. AmEx* and the Durbin Amendment's failure to move credit interchange both support it. Online Appendix **??** estimates two alternatives that relax it, one allowing substitution with incremental debit sales relative to cash and another treating debit as generating no incremental sales. These alternatives generate counterfactual predictions for the drivers of merchants' credit card acceptance decisions. Nonetheless, welfare results are close to the baseline for every counterfactual other than uncapping debit fees, where the substitution channel by construction matters most.

> Relatedly, the modelling of rewards is unnatural. Consumers enjoy a boost to their income for adopting a card even if they never use the card.
>
> I think that the author may have tried to respond to this comment with the text on page 21 reading "The pecuniary utility for single- homing agents is micro-founded in the consumer's optimal consumption problem across merchants. The optimized value of log utility for a consumer with CES preferences that generate the demand curve in Equation 28 is approximately log(U)." This requires more explanation; the microfoundation is not explicitly provided and is not clear to me.

Reply: In the model, consumers always use their preferred payment method at merchants that accept it. There is no choice about whether to use the card conditional on adoption. The concern about consumers receiving rewards without using the card does not arise in equilibrium: merchants accept cards, and consumers always use their preferred payment method at accepting merchants.

From the consumer's perspective, the only difference between a per-transaction reward and a lump-sum reward of equal monetary value is that a lump-sum reward has no effect on the allocation of spending across merchants. This is consistent with the Discover reduced-form evidence (Online Appendix **??**). When Discover offers higher rewards, consumers switch payment methods but do not reallocate shopping trips across stores.

Section **??** now presents the consumer's CES utility function (Equation **??**) and derives the demand function (Equation **??**) and pecuniary utility (Equation **??**) directly. Rewards f^w enter as a lump-sum income boost: total expenditure is $y(1 + f^w)$, and log indirect utility reduces to $\log U^w = \log y + f^w - \log P^w$. By Roy's identity, whether f^w is paid out lump sum or on each transaction has no effect on the optimized value of indirect utility.

"Another concern relates to merchant pricing. In general, markups and pass-through depend on the slope and curvature of demand, respectively. CES demand has one parameter that governs both markups and pass-through."

Mostly unsatisfactory response: in Section IV.F.2, the author claims that "Appendix F.6 extends the model to allow for an arbitrary degree of pass- through." This is not true. Appendix F.6 provides analysis of a case with no pass-through. I did not see anything in Appendix F.6 about allowing an arbitrary degree of pass-through; the misleading statement in the main text is the main reason why I have labelled the response as "mostly unsatisfactory." With that said, the additions in Appendix F.6 with no pass-through are much appreciated and interesting. I would suggest that the author acknowledges the problems associated with full pass-through and explains what happens in the opposite case with no pass-through. Upon making these changes and removing the false statement from the text, I would update my assessment to "Fully satisfactory response."

Reply: I have clarified in the main text that Online Appendix **??** considers the polar opposite case of no pass-through, as you suggested. I also discuss why data limitations prevent me from directly estimating pass-through.

"Last, there is a concern in the "increased competition" counterfactual that adding a new credit card network mechanically increases credit card usage by giving consumers new logit shocks for credit cards. Given that credit card usage is welfare reducing in the model, adding a new card is thus welfare reducing before the pro-competitive effects of entry (changes in prices/fees) are considered. It would be good to take note of this concern, even if it is not addressed via a change in the counterfactual."

No response: I could not find a response to this comment in the article. Although the "increased competition" counterfactual has been removed in favour of a focus on monopolization, my original comment also applies to a comparison of a monopoly regime with the baseline regime.

Reply: Your original concern about the entry counterfactual was valid: adding a new credit card network introduces new logit shocks that mechanically increase credit card adoption. When credit cards are welfare-reducing, this mechanical effect creates a bias against entry in the counterfactual.

However, the concern does not apply to the monopoly counterfactual. Monopolization changes ownership (one firm sets fees for all networks to maximize joint profits) but does not change the set of available products nor the realized logit shocks. Any changes in adoption patterns are driven entirely by changes in equilibrium fees and rewards, not by mechanical changes to preferences.

New Comments from Current Round

> It would be interesting to decompose the channels by which the fee cap affects welfare by computing the counterfactual (i) holding fixed prices and merchant adoption, (ii) holding fixed prices, (iii) holding fixed merchant adoption, and (iv) holding fixed neither (which, of course, is what the author is currently doing). I suppose that merchants compete away most of their gains from fee caps by reducing prices following the fee cap. Given the multi-sided nature of the market, it would also be interesting to see how participation changes on the merchant side affect consumer welfare and whether this dynamic is relevant at all for the welfare results.

Reply: I now include a full decomposition in Table **??**, which sequentially allows merchant responses to fee caps (Section **??**).

Your intuition is exactly right: merchants compete away nearly all of their direct gains from lower fees. With prices and acceptance fixed (first row), merchants pocket $50Bn in fee savings, but once they re-optimize prices (second row), competition dissipates $49Bn of those savings into lower consumer prices.

The more surprising finding is *why* consumers gain $28Bn overall. Fees and rewards fall by roughly equal amounts, so the direct transfer to consumers is small. The welfare gains come from reduced credit card adoption. Under price coherence, each credit card user imposes an externality through higher retail prices, so the marginal consumer's rewards are a cross-subsidy funded by all shoppers. Fee caps shrink this cross-subsidy, eliminating real costs such as debt aversion, that rewards were compensating for.

Total welfare in the first row rises by only $14Bn because lower rewards contract consumption when retail prices are fixed. Since margins are positive, reduced consumption creates deadweight loss. Merchant price cuts in the second row reverse this contraction, so the full benefit of reduced credit aversion materializes only across both rows together. The third row (Merchant Adoption) has a negligible net welfare effect.

> The results regarding the effects of competition on platform fees in Teh et al. suggest that multihoming plays a crucial role in determining whether competition tends to raise or lower merchant fees. To my understanding, there is no empirical paper that estimates how multihoming affects the extent to which competition shifts the relative balance of consumer fees (or rewards) and merchant fees. The author has an opening to do exactly this by running the monopoly counterfactual under alternative assumptions regarding consumer multihoming. This exercise would also suggest how monopolization would affect welfare in payment card (or, more generally, platform markets) with different levels of multihoming than the US payment card market.

Reply: I have added a "Dual Routing" counterfactual that increases the share of credit card holders who multihome from approximately 60% to approximately 80%.

Tables **??** and **??** show the results. Consistent with the theoretical predictions of **Teh.etal2022<empty cit** increased multihoming leads networks to compete more intensely for transactions rather than for exclusive relationships. Credit card fees decline by 13 bp (SE 3.2) and rewards by 15 bp (SE 3.4), because networks have less incentive to offer generous rewards when targeted cardholders already hold competing cards. Despite lower rewards, consumer welfare still increases by $3.3 billion (SE 1.4), net of the mechanical χ^{22} adjustment described in Section **??**.

The exercise varies multihoming under the competitive market structure rather than under monopoly. The Credit Card Competition Act is the closest policy analogue and would increase multihoming without changing market structure.

> How was the large retailer in Section III.B chosen? Are there any other retailers in the sample period that began accepting credit cards, to provide a sense of effects beyond a single retailer and beyond the grocery category?

Reply: At the time of the event studies, nearly all large retailers already accepted credit cards, so the grocers I study are highly selected. The event study is not meant to be representative of the average retailer but rather the marginal merchant.

To identify stores, I screened store-level changes in the share of payments on credit cards over time and verified that these shifts correspond to changes in credit card acceptance reported in news articles and press releases. I excluded events involving warehouse stores because those typically involve changes in *which* credit cards are accepted (accompanied by coordinated card-switching campaigns) rather than *whether* credit cards are accepted, which would confound the effect I am trying to measure. The Kilts data use agreement requires retailer anonymity. Only two events in the sample period meet these criteria, which is a limitation of the design.

> The article uses the number of trips as the outcome when studying merchant benefits from card acceptance, claiming that revenue is unsuitable for use due to its fat tails. But the retailer considers sales effects, not foot traffic effects, when choosing whether to adopt a payment card. Wouldn't a log transform of revenue deal with the fat tails?

Reply: The main text regression now uses dollar purchases at the consumer level as the main outcome to capture both trips and spending per trip.

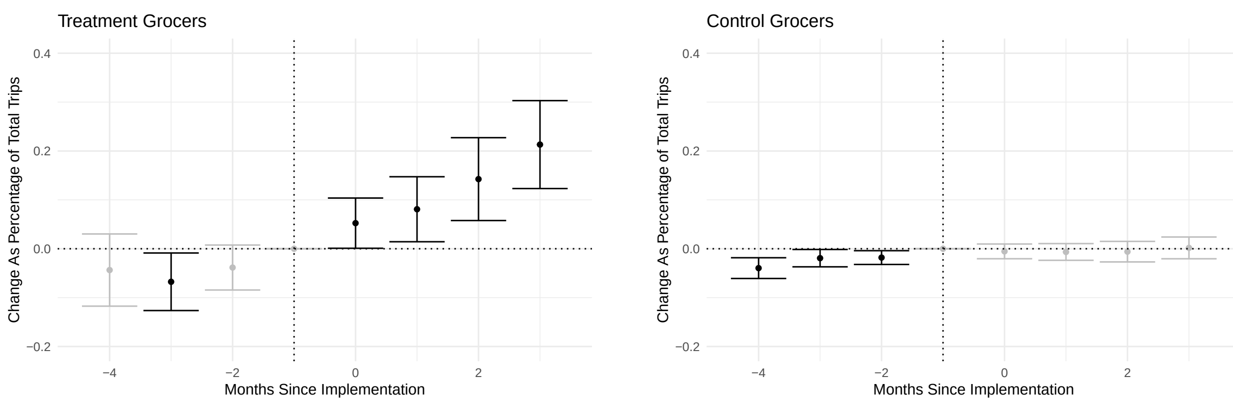
Why use a relatively exotic Poisson model as opposed to a more standard linear model in the analysis of merchant benefits from card acceptance? Are the results from a linear model qualitatively similar?

Reply: I use a Poisson specification following **Cohn.etal2022<empty citation>**, who show it is preferred for difference-in-differences designs with count-like outcomes bounded below by zero. OLS in levels estimates absolute changes, not percentage changes, and OLS in logs is not feasible because the outcome contains many zeros.

An appropriately designed linear model does yield similar estimates. Figures **??** and **??** run separate OLS difference-in-differences for the focal and control grocers. Subtracting the percentage estimates (using pre-period means) yields approximately the same result as the Poisson triple-difference. Poisson is more convenient because it directly estimates percentage effects and scales to multiple events without the two-step conversion.

Figure A.1: OLS difference-in-differences estimates for credit card acceptance effects

(a) Focal grocer (accepted credit cards) **(b)** Control grocers (always accepted)



Notes: OLS-style event study coefficients for credit card users relative to other consumers. The left panel shows results for the focal grocer that began accepting credit cards, and the right panel shows results for control grocers that always accepted credit cards. Subtracting these percentage estimates yields an effect comparable to the Poisson triple-difference estimate in the main text.

Why isn't the credit card share at the large grocery equal to zero before it began accepting credits cards (Figure 4a)? If there is measurement error in the data, how does this affect the author's estimation procedure?

Reply: I infer acceptance from large shifts in credit card payment shares and classify a merchant as accepting credit cards only if more than 5% of its sales are paid by credit. The nonzero credit share before the acceptance change reflects misreporting: consumers sometimes confuse signature debit cards with credit cards when responding to surveys (**Rysman.etal2025**). As shown in Appendix Table **??**, self-reported payment shares in HomeScan align well with aggregate payment volumes, so this individual-level noise does not bias the estimation of the payment patterns the model targets. I have also added a direct cross-dataset check of the multi-homing statistics in Appendix Table **??**: the usage-based Homescan multi-homing rate (56.7%) matches the ownership-based rate from the Diary of Consumer Payment Choice survey (57.7%) to within one percentage point over 2017–2019, so the Homescan single-homing share is not driven by usage-based misclassification.

> The first sentence of Section III.B.2, that "Credit card users shop more at the grocer because consumers value the flexibility of paying with credit — not because of rewards" is imprecise and not entirely clear to me. The author showed that consumers are sensitive to rewards, which are thus likely a large motivator of credit card usage. Thus, the flexibility of paying with credit is likely valued by consumers because credit card usage entails rewards. The author also states that the flexibility of paying with credit may be valued by consumers because consumers like to make large purchases with credit, but this begs the question. There must be some reason why consumers like to make large purchases with credit (perhaps because they dislike using credit, as the author argues, but the rewards accumulated on large purchases are especially high). The conclusion that I draw from Figure 5 is that credit cards are highly substitutable with each other but weakly substitutable with other payment methods.
>
> The author concludes from Figure 5(a) that higher Discover rewards do not affect consumer choices about which stores to visit. I have two comments on this conclusion:
>
> The author focuses only on grocery stores. Perhaps consumers' food shopping is not impacted by rewards given that food is a necessity, but other areas of spending would be affected. Can the author repeat the analysis for other categories?
>
> I am not even sure that the conclusion is correct. It looks like there are upticks in the black curve during the periods in which Discover offers higher rewards for groceries. Then again, the dashed line also seems to increase. It would be helpful for the author to estimate the differential impact of higher Discover rewards on Discover users versus non-Discover users using a fixed effects regression; this would allow the author to conduct inference on the impact of higher rewards on the number of trips.

Reply: The Discover evidence shows that rewards affect payment method choice but not store choice. Consumers switch to Discover to earn rewards but do not shift spending toward grocery stores. You are right that this does not "rule in" a particular mechanism for why credit card acceptance increases sales.

I have added a fixed effects regression in Online Appendix **??** with household and time fixed effects, clustering standard errors at the household level. The key coefficient on "Discover HH $\times$ Grocery Reward Month" is small and economically negligible for the grocery share of total household spending (0.002, SE 0.001), but the Discover share of grocery spending increases by 7.6 pp (SE 0.003). Consumers shift payment methods *within* grocery stores toward Discover but do not change *where* they shop.

It is not possible to study other categories. A credible design requires cases where Discover pays rewards at some stores but not at adequate substitutes. Grocery stores are a target of the rewards program, and **Ellickson.etal2020<empty citation>** show that grocers and warehouse stores are substitutes for food purchases. No other category offers this structure.

> In Figure 6, a line labelled "OptBlue" appears, but "OptBlue" is never discussed in the main text (although it is summarized in a figure note). I fail to see the point of including this second line as it does not influence the author's argument in Section III.C.1.

Reply: The OptBlue line serves two important roles. First, it explains why the model treats AmEx acceptance as comparable to Visa acceptance, despite the historical impression that AmEx is accepted less often. The OptBlue program cut AmEx merchant fees for small businesses starting in 2015, and Figure **??** shows that the Visa-AmEx acceptance gap closed shortly after. By the 2019 calibration year, nearly all merchants accept either all credit cards or none. Second, I use the OptBlue episode as an out-of-sample validation of the model's merchant fee sensitivity estimates in Section **??**. The model's predicted acceptance response to the fee cut matches the observed convergence. I have clarified this dual purpose in the text.

> What does the author mean "Since every Visa/MC merchant also accepts AmEx"? From experience, many merchants refuse AmEx but accept Visa and MC. Also, doesn't this statement clash with the author's preceding statement that merchant adoption strategies tend to follow a hierarchical pattern? Perhaps the author meant to say that "every AmEx merchant also accepts Visa/MC"? Even if so, it seems hard to believe that there is not a miniscule share of merchants that accept AmEx but not Visa/MC.

Reply: You are correct that historically many merchants refused AmEx but accepted Visa and MC. However, as Figure **??** shows, this gap has largely closed between 2016 and 2019. The number of merchants accepting AmEx now approximately equals the number accepting Visa. I have revised the text to clarify that the statement refers to the current state of the market, not historical patterns, and to note that this represents a change from the past.

> Figure 6 does not directly tell us about the extent of merchant multihoming. What we would need to assess this extent are additional lines showing the number of merchants on various combinations of the networks (Visa + MC, Visa + Amex, etc.).

Reply: I agree that detailed data on every possible acceptance subset would be ideal, and that Figure **??** alone is not sufficient to fully assess merchant multihoming.

However, unlike platform markets such as food delivery, where merchants routinely accept some platforms but not others, the card payment industry does not exhibit this pattern. Merchants that accept credit cards overwhelmingly accept all major networks, and very few accept Visa but not MC or vice versa. Precisely because combinatorial acceptance is not a feature of this industry, no data source systematically records the acceptance subsets of individual merchants.

The best available evidence comes from Yelp reviews in Online Appendix **??**. Using approximately 3,000 reviews that mention at least two payment methods, I find that merchants progressively add payment methods (cash $\rightarrow$ debit $\rightarrow$ Visa/MC $\rightarrow$ AmEx) in a hierarchical pattern rather than specializing in particular networks. Combined with Figure **??** showing that the number of AmEx-accepting merchants now approximately equals the number accepting Visa, the data support a world with limited heterogeneity in acceptance bundles.

> It does not immediately follow that large effects of card adoption on sales mean that merchants should be insensitive to fees. As the author notes in Section III.B.3, whether the sales increase from card adoption makes adoption worthwhile to a merchant depends on the merchant's margins (as well as the merchant's fixed costs of adoption). If, e.g., there was a large mass of merchants with margins around 20% and no fixed costs of adoption, then small changes in fees would change the adoption decision of many merchants. The elasticity of adoption with respect to fees similarly depends on the distribution of fixed costs of card adoption.

Reply: I agree. The sales increase from card acceptance is not sufficient to pin down the merchant fee elasticity. The elasticity depends on the distribution of merchant margins and fixed costs of adoption, which I do not directly observe. I have revised the text to be more transparent about this limitation and to clarify that the merchant fee sensitivity is inferred from the model's equilibrium conditions and validated against the OptBlue episode, rather than being directly estimated from the reduced form evidence. In the model, I assume all merchants have the same profit margin through one CES elasticity σ.

> Is anything lost by assuming that card users never prefer to pay with cash? As the author notes, the consumer's random selection to use either the primary or secondary card could arise from a model with preference shocks at the level of a transaction and payment method. It would seem natural for there to be some shock for using cash as well. Perhaps the author may overstate the benefits to consumers of payment cards by ruling out the possibility that sometimes consumers would prefer to use cash?

Reply: Yes, the model overstates the welfare gap between pure card users and pure cash users by polarizing consumers into extreme types. A richer model with transaction-level cash preference shocks would place consumers on a continuum. In the HomeScan data, roughly 25% of purchases are made with cash regardless of how many cards consumers carry, consistent with such shocks.

The aggregate welfare results are robust to this concern. Several offsetting forces preserve their direction. Credit aversion per card holder is overestimated: card holders use cards less often than modeled, so they bear credit aversion costs less frequently. But the number of card holders is underestimated, because the richer model needs more card holders to match observed spending shares. The rewards loss per card holder from a fee cap is similarly overstated, but this overstatement is spread across a larger number of card holders. These forces partially offset, so the main conclusions (that fee caps benefit consumers on net, with progressive distributional effects) are directionally robust.

> Page 18 reads "Second, consistent with the evidence in Section III.B.1, rewards do not affect relative consumption choices across merchants." But it seems that Section III.B.1 ("The Effects of Accepting Credit Cards") is about how credit card acceptance by a single large merchant affected sales at the merchant, not about rewards or a comparison of consumption across merchants. Perhaps the author meant to refer to Section III.B.2.

Reply: I have corrected the cross-reference to refer to the Section on the Discover evidence.

> Section V.C.1 states that "my estimated retail margin of 15.6 percent is also similar to the aggregate markups of 15–20% used in macro studies of misallocation." However, Section III.B.3 reads "credit card acceptance is profitable as long as margins exceed 20%. Census statistics indicate gross margins in the grocery industry were around 27% during this period." The author should fix the consistency between these parts of the paper, as the earlier argument that merchant should be insensitive to fees seems to rely on margins exceeding 20% (although, as discussed, this argument is flawed). Also, the author should also be consistent in writing out "percent" or using the "%" sign.

Reply: I have removed the back-of-the-envelope margin calculation from the reduced-form section. You are correct that the earlier argument was both inconsistent and relied on strong assumptions on the distribution of merchant margins. The reduced-form evidence is not sufficient to pin down the fee elasticity, so I have removed the attempt to do so from that part of the paper.

In the estimation section, I now clarify that the margin is estimated structurally: the model recovers a retail margin of 17.8 (SE 5), which is close to the 13–17% margin range implied by the 15–20% markups cited in macro studies of misallocation. I also clarify the identification logic: because credit and debit are segmented, consumers who would use credit instead pay cash when credit is not accepted, and debit transactions are unaffected. The marginal merchant therefore weighs credit fees against sales that would otherwise be lost to cash, and it is this credit-versus-cash gap that pins down the retail margin. I have also standardized notation to use % consistently throughout.

> The claim that "Capping fees at the cost of cash also simulates the effects of merchants freely surcharging consumers for the cost of card acceptance" in Section VI.A requires greater explanation.

Reply: As shown by **Zenger2011<empty citation>**, capping merchant fees at the cost of cash is theoretically equivalent to allowing merchants to freely surcharge consumers for the cost of card acceptance. However, the revised draft no longer emphasizes this equivalence in the main text. The primary counterfactual now caps credit fees at 120 bp, an intermediate cap that keeps the analysis close to observed fee levels. Because this cap exceeds the cost of cash, the surcharging equivalence does not apply directly. For completeness, Online Appendix **??** compares the intermediate cap to the cost-of-cash benchmark, where the surcharging equivalence holds: perfect surcharging makes the allocation of fees between merchants and consumers irrelevant, so the two equilibria coincide only at that specific cap level.

> A more appropriate name for Section VI.C would be "Reducing Competition Between Private Networks."

Reply: I have revised the section title to "Welfare Effects of Reducing Network Competition," which more precisely conveys the economic content.

Remaining Follow-up Points from Original Report

The following points were assessed as satisfactory, but I address some of the ones that I think require further clarification.

> For information on payment method choice, the article uses the Atlanta Federal Reserve's Diary of Consumer Payment Choice and the Survey of Consumer Payment Choice. How were the data combined? Do they include the same consumers, or did the author pool together data from two distinct sources?

Reply: There is complete overlap between the diary and survey respondents within each wave: the same individuals complete both the diary and the survey in a given year. I have clarified this in Section **??** by stating that "the diary and survey share a common panel of respondents" rather than the ambiguous "common set of identifiers."

> There is little said in the data section about the availability of data on merchants. It is unclear how the author can credibly estimate a model of merchant card acceptance without direct data on merchant acceptance.

Reply: The Diary of Consumer Payment Choice measures merchant acceptance transaction by transaction. For each purchase, the respondent records which payment methods the merchant would have accepted. Because the diary asks respondents to log every transaction over a three-day window in real time, this is not a retrospective recall question but a record made at the point of sale. The resulting acceptance rates are sales-weighted averages across the merchants each consumer visits. The model requires exactly this object: the probability that a randomly drawn transaction occurs at a merchant accepting a given network. A new footnote in Section **??** acknowledges that the measure is imperfect. It excludes merchants the consumer did not visit and relies on the consumer correctly observing acceptance. As a cross-check, Online Appendix Table **??** compares DCPC acceptance rates to Yelp business attributes by sector. The two sources agree closely in aggregate (95.8% vs. 96.3%), which suggests that consumer reporting in the diary is reasonably accurate. Transaction-level merchant acceptance data are rare, and the diary's real-time design together with the Yelp validation mitigate the most serious recall concerns.

> The author has revised the procedure for determining first and second cards as detailed by Appendix A.3.1 in light of the comment. However, this appendix is missing details. Several parts are incomplete (the first sentence of the appendix is incomplete; the final sentence of the "Cash-only consumers" section is incomplete; the section describes various cut-offs without mentioning their numerical values).

Reply: I have revised Appendix **??** to fix the incomplete sentences and include all numerical cutoff values used in the classification procedure.

> The author states "I hold fixed merchant actions because my difference-in-difference estimates remove any equilibrium effects of merchant actions." Some additional explanation would be helpful.

Reply: The intuition is as follows. The difference-in-differences design compares debit card volumes at regulated versus exempt issuers. Both groups of issuers' cardholders face the same merchant acceptance environment. Any changes in merchant behavior after Durbin affect all consumers equally and are differenced out by the control group. The model-simulated Durbin moment therefore holds merchant acceptance fixed when predicting how a reduction in debit rewards would change debit card volumes, because the empirical target has already purged this channel. I have expanded this explanation in Section **??**.

Dear Referee,

Thank you for carefully reading the previous version of the paper and providing me with great comments and suggestions. I think that the paper has improved substantially thanks to your suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.
2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss the paper modifications and extended analysis that address it.

Sincerely,

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification.

2. **Modeling.** A new "Key Assumptions" discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed.

3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover's rewards experiment shows some substitution at the margin. Appendix **??** estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit.

4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results ($31 bn vs. $27 bn).

5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout.

Detailed Response to Referee 2

For clarity, I group my responses thematically rather than following the original numbering. I first address merchant-side concerns (points 1a and 2c on acceptance costs, and point 5 on merchant competition), then consumer-side modeling (points 1b and 6 on heterogeneity, multihoming, information sets, and debit-credit substitution including the Discover evidence), then market structure (point 1c on three- vs. four-party systems and points 2a–2b on counterfactual robustness), then data and measurement (points 2d, 3, and 4). Additional sub-points from your report are collected at the end.

> Merchant acceptance costs: Unfortunately, not much has been done for the merchant side of the model. While equipment costs may be low, total acceptance costs matter. These can include installation fees, POS system integration, training, chargeback risks, and long-term contracts. Evidence suggests fixed costs are important, especially for small retailers. The paper should model or at least discuss these costs explicitly, as they can change equilibrium predictions dramatically.

> Merchant acceptance costs and equilibrium fragility: With fixed costs, even minor changes could cause disintermediation. The paper should report a sequence of equilibria as fees are gradually reduced toward the cost of cash.

Reply: Fixed costs of card acceptance cannot be identified separately from heterogeneity in sales benefits (γ) without exogenous variation in consumer card adoption that shifts merchant acceptance. This is a fundamental limitation of the approach that I now acknowledge explicitly.

Your concern about equilibrium fragility—that lower rewards could reduce consumer adoption enough to trigger cascading merchant disintermediation—directly shaped the choice of counterfactual. Rather than cap fees at the cost of cash, the main counterfactual now caps credit fees at 120 bp, roughly in line with Canadian merchant fees (**Innovation2023**). This keeps the analysis in a range where acceptance cascades have not materialized in practice, so the predictions do not depend on extrapolating merchant behavior far from current equilibrium conditions.

In response to your suggestion to report a sequence of equilibria, Online Appendix **??** compares the moderate cap to more aggressive caps, including the cost-of-cash benchmark and the Ramsey planner's solution. The relationship between cap level and welfare is concave: the 120 bps cap captures most of the planner's welfare gains, and the cost-of-cash cap delivers only about 7% more. This concavity is reassuring from the perspective of your cascade concern, because it means the welfare conclusions do not depend on pushing fees into territory where disintermediation effects would dominate.

> How does merchant competition in acceptance work if the benefit of accepting is the same for the first and last adopter?

Reply: Merchants differ in γ, which determines how much card acceptance increases sales to card-holders. A merchant accepts a card when the sales benefit from card-paying consumers exceeds the merchant fee, and this benefit is proportional to γ (Equation **??**). The "first" adopter is the highest-γ merchant, for whom the sales gain far exceeds the fee; the "last" is the marginal merchant whose gain just covers it. There is an equilibrium in pure strategies where merchants above the threshold accept and those below stay cash-only, so competition works through the extensive margin of merchant acceptance rather than through identical merchants racing to adopt.

> Consumer utility function and shopping choice: Why does mean unobserved utility of a card not vary with consumer type? Allowing heterogeneity only along monetary variables may bias results. Higher-income consumers likely have different bundles, and thus different unobserved utility. This could drive the finding that reward sensitivity rises with income. The model should allow $\Xi_{\omega i}$ to vary by income.

> Why doesn't χ vary with income or reward sensitivity? High-income and high-sensitivity consumers should gain more from multihoming.

Reply: I address these two comments together. The model allows unobserved utility to vary with consumer income through two channels. First, the random coefficient on card characteristics, $\beta_i \sim N\left(\beta_y \cdot \log \tilde{y}, \Sigma\right)$, means higher-income consumers have systematically different preferences for card characteristics. This allows the model to match average income differences between credit, debit, and cash users. What the model does not capture is whether high-income consumers have systematically different preferences across specific networks like Visa, MC, and AmEx, because the most representative data with both income and payment choice (the Diary of Consumer Payment Choice) has limited AmEx coverage.

Second, in response to your concern about income-dependent multihoming, I have added income-dependent complementarity parameters $\chi_i^{lm} = \chi^{lm} + \chi_y^{lm} \log \tilde{y}$. This allows higher-income consumers to derive different value from holding multiple cards. This is the dimension of income heterogeneity that matters most for the welfare analysis. Multihoming patterns by income determine the competitive pressure merchants face and thus the equilibrium fees networks can charge. Allowing Ξ to also vary with income on top of β_y and χ_y would be difficult to identify, because the data are not rich enough to distinguish income differences in baseline network preferences from the variation these other channels already capture.

> Information sets: The paper needs more discussion of what consumers know about merchant prices and acceptance decisions. If consumers observe both, why don't they sort perfectly across merchants? In the current model, sales benefits are fixed and independent of how many merchants accept. In reality, business stealing should reduce benefits as more merchants accept cards.

Reply: The model assumes consumers have complete information about merchant prices and acceptance decisions. This is now stated in Section **??**: "Consumers observe merchants' acceptance decisions and prices before choosing where to shop."

Consumers do not sort perfectly across merchants because of CES preferences, not information frictions. All consumers carry cash. Payment acceptance only changes the intensive margin of where consumers shop. Consumers value variety across merchants, so even with full information they spread their spending rather than concentrating it at a single store.

On business stealing: the model does capture this. The sales benefit to an individual merchant from accepting cards depends on how many other merchants accept. When few merchants accept cards, a merchant that accepts captures a large share of card users' spending. As more merchants accept, card users spread their spending across more stores, reducing the benefit to any individual merchant. This shows up through the price index P^w, which depends on other merchants' acceptance decisions.

> Debit-credit substitution: It seems unrealistic that consumers never substitute between debit and credit cards at the POS. Both are electronic, and often closer substitutes than either is with cash. Figure 4 suggests substitution from cash/debit to credit. The model should capture this, or at least acknowledge it.

Reply: This is a limitation. While incorporating credit-debit substitution would make the model more realistic along some dimensions, it also generates predictions at odds with the empirical evidence, so the main text keeps the assumption of point-of-sale segmentation.

The evidence on credit-debit substitution is mixed. On one hand, antitrust testimony in *Ohio v. AmEx* and consumer surveys describe credit and debit as distinct products at the point of sale (Online Appendix **??**). On the other hand, Discover's rewards shock (Online Appendix **??**) shifts consumer spending onto Discover from both cash and debit, consistent with some substitution at the point of sale.

Allowing substitution generates two equilibrium predictions (Online Appendix **??**) at odds with industry commentary and data. The first is that credit acceptance would depend on credit-debit multi-homing rates, contradicted by *Ohio v. AmEx* testimony in which merchants discipline credit fees by threatening to route transactions to rival credit cards rather than debit. The second is that capping debit fees would lower credit fees, contradicted by Durbin's null on credit interchange (Figure **??**).

Segmentation prioritizes the macro facts on merchant acceptance and pricing at the cost of imperfect fidelity to the micro evidence on consumer substitution. Online Appendix **??** estimates two alternative specifications that navigate this trade-off differently. One allows point-of-sale substitution and incremental debit sales relative to cash (Online Appendix **??**) and matches the micro evidence most closely. The other treats debit as generating no incremental sales relative to cash (Online Appendix **??**) and is silent on Durbin pricing because debit fees are zero by construction. Total welfare effects are similar across all three for the credit fee cap ($23–28B) and the merger ($8–15B).

> I do not understand footnote 12, which suggests that short-term transaction-specific incentives do not affect consumer usage decisions of credit versus debit cards, while rewards on all debit transactions do affect the adoption choice. Does Figure 5 apply only to consumers who can potentially substitute between Discover and debit (i.e., face the choice between them and cash) or is it an aggregate pattern across all consumers? Is it possible that the pattern on the figure is generated by two groups of consumers? Consumers in one group do not have any credit cards to substitute away from, while consumers in the second group have another credit card in addition to Discover making it easy to switch without using their debit cards. This question can probably be addressed by reporting the probability of observing consumers with a debit card and Discover credit adoption combination only (i.e., without other cards from Visa or MC). If this group is relatively small or cannot be isolated for the purpose of the exercise, the evidence cannot be used in support of no substitution between debit and credit cards.

Reply: The role of the Discover event study in Online Appendix **??** is now narrower than in the previous draft. Its primary role is to show that consumers do not change where they shop in response to rewards, and that the null store-choice result is not driven by inattention: Table **??** confirms that consumers actively shift spending toward Discover during reward quarters (a 7.6 pp. increase) while their grocery spending share is flat.

To address your concern about consumer heterogeneity: the table below repeats the analysis for households whose primary or secondary card is Discover and who hold no other credit card among these top two cards, so they cannot easily substitute to another credit card. These households may own additional credit cards that they use less frequently. "Discover only" means Discover is the sole credit card among the primary and secondary cards recorded in the data. The grocery spending share is again flat. The Discover spending share at grocery rises by only 2.5 pp ($p < 0.001$), compared to 7.6 pp in the full sample, consistent with debit being a much weaker substitute for credit than other credit cards are.

Effect of Discover cashback rewards on payment composition: Discover-only credit card holders

	Grocery	Cash	Debit	Other Credit	Discover
Discover HH × Grocery × Reward Month		-0.005+	-0.017***	0.002	0.025***
		(0.003)	(0.004)	(0.002)	(0.003)
Discover HH × Reward Month	-0.004+				
	(0.002)				
Observations	2741191	4586461	4586461	4586461	4586461
R^2	0.595	0.803	0.884	0.893	0.909
HH × Grocery FE		Yes	Yes	Yes	Yes
HH × Time FE		Yes	Yes	Yes	Yes
Grocery × Time FE		Yes	Yes	Yes	Yes
Household FE	Yes				
Time FE	Yes				
Clustered SE	HH	HH	HH	HH	HH

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Same specification as Table **??**. Treatment: households whose only credit card among their primary and secondary cards is Discover (they may hold additional, less-used credit cards). Control: households with exactly one non-Discover credit card among their primary and secondary cards.

Also, would it be possible to enforce debit card adoption if a consumer chooses to adopt a credit card such that the choice set contains options of being (1) cash-only

> user, (2) cash and debit user, (3) debit and one credit card user, and (4) debit and two credit cards user?

Reply: Your suggested choice set would eliminate consumers who use credit cards and cash but do not use debit cards. While most credit card holders technically own a debit card, the model captures usage patterns rather than ownership. The NielsenIQ Homescan data shows a meaningful share of consumers who use credit at virtually all stores but never use debit, even though they presumably have a bank account. Section **??** ("Two-Card Restriction" paragraph) explains the data motivation: consumers put around 95% of their card spending on just two networks (Online Appendix Table **??**), so the two-card model closely matches observed behavior.

> Three- vs. four-party systems: Assuming away issuer/acquirer markets is not innocuous. AmEx may be more efficient as a three-party system. With regulation, Visa/MC could exit. The author should compute welfare if only AmEx survives after regulation.

Reply: Vertical relationships between networks, issuers, and acquirers are an important feature of payment markets. My model treats networks, issuers, and acquirers as a single entity that maximizes joint profits. As I now discuss in Section **??**, this is motivated by the long-term contracting arrangements that tie these parties together in the U.S. market and by the DOJ's 2024 antitrust suit against Visa, which alleges that the network deploys payments to issuers and potential rivals to coordinate the chain. Double marginalization and other vertical frictions between the three parties could still move fees and rewards quantitatively; a fully separated vertical model is an important direction for future work.

I do not compute the AmEx-only scenario directly because eliminating Visa and MC changes the number of logit draws consumers face, creating a variety-channel welfare effect that confounds the competitive mechanism. The merger counterfactual isolates competition cleanly: it holds the set of payment technologies fixed and changes only the competitive structure among networks. The Australian experience also speaks against exit: Australia imposed interchange fee caps on four-party networks in 2003, and neither Visa nor MC left the market, suggesting fee regulation reduces rents without pushing four-party networks out.

> Counterfactual: What if credit cards are eliminated entirely? Is welfare higher than in the factual equilibrium? Edelman & Wright (2015) suggest consumers may

be better off without intermediation. The author should explore this empirically, or at least trace equilibria as fees fall until cards vanish.

Reply: My model estimates imply that consumers are worse off by 13 billion dollars relative to an economy without card payments, consistent with **Edelman.Wright2015<empty citation>**. But this number should not be taken literally: the random coefficients specification does not capture inframarginal consumers who place very high value on card features, so the welfare loss from full elimination depends heavily on modeling assumptions.

Rather than tracing equilibria all the way to zero, I compare the 120 bps cap to the cost-of-cash cap and the constrained social optimum. Online Appendix **??** shows that the cost-of-cash cap delivers only about 7% more welfare than the 120 bps cap, and the 120 bps cap captures roughly 77% of the Ramsey planner's welfare gains. The conclusions therefore do not require extrapolating to a world without cards.

Unobserved quality: Why should card quality remain fixed under regulation? Networks may cut services, degrade quality, or withdraw products. The author could compute how much deterioration would push usage near zero.

Reply: The model represents a short-run equilibrium in which unobserved card characteristics are held fixed. It is entirely fair that networks could adjust quality in response to regulation.

In practice, the Australian experience suggests that non-rewards annual fees did not rise after the fee caps, and the spread between credit card rates and unsecured term loans was unchanged (Appendix Figure **??**). However, I cannot rule out that networks adjusted non-price characteristics that I do not measure, such as customer service, fraud protection, or card benefits. I have added language in Section **??** acknowledging this limitation and noting that the counterfactuals are best interpreted as short-run predictions.

Endogenizing product characteristics is a hard problem in its own right. **Wollmann2018<empty citatio** studies endogenous product entry in the truck market, and even in that simpler setting it requires substantial additional modeling and data. In the payment network context, it would require specifying how networks allocate resources across fraud protection, customer service, rewards structures, and other dimensions, which is beyond the scope of this paper.

New payment methods: If debit and credit are poor substitutes, a new method may not help. It would be good to discuss how such an entrant could gain traction.

Reply: You are correct. If debit and credit cards are poor substitutes, then a new payment method that resembles debit (such as many proposed public payment platforms) is unlikely to displace credit cards simply by offering lower fees. I clarify this in Section **??**. This is consistent with the limited success of lower-fee payment alternatives in markets where credit cards are well-established.

> Why is cash assumed cheapest? With contactless features, cards may be cheaper. The reliance on signature card data could bias unobserved quality estimates.

Reply: The model does not infer costs from the physical qualities of cards. Network costs are backed out from the networks' first-order conditions, and the cost of cash is taken from **Felt.etal2020<empty citation>**, who measure resource costs of payment instruments in both Canada and the U.S.; I use their U.S. estimate.

> The paper should clarify the years covered by each data source to ensure consistency.

Reply: The estimation appendix now states the years covered by each data source inline. Table **??** maps each data source to the parameters it identifies and lists additional sources used for validation.

> In general, I would expect consumer price sensitivity to decline as income increases. Higher responsiveness of high-income consumers to rewards does not seem to require higher absolute price sensitivity if high income consumers spend substantially more on their cards. Due to the high total value of all transactions, the amount of pecuniary rewards is naturally higher for richer individuals. Hence, even with low price/reward sensitivity high income consumers should be gaining more from adopting and using cards than their low income counterparts.

Reply: Your point is correct that higher-income consumers earn more dollar rewards because they spend more, but α_i is sensitivity to the reward *rate*, not to dollar rewards. Pecuniary utility (Equation **??**) depends on the reward rate f^j—a consumer who spends \$10,000 and one who spends \$1,000 face the same f^j. The income elasticity $\hat{\alpha}_y = 0.20$ (SE 0.06) is therefore identified from how wallet choices vary with income conditional on the reward rate, not from the mechanical correlation between income and total dollar rewards. This is consistent with evidence that higher-income consumers are more attentive to financial product terms (**Hastings.etal2017**), even though they may be less sensitive to retail prices (**Sangani2024**).

> Figure 4, panel (b) "Effects on the number of trips" only illustrates the change in trips by credit card consumers. What happens to the number of trips by cash and debit card consumers? Did they decline by the same amount? This evidence is important in understanding where the increase in sales is coming from.

Reply: The triple-difference design conditions on consumers who use credit cards at other stores (treatment) versus those who do not (control), where the control group includes both cash-only and debit-only consumers. The coefficient therefore already measures the *differential* change in trips for credit card users relative to non-credit-card users at the same store, netting out any common trends, including changes in store-level traffic driven by concurrent pricing or demographic shifts.

For the structural model, the key moment is how credit card users respond to acceptance, which is what the triple-difference identifies regardless of whether cash and debit trips moved in parallel. Decomposing the raw trends by payment type would not be informative here, since those trends confound the store-level demand shocks that motivated the triple-difference design in the first place.

> Differences between 3- and 4-party systems: the exercise of computing welfare if only AmEx survives after regulation could also illustrate the role of competition in rewards between card providers for the level of excessive card adoption.

Reply: Eliminating Visa and MC changes the number of logit shocks consumers draw over payment options, which affects welfare through a pure variety channel that is separate from the competitive mechanism the referee highlights. The merger counterfactual avoids this issue: it holds fixed the set of available payment technologies and changes only the competitive structure among networks, so the welfare comparison isolates the role of competition cleanly.

> The author should be able to trace a sequence of equilibria for different levels of merchant fees starting from the factual level all the way until credit cards are no longer used in equilibrium. This potentially may require a reversal of the fees, with consumers instead of receiving rewards having to pay merchants for the opportunity to use credit cards.

Reply: Appendix **??** traces exactly this kind of sequence: I compare equilibria under the 120 bps cap, the tourist test cap (where merchant fees equal the cost of cash), and the Ramsey planner's solution. As you anticipated, at sufficiently low fee caps the networks do charge consumers negative rewards — that is, consumers pay for the use of cards. The welfare relationship is concave: most gains come from the initial fee reductions, and tighter caps yield diminishing returns.

> Using data from only signature cards, which are less convenient and costlier to use than contactless ones, may lead to incorrect estimates of unobserved characteristics. If the data distinguishes between contactless and signature cards, it would be nice to see different estimates. Otherwise, the author may want to allow for some time variation in the unobserved product characteristics.

Reply: The data do not distinguish between contactless and signature transactions, only between debit and credit. The unobserved characteristics Ξ are estimated to rationalize observed market shares in a 2019 cross-section, so they absorb whatever mix of contactless and signature use prevailed at that time. Since the goal is to model the market structure as it stood in 2019, time variation in Ξ is not relevant to the exercise.

Dear Referee,

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This letter has two sections:

1. The first section summarizes the key changes to the manuscript.
2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss the paper modifications and extended analysis that address it.

Sincerely,

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1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification.

2. **Modeling.** A new "Key Assumptions" discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed.

3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover's rewards experiment shows some substitution at the margin. Appendix **??** estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit.

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5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout.

Detailed Response to Referee 3

The assumption that consumers who hold both debit and credit cards never substitute between them is unintuitive. Even with the π probability formulation, it is unclear why this assumption is not important for results. Why is γ not enough to drive merchant adoption when debit is available as a backup? Please expand the discussion of consumer substitution (top of page 17).

Reply: γ is enough to drive merchant credit acceptance under segmentation. The challenge is that allowing credit-debit substitution generates equilibrium predictions at odds with the empirical evidence, so the main text keeps the assumption of point-of-sale segmentation.

The evidence on credit-debit substitution is mixed. On one hand, antitrust testimony in *Ohio v. AmEx* and consumer surveys describe credit and debit as distinct products at the point of sale (Online Appendix **??**). On the other hand, Discover's rewards shock (Online Appendix **??**) shifts consumer spending onto Discover from both cash and debit, consistent with some substitution at the point of sale.

Allowing substitution generates two equilibrium predictions at odds with industry commentary and data. The first is that credit acceptance would depend on credit-debit multi-homing rates (Online Appendix **??**, Equation **??**). Expert witnesses in *Ohio v. AmEx* instead describe merchants disciplining credit fees by threatening to route transactions to rival credit cards rather than debit, and networks report pricing credit against credit while ignoring debit. The second is that capping debit fees would lower credit fees, but the Durbin Amendment cut regulated debit interchange by roughly half without moving credit interchange (Figure **??**).

Online Appendix **??** estimates two alternative specifications that navigate this trade-off differently. One allows point-of-sale substitution and incremental debit sales relative to cash (Online Appendix **??**) and matches the consumer-side substitution evidence most closely. The other treats debit as generating no incremental sales relative to cash (Online Appendix **??**), removing the multi-homing channel from the credit-acceptance threshold so credit acceptance depends only on credit fees. It is silent on Durbin pricing because debit fees are zero by construction. For credit fee cap counterfactuals, both alternatives deliver welfare similar to the baseline ($23B and $28B vs. $27B).

The treatment effects estimation relies heavily on a single grocer adopting credit cards. If this grocer engaged in advertising or a promotion, that could bias results.

> Please discuss whether such events occurred, and acknowledge the limitation of relying on one case study.

Reply: Acceptance changes are rare (I find only two clear cases in the Homescan panel), and the estimates should be interpreted with this limitation in mind. That said, both events show similar patterns, suggesting the results are not driven by idiosyncratic features of any one retailer.

The main text now frames the merchant-type recovery as closer to a calibration exercise than a conventional estimation, given the limited variation in merchant adoption in a mature payments market. It also clarifies the identifying assumption and what would violate it. The key assumption is that merchants do not differentially target credit card users relative to other consumers of similar income levels. A confounding event would be one that simultaneously shifts merchant acceptance and consumer card holdings, for example a co-branded credit card campaign that onboards consumers onto a new card at the same time the merchant changes which cards it accepts. I excluded such events and verified that no similar campaigns accompanied the events I study. I verified against news reports that the acceptance changes were not accompanied by major advertising campaigns or promotions. I cannot name the retailers in the main text because the Homescan data use agreement requires retailer identities to remain anonymous.

> Clarify the discussion on pages A-34 to A-35. My understanding is that consumers do not observe debit vs credit preferences at adoption (preferences γ_{it} are integrated out). If a consumer knew she had bifurcated preferences, she would adopt both cards. Please explain this assumption more clearly.

Reply: The appendix microfounds the complementarity parameter χ from the main model. The idea is that consumers draw transaction-level shocks that determine whether a given purchase suits credit or debit. These shocks are realized at the point of sale, not at adoption. At adoption, consumers integrate over the distribution of future transaction shocks when choosing which cards to carry.

You are exactly right that if a consumer knew she would have bifurcated preferences, she would want to adopt both card types. This is the case of a large variance Σ of transaction-specific preferences. This is precisely one microfoundation for why the complementarity parameter χ is positive for credit-debit pairs. Consumers who expect to sometimes prefer credit and sometimes prefer debit have strong option value from holding both. A consumer who instead knows she strongly prefers credit has less option value from carrying debit and may benefit more from holding two credit cards, since there is also option value within a card type (e.g., different reward categories suit different purchases). The parameter χ in the main model captures this trade-off in reduced form. I have revised Online Appendix **??** to make these two points explicit.

> I was pleased with the passthrough discussion. Can the paper produce the counterfactuals underlying Table A.12?

Reply: Yes. Online Appendix **??** now re-estimates the model under zero passthrough and reports counterfactuals in Table **??**. Under no passthrough, merchants absorb fee changes entirely within their margins rather than adjusting retail prices.

For counterfactuals with small fee and reward changes, welfare effects are close to baseline. Uncapping debit raises total welfare by \$8 billion (versus \$7 billion), dual routing yields \$2.9 billion (versus \$3.8 billion, both net of the mechanical χ^{22} adjustment described in Section **??**), and CBDC yields \$1.4 billion (versus \$1.6 billion).

For counterfactuals with large net fee changes, deadweight loss creates differences. The fee cap welfare gain falls to \$17 billion from \$27 billion under full passthrough, and the monopoly gain falls to \$8 billion from \$15 billion. Under full passthrough, positive retail markups mean that cutting consumption raises deadweight loss, but merchant price cuts after a fee cap reverse this distortion. Under zero passthrough, the baseline economy has less deadweight loss because rewards expand consumption while merchant fees do not raise prices. Fee caps and monopoly both reduce consumption (through lower rewards) without any offsetting price reductions, so deadweight loss rises and total welfare gains shrink.

The surplus distribution under the fee cap also shifts: merchants retain the fee savings rather than passing them to consumers, so merchant surplus rises by \$49 billion while consumer surplus falls by \$31 billion (versus gains for both groups under full passthrough).

> Table 1 should indicate which dataset it uses and the number of observations.

Reply: The updated table caption refers to the Diary of Consumer Payment Choice. I have added observation counts to Table **??** to clarify the sample sizes underlying the summary statistics.

Estimation Equation 2 does not reference recent treatment effects literature (e.g., Sun and Abraham). If these methods do not apply to triple-differences with a continuous treatment, please acknowledge explicitly.

Reply: Recent work on treatment effect heterogeneity under staggered adoption (Sun and Abraham, Callaway and Sant'Anna) addresses bias when already-treated units serve as controls. This concern does not apply here. The control group consists of grocery stores that accepted credit cards throughout the sample. The two merchants that changed acceptance are the treated units, and they are never used as controls for each other. The estimated effect is a simple average treatment effect across the two events, not a staggered difference-in-differences. I have added language to Section **??** clarifying this design.

In Section III.C.2, how is "carrying" a card measured? Homescan likely does not record this. Please clarify.

Reply: I measure "carrying" based on observed usage in the Homescan data, not self-reported ownership. A consumer is classified as carrying a card if they use it during the sample period. Among non-cash consumers, I define the primary card as the one with the most trips and the secondary card (if any) as the one with the second-most trips. This usage-based measure is appropriate for the model, which captures payment behavior rather than ownership. A credit consumer who owns a debit card but never uses it is effectively a credit-only user for merchant acceptance decisions.

Why does P^w have a superscript "w"? Does the price index depend on a consumer's wallet, or is it being distinguished from another P?

Reply: Yes, the price index depends on the consumer's wallet. I have clarified this in the text following Equation **??**: "The price index depends on w because consumer utility depends on the cards they carry." Intuitively, a consumer carrying Visa sees a lower effective price at Visa-accepting merchants than a cash-only consumer does, so P^w aggregates prices weighted by how much utility a consumer with wallet w derives from each merchant's acceptance decisions.

In the displayed equation near the bottom of page 21, should y be indexed by i?

Reply: I appreciate the suggestion. The notation suppresses the i subscript on y to avoid cluttering the equations. Throughout the model section, y represents a generic consumer's income rather than being indexed by i. The context makes clear that y refers to the income of the consumer under consideration. I have opted to keep the notation as is to improve readability.

> The acronym PCE should be spelled out the first time it appears.

Reply: Fixed. I have spelled out Personal Consumption Expenditures (PCE) at first use.

> The first sentence of Section A.3.1 needs revision. Likewise, the last sentence of the "Cash-only consumers" paragraph on the same page is incorrect and needs fixing.

Reply: I have revised the first sentence of the "Defining Payment Preferences" subsection in Appendix **??** to clarify the purpose of the data cleaning steps. I have also corrected the last sentence of the "Cash-only consumers" paragraph.

> The first sentence of the proof of Theorem 1 (page A-24) requires revision.

Reply: I have revised the first sentence of the proof of Theorem **??** (the quasi-profit approximation theorem) to fix the grammar issue.

> In Figure A.6, the two lines appear indistinguishable. Are they always perfectly overlapping?

Reply: The lines are very close at observed fee levels, and the difference becomes visible only for fees substantially higher than current equilibrium.

> In Section C.4.0, the paper should explicitly name the figure being described.

Reply: I have added an explicit reference to the figure of the upper envelope in Figure **??**.

> In references, "et al." is fine in-text, but please list all coauthors in the bibliography.

Reply: I have verified that the bibliography lists all coauthors for every reference. The "et al." abbreviation appears only in the in-text citations (as is standard for papers with many authors), while the bibliography entries include the complete author lists.

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Detailed Response to Referee 4

Below, I group together three related comments about the substitution between debit and credit.

> The interpretation of the Durbin regulation is problematic. The 29% decline in signature debit volume for regulated issuers is taken as evidence of reward sensitivity, but debit use continued to grow overall, largely replacing cash, while exempt issuers promoted debit more aggressively post-Durbin. The difference-in-difference estimate does not imply reward-driven substitution to credit, and the claim of high reward sensitivity is overstated.
>
> The claim that Durbin reduced welfare by shifting debit users to credit is likewise speculative. Debit and credit cards are not interchangeable for many consumers—credit constraints, budgeting preferences, or maxed-out credit lines prevent substitution. The paper argues that debit and credit may be substitutes at the adoption stage, but this is only a theoretical possibility, without empirical backing, and no existing studies have documented substantial migration from debit to credit following Durbin.
>
> Without a convincing interpretation, the model likely misestimates reward sensitivity and debit-credit substitutability, risking misspecification and flawed policy conclusions. If debit and credit are not substitutes, Durbin may have improved welfare.

Reply: You are right that cleanly identifying debit-to-credit substitution is difficult, and no single piece of evidence is dispositive. The core question is whether credit constraints, budgeting preferences, or maxed-out credit lines prevent debit users from substituting toward credit.

I am not the first paper to suggest that Durbin caused migration from debit to credit. **Mukharlyamov.Sarin2025<empty citation>** exploit geographic variation in Durbin exposure and find that more-affected areas saw larger increases in credit card usage, providing direct evidence of debit-to-credit migration at the margin.

Two independent datasets suggest that many consumers choose debit as an active choice and not because they are constrained. In the NielsenIQ Homescan panel, around 75% of primary debit card holders also use a credit card (Figure **??**). In the DCPC, Table **??** shows that 79% of debit-primary consumers hold a credit card, 53% hold a rewards credit card, and their average credit utilization is only 27%. These consumers have the option of paying with credit but choose debit, so access is not the key barrier.

Even among the minority who do not hold a credit card, marketing surveys suggest that the inability to qualify is not the primary barrier. Around 37% cite debt aversion ("prefer not to carry any debt"), versus only 26% who report not qualifying (**Boehm2018**). In the second-choice survey (Table **??**), 47% of debit users report they would switch to credit if debit were unavailable.

Even if constraints matter, they do not affect the estimated welfare results. Online Appendix **??** re-estimates the model with binding credit constraints calibrated from credit score data. The counterfactual welfare effects are similar to baseline, because the consumers on the debit-credit margin are predominantly those with available credit capacity.

I agree that debit card volumes have grown dramatically over the past 10 years, but that too is consistent with my model. Aggregate trend data are descriptive only, but the pattern is consistent with substitution. Figure **??** shows that debit growth slowed relative to its pre-Durbin trajectory while credit exceeded its trend. Post-Durbin volumes confound the fee cap with concurrent changes—dual routing mandates, the premium credit card expansion (e.g., Chase Sapphire Reserve in 2016), and AmEx's merchant fee cuts—which is why the within-issuer difference-in-differences is the preferred identification strategy.

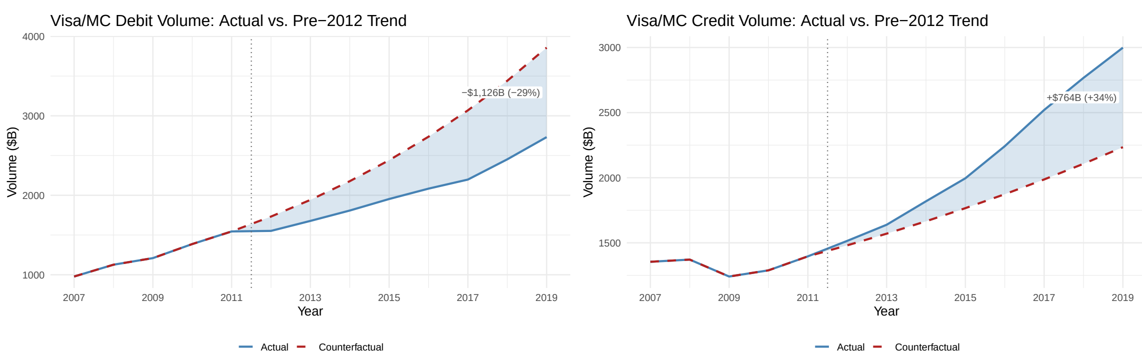

Figure A.2: Aggregate debit and credit card volumes over time, with pre-Durbin trend extrapolations

Australia provides a rough external plausibility check on the magnitude of migration between debit and credit under tighter regulation. Australia caps both credit and debit interchange and has a debit:credit ratio of roughly 59:41 (**Caddy.etal2020**), compared with 45:55 in the United States under current regulation. The closest analog in the model is the tourist test counterfactual (Online Appendix **??**), which caps both credit and debit at the cost of cash. It implies a card-only debit:credit ratio of roughly 69:31, somewhat more debit-heavy than Australia. The magnitudes are in the same ballpark, and the small remaining gap is in the direction the model predicts: the tourist test caps credit more tightly than Australia does, which pushes the mix further toward debit.

As a direct test of whether the estimated reward sensitivity is overstated, Online Appendix **??** halves the targeted Durbin moment, so the model matches only half the observed decline in debit volumes. This reduces reward sensitivity from 6.74 to 5.92, but the halved moment produces negative estimated marginal costs for Visa, which are difficult to reconcile with accounting data. The qualitative welfare conclusions are nonetheless preserved: capping merchant fees still improves welfare, and uncapping the Durbin Amendment would also improve welfare. Reducing reward sensitivity makes the merger-to-monopoly counterfactual harmful because networks have more market power over consumers.

> Before proceeding to quantitative counterfactuals, the paper should establish key model properties. Define a socially optimal benchmark and illustrate divergences from it. This would clarify mechanisms and improve interpretation.

Reply: The theoretical literature provides the relevant benchmark in simpler settings. **Rochet.Tirole2011<empty citation>** show that the welfare-maximizing interchange fee equates the merchant fee to the cost of cash when networks lack market power. **Edelman.Wright2015<e** show that intermediation reduces welfare when platforms extract surplus through price coherence. Both results rest on the same mechanism: consumers do not internalize the merchant fee's effect on retail prices. That mechanism is present in my model, so the cost-of-cash cap retains its interpretation as the social benchmark.

The cost-of-cash cap (approximately 30 bp) replicates the Rochet-Tirole tourist test, and Online Appendix **??** confirms that even the more moderate 120 bps cap captures roughly 77% of what the Ramsey planner's solution delivers. The contribution of the paper is not to characterize the optimum but to quantify the welfare effects in a setting with realistic consumer heterogeneity, multi-homing, and network competition.

> The counterfactuals are incomplete. The paper shows capping both debit and credit card fees helps, and that capping debit alone hurts. But what about capping credit card fees only? What is the optimal cap level? These questions remain unanswered.

Reply: The revised draft adopts this suggestion: the main counterfactual caps credit card merchant fees at 120 bp while leaving the Durbin Amendment cap on debit unchanged. Together with the appendix analysis, the paper now covers the full space of fee cap policies. The credit-only cap at 120 bps generates $27 billion in welfare gains (Section **??**).

The tourist test in Online Appendix ?? caps both credit and debit fees at the cost of cash (roughly 30 bps), adding only $2 billion more. The Ramsey planner provides the optimal cap level computationally: it achieves $35 billion in total gains, so the 120 bps credit-only cap already captures roughly 77% of the planner's welfare gains without directly regulating rewards.

> The finding that competition reduces welfare is based on an extreme case (merging all networks into one). Yet the paper also shows that adding a public debit network helps. What is the optimal number and composition of networks? Please explore or discuss.

Reply: The model is not well-suited to determine the optimal number of networks. Introducing a new network requires specifying its unobserved characteristics and generates new logit shocks that mechanically affect adoption patterns. Without a strong microfoundation for what these additional characteristics represent, the optimal network count would hinge on assumptions that the model cannot discipline.

The public debit counterfactual is not intended as a statement about optimal network composition. It addresses a specific policy question—whether expanding public presence in payment markets, as several countries have considered, would improve welfare.

> The assumption that debit and credit card features are fixed and unaffected by fees is vulnerable to the Lucas critique. Post-Durbin, banks changed checking account terms (fees, minimum balances). Similarly, if credit fees were capped, issuers could tighten criteria or change service quality. Without accounting for issuer responses, the welfare analysis is incomplete.

Reply: You raise a valid concern. The model is best interpreted as providing short-run predictions in which card characteristics are held fixed. In the longer run, issuers could respond to fee regulation by adjusting credit criteria, reducing non-pecuniary rewards or service quality in ways the model does not capture.

The Australian experience provides some reassurance on this point. Australia has had interchange fee caps since 2003, and Appendix Figure ?? shows that non-rewards annual fees did not rise, and the spread between credit card rates and term loans was unchanged following regulation. However, I cannot rule out that issuers adjusted non-price dimensions of quality. I have added language in Section ?? acknowledging this limitation and noting that the welfare counterfactuals are most reliable as short-run predictions. Section ?? contextualizes this limitation alongside other sensitivity analyses in the main text. To the extent that issuers would respond to fee caps by reducing card quality, the model would overstate consumer welfare gains from regulation.